the end of the penis.

Answer : D

5.2.3 Gametogenesis

Gametogenesis is the process of gametes formation (oocytes / sperm) from bi - potential **primordial germ cell**. A haploid daughter cell (n) is formed from a diploid mother cell (2n) through meiosis.

- In male it is called **spermatogenesis**, produces spermatozoa
- In female it is called oogenesis, produces ova

Spermatogenesis

Spermatogenesis is the process of producing sperm cells in the testes. Haploid (n) spermatozoa develop from germ cells in the seminiferous tubules of the testis, it starts at puberty. The testes also produce the male hormone called androgen (testicular hormone) .

Germinal epithelium cells are **primordial germ cells** (PGCS), lining seminiferous tubules ; at sexual maturity **PGCs** are mitotically divide to produce **spermatogonia** (2n).

Steps of spermatogenesis:

- Spermatogonia (2n) $\xrightarrow{\text{mitosis}}$ **Primary spermatocytes (2n)**
- Primary spermatocytes (2n) $\xrightarrow{\text{mitosis}}$ two haploid secondary **spermatocytes (n)**
- Second meiotic division resulting in production of four equal and haploid **spermatids**.
- The **spermatids (n)** $\xrightarrow{\text{spermiogenesis}}$ matured **spermatozoa** (sperm).
- Maturation of the sperm cells takes place in the epididymis.

Some of PGCs cells are tall and somatic called **Sertoli cells**, used to nourish (nursing) developing sperms.

Oogenesis

Oogenesis is the process of producing the female egg cell. The primordial germ cells in the outermost layers of the Ovaries divide mitotially to produce oogonium. About 1 - 2 million of cells are formed in the embryogenesis before birth. Steps of oogenesis :

- **Oogonium** undergoes mitotic division to produce primary oocyte (2n), which is arrested until puberty.

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- Primary oocyte (2n) undergoes meiosis to produce secondary oocyte (n) and a smaller polar body (n).
- Then the secondary oocytes are arrested in the ovary until ovulation date, and then developed in to oocyte and polar bodies. Polar bodies are non - functional egg cells that can't be fertilized.
- At the end of meiosis II, one oocyte and three polar bodies are produced

5.2.4 Menstrual Cycle

Menstrual cycle is the process of discharging blood and other things through the vagina. The blood is due to shedding of the lining of **endometrium** which occurs every month from puberty to **menopause** excluding pregnancy if ovum is not fertilized.

- It includes changes that occur in the ovary and the uterus, the level of hormones in the blood.
- The duration of the cycles on average is about 28 days
- In some women it can range from 20 days to 45 days, the difference in the duration is associated with decreased fertility.

Phases of the Menstrual Cycles :

There are four phases of menstrual cycles. These are: follicular phase, ovulation phase, luteal phase and menstrual phase.

1. The follicular phase :

- It is the event when the anterior pituitary produces follicles stimulated hormones (FSH), which stimulates.

Follicular maturation and development of endometrium.

- Happen at day 5 - 13 after menstruation

2. The ovulation phase :

- Takes place in the midway (days 12 - 14)
- Estrogen stimulates hypothalamus to produce Gonadotropin Releasing Hormones (GnRH), then
- GnRH stimulates the anterior pituitary to secret **luteinizing hormone** (LH) to release an egg. It is an example of **a positive feed back control**

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3. The Luteal phase:

Luteal phase is the phase that follicle erupted and become **corpus luteum** to secret high **progesterone** and lower **estrogen** hormones. It ranges from 15 - 28 days after menstruation. The functions of estrogen and progesterone hormones are :

- These hormones stimulate **thickening of the endometrial lining** in preparation for pregnancy
- These hormones inhibit the secretion of **FSH** and **LH** to prevent any follicle from developing. It is an example of a **negative feed back control**

Note : A feedback control mechanism is a physiological regulation system in the body that works to return the body to its normal internal state, commonly known as homeostasis.

4. The Menstruation phase:

The menstruation phase is the last phase of the menstrual cycle, which is preceded by shedding of endometrium and discharge of blood via the vagina or continuation of pregnancy. There are two possibilities :

- If fertilization occurs, the embryo develops, implants in the endometrium and release hormones to sustain the corpus luteum.
- If fertilization fails, the corpus luteum **degenerates**. estrogen and progesterone levels drop and forms **corpus albinos.** Then endometrium is shed and eliminated from the body in the form of menstrual blood or period.

Luteinizing Hormone (LH) and Follicle Stimulating Hormone (FSH)

LH and FSH are very import hormones to control and regulate fertility in both male and female

Luteinizing Hormone (LH)

- LH from pituitary gland regulates the function of the testes in men and ovaries in women.
- In **men**, it stimulates cells in the testes to produce **testosterone** to support **sperm production.**
- In women, it stimulates the ovarian follicles to produce the female sex hormone **(estrogen)** and release of a mature **oocyte** (egg) from the

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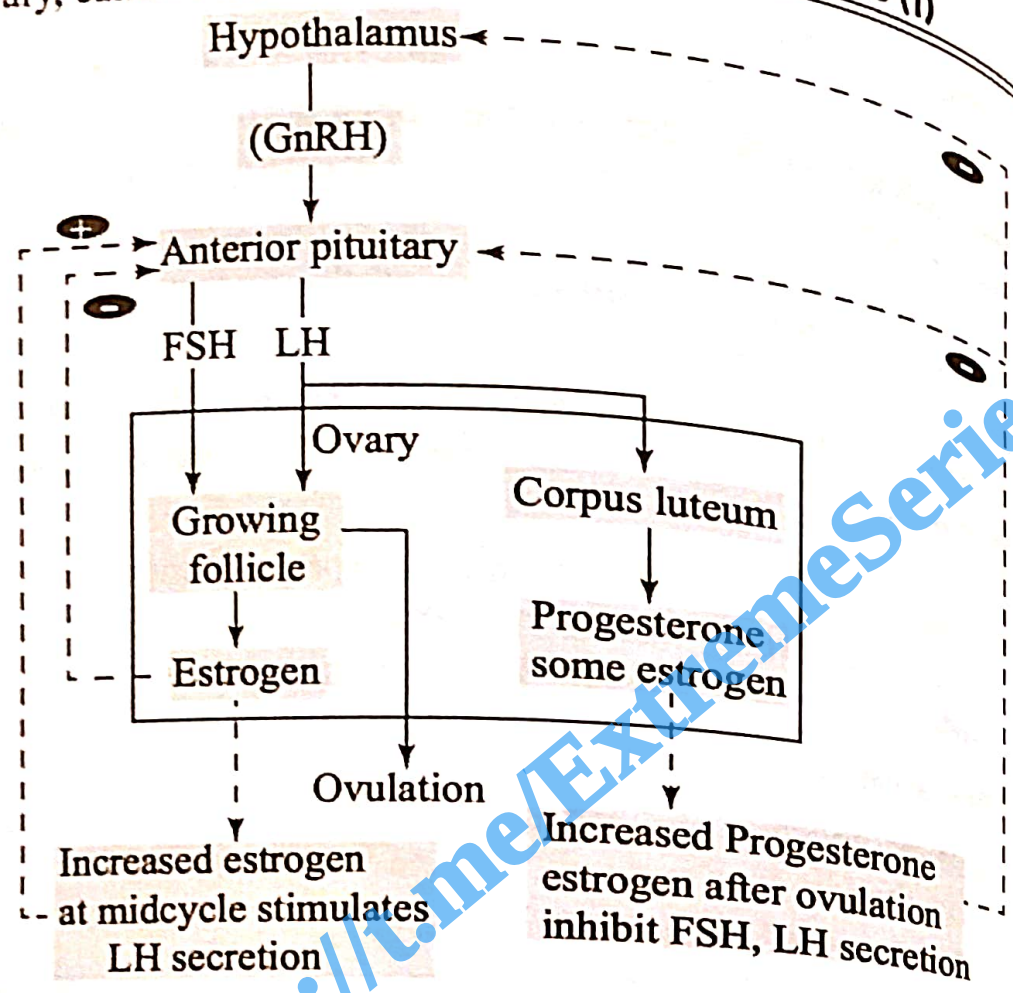
ovary, called ovulation.



Follicle Stimulating Hormone (FSH)

- FSH plays a role in sexual development, reproduction and affe ovaries and testicles.
- Women are born with all the eggs (1 - 2 million) at birth, of wh around 400, 000 eggs reach puberty, 400 eggs mature and ovul throughout her reproductive years.
- The rise in FSH stimulates the maturation of the follicle (egg) in t ovary, produces **estrogen** and reduces the release of gonadotrop hormone by hypothalamus as negative feedback.

Fertilization and Pregnancy

Fertilization: is the union of egg and sperm cells in the fallopian tub oviduct to produce zygote. After 6 days of fertilization, embryo attaches the wall of uterus, it is called **implantation.**

Pregnancy : is a period in which the fetus develops inside the uterus;

- It lasts about 40 weeks until delivery
- It is counted from the last period

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- A pregnancy with more than one fetus at a time is called **multiple pregnancy** (twin or triple)

Illustrative Example

20. Arrange the following stages into correct sequence

1. Spermatogonium
2. Spermatids
3. Primary spermatocyte
4. Secondary spermatocyte
5. Sperm

Explanation :- The steps of spermatogenesis (production of sperm) is : spermatogonium → primary spermatocyte → secondary spermatocyte → spermatids → sperm

21. The main difference between spermatogenesis and oogenesis is that oogenesis is :

A. The process begin before birth
B. Serotoli cells nourish immature sperms
C. Takes place in the testes
D. Produces sperm cells

Explanation :- The main difference between spermatogenesis and Oogenesis are :

Feature	Spermatogenesis	Oogenesis
Sex cell / gamete	sperm	Ova / egg
Takes place in	testes	ovary
Time, the process begins	After puberty	Before birth
Maturation / nursing of immature gametes is by	Serotoli cells	Graphian follicle
Total number of gametes at the end of meiosis II	4 sperm cells	1 oocyte and 3 polar bodies

Answer: A

22. Which one of the following is not true about the phases of menstrual cycle ?

A. Follicular phase - maturation of egg
B. Ovulation phase - release of egg from ovary
C. Luteal phase - the secretion of testosterone and progesterone

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D. Menstrual phase - corpus Leuteum degenerated

Explanation :- the four phases of menstrual cycle is :

Phases of menstrual cycle	Events / activities	Days
Menstrual phase	Corpus leuteum deenerated, menstruation occur	1 - 4
Follicular phase	Follicle / egg maturation, endometrium developed	5 - 13
Ovulation phase	Ovary release egg in to oviduct	14
Luteal phase	Follicle becomes the corpus luteum ; produce high progesterone and low estrogen ; endometrium developed	15 - 28

Answer : C

23. An example of positive feedback control by estrogen, produced from matured follicle is ____?

A. To increase LH
B. to increase LH and FSH
C. to decrease GNRH
D. to decrease LH and FSH

Explanation :- an example of a **positive feedback control** at follicular phase of menstruation :

- Matured follicle produces high amount of **estrogen** ; then
- **Estrogen** stimulates hypothalamus to produce **Gonadotropin Releasing Hormones** (GnRH), then
- GnRH stimulates the anterior pituitary to produce **luteinizing hormones** (LH) to release an egg

Answer : A

5.2.5 Mechanism of Action of Contraceptives

Contraception is the Intentional (deliberate) process of preventing pregnancy. The major contraception methods are classified as : **Barrier** and **non - barrier** methods

I. Barrier (physical) method

Barrier (physical) method of contraception prevents the union of sperm and ovum (egg). This includes : Condom, Diaphragm and Cervical cap

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i. **Condom :** is a sheath that is used to prevent the sperm from entering into the vagina.
   - ➥ It can be a male and female condom, diaphragm, and cervical cup

ii. **Diaphragm :** is a small and soft silicon domed placed in the vagina
   - ➥ Covers cervix and surrounding area
   - ➥ Stops sperm cell reaching the egg
   - ➥ Reusable, removed within 6 hrs of stay after sex
   - ➥ Needs spermicide for effectiveness
   - ➥ No protection for STI

iii. **Cervical cap :** is a small cup shaped disk, which cover the bottom of cervix.
   - ➥ Smaller than diaphragm
   - ➥ Reusable up to one year ; can be used up to 48 hrs once placed
   - ➥ Needs spermicide for effectiveness

2. Non - barrier methods :

Non - barrier methods of contraception include natural method, hormonal methods, IUD, sterilization, and spermicides.

I. Natural method

It is carried out naturally with out using any physical barrier or hormonal methods These are :

- ➥ **Rhythm or calender method :-** It is based on the understanding of the menstrual cycle and prediction of the ovulation time. This method works perfectly for a woman having a regular menstrual cycle.
- ➥ **Breast feeding :-** A breast feeding woman does not ovulate while she is feeding breast ; because the level of progesterone is the blood is high to regulate the feeding breast
- ➥ **Cirvical mucus method:** It is based on cyclical vaginal secretions change throughout cycle.

just before and after ovulation you have clear and slipper mucus, stretched between your finger ; not safe days (pregnancy can happen)

after ovulation you have less mucus which looks like cloudy and sticky, which is safe days (no pregnancy)

- ➥ **Basal body temperature :** it is based on changes in body temperature

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around ovulation time

Just after ovulation, the release of progesterone causes slight increase in temperature and drop immediately before ovulation.

- **Withdrawal method :** during orgasm the male ejaculates the semen outside of the female genitals.

II. Hormonal method :

Hormonal method of contaception is a type of birth control that uses hormones to prevent pregnancy. There are different types of hormonal contraception. These are : combined hormone pill, progesterone only pills, hormone implants, post pill and contraceptive ring (vaginal ring)

i. Combined hormone pill :

- It is oral contraception
- Contains estrogen and progesterone hormone
- Rising of estrogen level in the blood is detected by the pituitary gland.
- This slows down the production of follicle stimulating hormone (FSH)
- This causes delaying of maturation of follicles

  An example of negative feedback control
- Progesterone increase the mucus in the cervix and vagina to prevent entry of sperm

ii. Progesterone only pills :

- It is an oral contraception
- Stop ovulation, thickening mucus of cervix to block sperm reach to an egg

iii. Post pill :

- It is an emergency contraception taken within 72 hrs. of open sex
- It is an oral contraception
- Contain levonorgestrel (progesterone) hormone
- It prevents ovulation, block fertilization, stop implantation

iv. Contraceptive ring (vaginal ring) :

- It is a small flexible plastic ring containing estrogen and progesterone and placed in the vagina
- Release the hormone into the blood continuously
- It stops ovulation

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- It is used for 3 weeks once inserted

v. **Hormone implants :**

- They are small flexible plastic / silicon capsule containing progesterone hormones placed under the skin of upper arm of women
- The hormones are released slowly into the blood and prevent pregnancy for three years
- They can be removed if the women want to get pregnant

vi. **Hormones injections**

- It is a method of injecting the hormone progesterone
- Also known as DEPO provera
- It prevent pregnancy by stoping ovulation
- It can prevent pregnancy up to three months

III. Sterilization or Surgical Contraception

Sterilization or surgical contraception is a permanent method of contraception. It can be applied to both men and women. It does no affect hormones, desire to sex and sexual functioning.

a) **Vasectomy:** is a male sterilization, it involves cutting and tying of vasdeferenes (sperm duct) to revents the passage of sperm along the urethra. Once cut and tied it is difficult to restore, thus, such a method requires a big decision.

- Ejaculations contain only semen but no sperm cells
- The stored sperm cells are absorbed by the body Thus, such a method requires a big decision

b) **Tubectomy** - is a female sterilization, it involves cutting and tying the oviduct (fallopian tube). This prevents the release of egg from ovary to fallopian tube where fertilization takes place.

IV. Intrauterine Device (IUD)

Intrauterine devices are T - shaped device, made up of plastic and copper, and progesterone hormone.

- It is inserted to the uterus by trained professional
- It prevents fertilization and implantation

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V. Chemical Barrier / Spermicides

Spermicides : are chemical that kills sperm. It include : foam, sponge, creams, gels, film and suppositories

- Not effective enough to be used without a condom or other method
- Applied before sexual intercourse
- No protection from STIs.

24. Explain the mechanism how combined hormonal pill contraception prevent pregnancy ?

Explanation : - combined hormonal pill

- Contains estrogen and progesterone hormones
- Rising of estrogen level in the blood is detected by the pituitary gland
- This slows down the production of follicle stimulating hormone (FSH)
- This causes delaying of maturation of follicles (egg)

25. Which of the following is **NOT** a natural birth control method ?

A. Breast feeding
B. Calendar method
C. Withdrawal method
D. Implant

Explanation : -

- The natural birth control is a contraceptive means without any artificial methods (barrier or hormones)
- It includes : breast feeding, calendar method, basal body temperature, cervical mucus, withdrawal and rhythm method.

Answer : D

26. Rearrange the following contraceptive methods according to increasing level of their effectiveness

1. Condom
2. Pills
3. Sterilization
4. Calendar method
5. Hormonal implant

A. 1, 2, 3, 4, 5
B. 5, 4, 3, 2, 1
C. 3, 2, 5, 1, 4
D. 3, 1, 5, 4, 2

Explanation :- effectiveness of contraceptive methods is :

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Contraceptive methods	Effectiveness
Sterilization (tubal ligation, vasectomy)	Closer to 100 %
Pills	99 %
Hormonal implant or injection	99 %
Condom, Diaphragm	97 %
Calendar method	90 %
Spermicides	

Answer : C

Question number 27 is based on the following list of contraceptive methods

1. Diaphragm
2. Condom
3. Intra uterine device
4. Combined pill
5. Cervical cap

27. Which contraceptive methods are physical barriers ?

A. 1, 2, 3 and 4
B. 3 and 4
C. 1 and 3
D. 1 and 2

Explanation :- Physical / barriers methods of contraception are : condom (male and female), cervical cap and diaphragm.

Answer : D

28. Which of the following contraceptive methods prevents fertilization and implantation ?

A. The diaphragm
B. The combined pills
C. The intra uterine device
D. Sterilization

Explanation :- Intra uterine devices (IUD) are contraceptive methods that are capable of preventing fertilization and implantation.

Answer : C

29. Burying a small plastic capsule contraception under the upper arm of a female is ____ ?

A. Mixed pills
B. Intra uterine device
C. Hormone injection
D. Hormone implant

Explanation :- Hormone implants are a flexible thin plastic capsules containing progesterone hormone, inserted under the skin of the arm.

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30. How does the diaphragm control unwanted pregnancy ?

A. Avoiding sex during the fertile time

B. Placing over the penis to collect semen

C. Inserting into the vagina to cover the cervix area before sexual intercourse

D. It uses synthetic female sex hormones

Answer : D

Explanation :-

- Diaphragm is inserted into the vagina before sexual intercourse
- It covers the cervix and prevents the passage of sperm
- Addition of spermicides (sperm killing cream) increases the effectiveness.

31. Chemical barriers / spermicides are available in the form of ?

A. Suppositories

B. Cream, foam

C. Gel, film

D. All

Answer : C

Explanation : -

- Chemical barriers / spermicides are applied in the vagina before sexual intercourse to kill sperm cells
- It includes foam, sponge, suppositories, creams, gels, film and suppositories.

Answer : D

5.2.6 Causes of Infertility in Humans

Infertility : is the inability of a person to reproduce by natural way.

- It is inability to get pregnant after unprotected sex in female
- It is a low sperm count, blockage, abnormal sperm production in male

Causes of Infertility in Women

The main causes of infertility in woman includes : thyroid problem, pelvis surgery, endometriosis, fibroids, pelvic inflammatory disease, sterilization, medicine and drug.

- **Thyroid problems :** Hypothyroidism (low thyroxin) or hyperthyroidism (high thyroxin) interferes with the ovulation process.

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- **Pelvic surgery :** damages and scars on the fallopian tubes and a cut on ovaries
- **Cervical mucus problems :** Harder mucus does not let the sperm to swim in to vagina.
- **Fibroids :** is the non - cancerous growths in the uterus lining the endometrium
- **Endometriosis :** is a condition where uterine like tissue grows outside the uterus
- **Pelvic inflammatory diseases :** are due to infections of the upper female genital tract.
- **Sterilization :** is due to a choice not to have children
- **Medicine and drugs :** have side effects on fertility

Causes of Infertility in Men

The main causes of infertility in men include :

- Production of low sperm count (bellow 15 million / mm of semen per ejaculation).
- Abnormal sperm cell production
- Blockage of the passage of sperm
- Chronic health problems, illness, injuries, inherited disorder
- Hormonal imbalance.

Note :The main risk factors of infertility in men includes smoking, drug and alcohol use, obesity, trauma on testes

5.2.7 The Major Sexually Transmitted Infections (STIs) in Ethiopia

Sexually transmitted diseases (STDs) are infectious diseases spread by sexual contacts.

- The major causative agents are bacteria virus, fungi and parasite
- It is a growing problem in Ethiopia and developing countries

The most commonly known STDs are :

1. **Gonorrhea (gonococcal infection)**

- The causative agent is **Neisseria gonorrhoeae** (bacteria)
- It affects the mucus areas of the body (the vagina, penis, throat and

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rectum).

- Transmission is through sexual contact, unprotected sex, having many sexual partner
- Symptoms : burning sensation while urinating and a yellowish - white discharge from penis or vagina.
- If a pregnant woman has untreated gonorrhea, she can pass the infection on to her unborn baby, which results in blindness.
- Prevention : use protective sexual contact (a male or female condom), be faithful to your partner.
- Treatment : it can be treated by using antibiotic.

2. Syphilis

- The causative agent is **Treponema pallidum** (bacteria)
- Transmission : through sexual contact and from mother to foetus
- Prevention / treatment : protective sexual contact and medical treatment (antibiotics).

The stage of syphilis development :

i. **Primary stage** (3 - 4 weeks later) : painless and non - itchy chancers around genital organ, rectum and mouth.

ii. **Secondary stage** (4 - 8 weeks later) : The common symptom is a rash on hands and soles of feet ; tiredness, fever, sore throat, swollen glands

iii. **Latent syphilis** : it is asymptomatic (no symptom),it happened within 1 year after infection)

iv. **Tertiary (late syphilis)** : it involves illness in the skin (gummatous), central nervous system (neurological) and heart (cardiovascular). It happens within 1 - 10 years after infection.

3. Chancroid :

- Causative agent is **Haemophilus ducreyi** (bacterial)
- It is more commonly seen in men than in women
- Transmission : unprotected sexual contact
- Highly contagious, sore ulcerations on the genitals
- Prevention / Treatment : protective sexual contact ; it can be treated by antibiotics

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4. Bacterial vaginosis :

- Causative agent : **Bacterial vaginosis**
- It is due to overgrowth of bacteria naturally found in vagina
- Have symptoms of vaginal discharge, burning during urination
- Post - surgical infection increases the risk
- Prevalence in Ethiopia is 2.8 - 19.4 %

5. Chlamydia :

- Causative agent : **Chlamydia trachomatis**
- Symptoms : painful urination, discharge from genital, painful sexual intercourse in woman, testicular pain in men
- Transmission : sexual contact
- Prevention / treatment : protective sexual contact ; medical treatment

6. Genital herpes :

Causative agents : herpes simplex viruses such as

a) **Human simplex virus 1 (HSV - 1)**

- Affect the mouth (sore around the mouth) and spread through saliva

b) **Human simplex virus 2 (HSV - 2)**

- Affect genital organs, anal and the mouth ;
- transmitted through sexual intercourse
- No cure for it, clear up by themselves

7. Genital warts :

- The causative agent : **Human papilloma virus (HPV)**
- It affects the skin, mucous membrane, throat, cervix, anus and mouth
- It increases the risk of cervical and throat cancer.

8. Molluscum contagiosum :

- It is a viral infection affecting the skin
- It affects both adult and children
- The causative agent is **Molluscum contagiosum**
- Symptoms : a small round bump
- Transmission : skin to skin contact (contagious)

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9. Human Immunodeficiency Virus (HIV)

Human Immunodeficiency Virus (HIV) is retrovirus that attacks the human immune system cells, such as T - helper cells / **CD4 - T lymphocytes** and organelles such as ribosomes, Golgi bodies and endoplasmic reticulum.

- This makes a person vulnerable to **opportunistic infections** such as TB, Kaposis sarcoma, pneumonia, etc
- When it gets worse over time it becomes AIDS

HIV infection in human body has three stages : Acute stage, chronic stage / clinical latency, and AIDS stage

i. Acute stage

- 2 - 4 weeks after infection
- Shows a symptom of flu, fever, headache
- Viral load increases ; CD4 cells are declined

ii. Chronic stage / clinical latency

- 1 - 7 years after infection
- Usually asymptomatic (no symptoms)
- Viral load sightly increases, CD4 cells are declined but keep replacing

iii. AIDS stage

- 10 years and above after infection
- The most severe stage is when CD4 cell decline to 200 / mm^3 of blood
- Thus, opportunistic infection increases

Note : You can get HIV through contaminated sharp materials, mother to child during (pregnancy, birth and breastfeeding), unprotected sex, blood transfusions, and organ / tissue transplants.

10. Hepatitis B:

It is a viral infection that attacks the liver. Acute hepatitis B causes liver cancer (**Cirrhosis and hepatocellular carcinoma**). 82000 deaths every year globally due to liver cancer or liver failure (WHO, 2019).

- ☑ The virus remains in the semen, blood and other body fluids
- ☑ **Transmissions are** through sexual contact ; infected mother to child

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during delivery or blood, body fluids contacts, and on - sterile equipment for injection and puncturing the skin with sharp objects.

☑ **Symptoms** : Yellowish of the skin and eyes (jaundice), dark urine, extreme fatigue, nausea, vomiting and abdominal pain

☑ Prevention / treatment : protective sexual contact and avoid contact of bodily fluids or contaminations and Vaccination

11. Crabs (public lice) :

- It is a parasitic disease
- Causative agent is crabs
- affect hair containing skin, and
- spread through close physical/ sexual contacts, bed linen or towel

12. Trichomoniasis :

- The causative agent is **trichomonas vaginalis** (parasitic) ;
- Have symptoms of vaginal discharge, pain during urination or sex, itching.
- Transmission is through sexual contacts

13. Scabies :

- The causative agent is sarcoptes scabiei
- It is a parasitic skin disease
- Spread by skin to skin contact, clothing, sheet, towel, furniture from infected person.

5.2.8 Epidemiology of STI in Ethiopia

Epidemiology : is the study the frequency, distribution and determinants of disease, other disease related issues and its control.

- Very little information on the prevalence and incidence of STI in Ethiopia.
- The highest reported rate of STI is in the range of 15 - 24 years old.

Risks of STIs Epidemic in Ethiopia

The main risk factors that can increase the chance of getting STI in Ethiopia are :

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- Unprotected sex
- Being sexually active
- Having history of STI
- Rape and assault
- Sharing contaminated sharp materials
- Being young
- Misuse of alcohol and drug

How do alcohol and drugs increase the chance of getting STI?

- Because it inhibits your mental judgement and lead to unprotected sex, unwanted pregnancy, and STI.

Illustrative examples

32. Among the sexually transmitted disease, which is not caused by bacteria

A. Syphilis
B. Gonorrhea
C. Chancroid
D. AIDS

Explanation :-

Pathogens of STIs	Examples of STI
Bacteria	Gonorrhea, syphilis, chlamydia, bacteria vaginosis, chancroid
Virus	AIDS, genital herpes, genital warts, hepatitis B, Molluscum contagiosum
Parasite	Crabs, trichomoniasis, scabies
Fungus	Candidiasis

Answer : D

33. At Which stage of syphilis, symptoms of illness in the skin, bones, central nervous system are developed ?

A. Primary stage
B. Secondary stage
C. Tertiary stage
D. Latent stage

Explanation :- syphilis infection has 4 stages :

- At primary stage : painless and non - itchy chancers around genital organ, rectum and mouth.
- At secondary stage : rash on hands and soles of feet ; tiredness, fever, sore throat, swollen glands
- At latent stage : asymptomatic (no symptom)

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➥ At tertiary stage : illness in the skin (gummatous), central nervous system(neurological) and heart (cardiovascular).

Answer : C

34. Which of the following disease and its causative agent is mis - matched
A. Gonorrhea - Neisseria gonorrhea
B. Syphilis - Treponema pallidum
C. Chancroid - Heamophilus ducrie
D. Chlamydia - human papilloma virus

Explanation : STI

Types of STI	Causative agents
Gonorrhea	Neisseria gonorrhea
Syphilis	Triponema pallidum
Chlamydia	Chlamidia trichomatis
Chancroid	Haemophilus ducreyi
Scabies	Sarcoptes scabiei
Genital warts	Human papilloma virus
Trichomoniasis	trichomonas vainalis
Bacterial vaginosis	Bacterial vaginosis
AIDS	HIV

Answer : D

35. Which one of the following is true about the stages of HIV / AIDS in the body ?
A. At acute stage the viral load and CD4 count increases
B. At clinical latency it has no symptoms
C. At AIDS stage CD4 and viral load decreases
D. At AIDS stage opportunistic infections increases

Explanation :- HIV / AIDS has three stages : Acute stage, chronic / latency stage and AIDS stage

➥ At AIDS stage the viral load increases, CD4 cells declined, opportunistic infections rises

Answer : D

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5.3 Harmful Traditional Practices

Harmful traditional practices are a violation of human right that put the reproductive health and right at greater risk, affect the physical and mental health of individuals. In Ethiopia, more than 20 different harmful traditional practices, and the FGM alone accounts for 73 %. Examples of Harmful traditional practices are :

I. Female Genital mutilation (FGM) : it is the partial or total removal of external female genitals. In Ethiopia it is classified in to three main categories.

These are Clitoridectomy, Excision and infibulation

Clitoridectomy : is partial removal of clitoris or total removal of labia minora. It is more common in Ethiopia

Excision : is the removal of part of clitory or part of the labia minora

Infibulation : is cutting the external genitals and stitching, just leaving a small opening.

What is the effect of FGM ?

- ➥ It lefts painful scar, and excessive bleeding
- ➥ Infections in the urinary tract
- ➥ Irregular flow off menstruation, difficulty in urinating,
- ➥ Difficult and unsatisfied sexual intercourse
- ➥ Prolonged child labor

II. Early marriage

- ➥ It is an act of forcing children to get marriage
- ➥ Have an impact of school drop, high risk of pregnancy and birth complication or maternal death

III. Kidnapping : it is an abduction to make a girl wife forcefully

IV. Gender - based violence : It is an unwanted sexual contact, sexual abuse or harassment within relative or marriage.

5.3.1 Family Planning

Family planning is the ability to anticipate and attain desired number of children, spacing and timing of birth, acquiring resources for raising the children, time, social, financing, and environmental conditioned.

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5.3.2 Risks Related to the Lack of Family Planning

- Maternal health during pregnancy, child birth and the post - partum period.
- Early pregnancy in teen increases a higher risk of complications and death

5.3.3 Family Planning Actions :

Prevention of unwanted pregnancy, safe from STIs, get resources to prevent pregnancy, waiting until 6 months after abortion, adoption, use birth control and assisted reproductive technology, use calendar method.

5.3.4 Family Planning Services :

Pregnancy test and counseling, pregnancy achieving services, counseling on infertility, counseling on STI, breast and pelvic examination, breast and cervical cancer examination.

Effects of alcohol use, chewing khat, cannabis and other drug uses on STIs transmission and unwanted pregnancy

Peoples who are addicted to alcohol use, chewing khat and other illegal drugs are highly vulnerable to STI including HIV / AIDS and unwanted pregnancy.

5.4 Effects Using of Alcohol, Khat, & Drug on STI & Unwanted Pregnancy

5.4.1 Effects of Alcohol Use :

The normal blood alcohol limit for **euphoria** (happiness) is 0.1 %.
Excessive alcohol use in people results in :

- Chronic disease such as high blood pressure, heart disease, stroke, liver disease, and digestive problems, dementia, sleepless and confusion, brain damage, cirrhosis,
- Violence, traffic accidents and **death** (6.2 % males and 1.1% females' globally).
- Risky sex and HIV transmission

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5.4.2 Effects of Chewing khat :

Khat is a shrubs plant known as **Catha edulis**, it is a popular leaf in the horn and East Africa, Europe, and North America.

- Contain a drug called **cathinone**, which stimulate CNS
- People chew the leaves as a recreational **stimulant**
- **The immediate effect** of chewing khat is alertness, concentration, confidence, euphoria, friendliness, increased (blood pressure, heart rate), insomnia, psychosis, lower appetite, talkativeness.
- Long term effect of chewing khat is depression, hallucination, heart attack, extreme psychosis, oral cancer and death.
- In Eastern Ethiopia it is chewed by 70 % adolescent boys and 30 % adolescent girls.

5.4.3 Effects of Drug Use :

Drugs are substances that change a person's mental or physical state. They can affect the way your brain works, how you feel and behave. Some drugs immediately affect the body and brain and other have long lasting health problem. Some of the examples of harmful drugs are marijuana, heroin, cannabis, cocaine and inhalants.

1. Marijuana :

- Extracted from a plant called **Cannabis sativa**
- The high potency strain is **hashish**
- It harms the brain, breathing system, heart, child development, muscular system.

2. Cocaine :

- Extracted from plant called **Erythroxylem coca**
- It harms the systems of breathing nervous, gastrointestinal and cardiovascular.

3. Heroin :

- This drug is made from morphine (a resin of opium poppy).
- The long term effects of using heroin are : liver, lung and kidney disease, mental disorder, and miscarriage,

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4. Inhalants :

- Inhalants are chemical found in ordinary households or work place and inhaled to get in high mood but have health problems.

Supplementary Exercise

36. The largest percentage of Harmful traditional practices in Ethiopia is ___ ?
   A. Mutilation of skin
   B. Early marriage
   C. Female genital mutilation
   D. Gender based violence

37. The immediate effect of chewing khat is _____
   A. Myocardial infraction
   B. Depression
   C. Talkativeness
   D. Hallucination

38. Which of the following female genital mutilation involves cutting the external genitals and stitching, just leaving a small opening ?
   A. clitorydictomy
   B. infibulation
   C. excision
   D. circumcision

39. Family planning is defined as _____ ?
   A. Spacing and timing of birth
   B. Pregnancy test and counseling
   C. Adoption
   D. All

40. Which of the following is true about the effect of drinking alcohol ?
   A. Risky sex
   B. Cirrhosis and digestive problem
   C. Traffic accidents
   D. All

41. Human papilloma virus (HPV) is the causative agent of ____ disease ?
   A. Genital warts
   B. Genital herpes
   C. Hepatitis B
   D. Molluscum contagiosm

42. Which of the following cell is haploid (n) ?
   A. Primary oocyte
   B. Secondary oocyte
   C. oogonium
   D. Corpus luteum

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43. The hormone that reaches its highest level after ovulation / luteal phase is
A. Estrogen
B. Progesterone
C. FSH
D. LH

44. Which of the following **is not** a hormonal type of contraception ?
A. Depo - provera
B. diaphragm
C. oral contraceptives
D. implant

45. _____ is a drug in khat, which brings a sense of stimulation in people ?
A. Ethanol
B. Cathinone
C. Nicotine
D. Caffeine

46. The long - term effect of chewing khat is _____ ?
A. myocardial infraction
B. arousal
C. alertness
D. concentration

47. Drinking alcohol leads to poor judgment, which resulted in ____ ?
A. risky sex
B. unwanted pregnancy
C. HIV infection
D. all

48. The largest percentage of Harmful traditional practices in Ethiopia is __ ?
A. Mutilation of skin
B. Early marriage
C. Female genital mutilation
D. Gender based violence

49. A type of cells in the bone, which is responsible for new bone formation
A. Osteoblasts
B. Osteocytes
C. Osteoclasts
D. Osteoprogenitor

50. A part of female reproduction part, where fertilization is taken place ?
A. Ovary
B. Oviduct
C. Uterus
D. Cervix

51. A contraceptive, which is taken as an emergency within 72 hrs. after unprotected sex ?
A. combined pill
B. vaginal ring
C. post pill
D. implant

52. Which one of the following is / are **NOT** the cause of infertility in male
A. Low sperm count
B. Repeated exposure to STI

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C. Endometriosis D. Chronic disease

53. The major risk factors, which is related to infertility in human is ___ ?
A. Exposure to toxin
B. being overweight
C. drinking alcohol and smoking
D. all

54. The female reproductive structure, which used to wipes the egg from ovary, is ____ ?
A. Fimbria C. labia majora
B. clitoris D. oviduct

55. Which part of Ethiopia khat is the most popular and chewed ?
A. Eastern C. Southern
B. Northern D. Western

56. The female menstrual phase, which is accompanied by secretion of high progesterone and low estrogen to build uterine wall, is ___ ?
A. Luteal C. menstrual
B. ovulation D. follicular

57. Clitoris of the female is _____ ?
A. Sensitive protrusion C. covered by prepuce
B. comparable to penis D. all

58. Which of the following is Not the major risk factor that increases the exposure to STI ?
A. Using alcohol and drug
B. Rape and assault
C. Unprotected sex between couples
D. Sexual contact with multiple partners

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Matching Item

A	B
59. Ossicles	A. Auditory
60. Thorax (Rib - Cage)	B. Holds head
61. Vertebral column	C. Protect heart and lung
62. Cranium	D. Holds up human body
63. Hyoid	E. Holds and protect brain

Short Answer

64. A part of a female reproductive part that holds the fetus in place, dilate during birth and separate uterus from vagina is called _____ ?
65. The release of egg from ovary is takes place at _____ phase of menstrual cycle ?
66. A gland located on the wall of vagina, which produces mucus and liquid is known as _____ ?
67. The type of joint located in shoulder and hip is _____ ?
68. The type of protein on the thin filament of actin in the muscle is _____ and _____ ?
69. Total or partial removed of the external genital organ of a girl is known as _____ ?
70. List the three stage of HIV infection ?
71. HIV targets and affects _____ types of WBC.
72. At follicular phase the level of estrogen increases, which stimulate pituitary gland to produce LH to release egg from ovary to oviduct this is an example of _____ mechanism ?
73. The number of CD4 cells per millimeter of blood in healthy adult is _____, while in AIDS patient is _____ ?
74. An Intentional (delibrate) process of preventing pregnancy through different material and technique is called ?
75. A parasitic STI, which attack the armpits, mustache or beard, is ___?
76. The most common FGM in Ethiopia is _____ ?
77. The cause of infertility in which woman due to the growth of tissue, which is similar to endometrium the upper genital area, is _____ ?

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Essay Type

78. Describe the role of the cervix ?
79. List the correct sequences of contraction of muscle
80. Explain the long term effects of chewing khat on health ?
81. List the functions of bone in human ?
82. Explain the reason why the total number of bones in adult humans is decreased to 206 ?
83. Discuss the means of Hepatitis B transmission ?
84. What is family planning ?
85. What are the harmful traditional practices in Ethiopia ?

Answer to Supplementary Exercise on unit 5

36. C
37. C
38. B
39. D
40. D
41. A
42. B
43. B
44. B
45. B
46. A
47. D
48. C
49. A
50. B
51. C
52. C
53. D
54. A
55. A
56. A
57. D
58. C
59. A
60. C
61. D
62. E
63. B

Short Answer

64. Cervix
65. Ovulation phase
66. Bartholins gland
67. The ball and socket
68. Troponin and tropomysin
69. Female genital mutilation
70. Acute stage, chronic / latent disease. AIDS stage
71. CD4 cells / T - helper cells

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72. Positive feedback mechanism
73. 500 - 1600 /mm of blood, 200 /mm of blood
74. Contraception
75. Crabs / pubic lice
76. Clitorydictomy
77. Endometroisis,
78. To hold fetus in place
79. Steps of muscle contraction :
    - Calcium ion binds to **troponin** on actin → The myosin head binds to actin, forming a cross bridge → power stroke to pull the actin filaments → energy is supplied to myosin from **ATP** → **sliding** of **myosin** over **actin** filaments → **sarcomere** is shortened, thus **muscle** is **contracted**.
80. Long term effects of chewing khat : depression, hallucination, heart attack, extreme psychosis, oral cancer and death.
81. The functions of bone : support body, protect internal organs, producing blood cells, storing and releasing minerals and fat.
82. Because many of the bones are fused between birth and the end of maturity.
83. Hepatitis B is transmitted through
    - Through sexual contact ;
    - Infected mother to child during delivery or blood or body fluids contacts.
    - None - sterile equipment for injection and puncturing the skin with sharp objects.
84. Family planning is : ability to anticipate and attain desired number of children, spacing and timing of birth and acquiring resources for raising the children, time, social, financial, and environmental conditioned.
85. The harmful traditional practices in Ethiopia :
    - Early marriage, female genital mutilation, gender based violence, kidnapping/ abduction of girls to make wife

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UNIT 6

6. Population and Natural Resources

6.1 Population Ecology

Ecology is the study of the interaction between living organisms with each other and with the physical environment.

Population ecology considers both the number of individuals of a particular species that are found in an area and the dynamics (changes) of the population

For example, it is a bout why or how the number of individuals are changed (increased or decreased) over time.

What are the levels of ecological organization from simple to complex?

Individual → population → community → ecosystem → biome → biosphere.

A population is a group of similar organisms of the same species, which can interbred together and produce viable offspring within a given time.

Community - all the organisms living and in a particular area.

Ecosystem - a biological community of interacting organisms and their physical environment

Biom - a large community of pants and animals occupying a major habitat

Biosphere - this is where all life on earth lives

Statistical measurements (demographic parameters) that used to investigate the size of population are:

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- Population size
- density
- dispersal
- Population growth.

6.1.1 Population Size

Population size is the total number of individuals present in a particular habitat where it is designated by the capital letter "N"

- Scientists use sampling technique to estimate the larger population in a particular habitat.
- They use one or more samples from the population and use these samples to make inference about the population as a whole.

For example, the quadrant and mark - recapture methods

Quadrant Method

- Quadrants are sampling areas of the same size and shape such as rectangle, and hexagons
- It is made from sticks and string and placed on the group to count organisms in each area.
- It is used to study organisms such as plant and small, slow moving organisms.

Many quadrants are located randomly throughout the habitat at several location. This is because:

- To make sure that the recorded number are representative for the overall habitat.
- The numbers in each quadrant are used to estimate the total population size in the area.

Calculated by a formula $N = A/a \times n$

Total population size (N) =

$$\frac{\text{Total study are (A)}}{\text{The area of the quadrant (a)}} \times \text{Number of organism per quadrant (n)}$$

Mark - recaptured method

This method is used to determine the population size of animals that move from one place to another The steps are:

- Capture a sample of animals, mark them, and release them
- Capture a sample of animals that includes both marked and unmarked individuals.

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- Estimate how many individuals there are in the total population.

Calculated by formula $N = \frac{A}{x} \times n$

Total population size (N) = $\frac{\text{Number of first catch (A)}}{\text{number of second catch (x)}} \times$ total number of marked second catch (n)

How do the sizes of population affect the survival of population?

- Large numbers of population have more genetic variation than in small populations.
- Thus, they have an advantage of adaptation in natural selection such as disease, predators, and competitors

Illustrative Example

1. The simplest form of ecological organisation is
   A. Community
   B. biome
   C. ecosystem
   D. biosphere

Explanation: level of ecological organisation is

Organism → population → community → ecosystem → biosphere.

Answer: A

2. The method of determining population size of animals that move from one place to another area is
   A. Quadrat
   B. mark - recapture
   C. population density
   D. population desertion

Explanation:

Mark - recapture method - is a method that used to determine the size of animals moving from one place in an area to another.

Answer: B

3. A student wants to count a total number grasses grown in $100m^2$ of a school. He prepared 5 quadratures each with $1m^2$ area and counted the number of grasses (100, 80, 60, 40 and 20) respectively. Calculate the total number of grasses in the school?

A. 4000 B. 5000 C. 6000 D. 8000

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Given

$A = 100m^2$, $a = 1m^2$

'n' is calculated as the average number of grasses from all quadrature

$$\frac{100 + 80 + 60 + 40 + 20}{5} = 60$$

Total number of population size (N) $\frac{A \times n}{a}$

$$\frac{100\,m^2}{1m^2} \times 60 = 6000 \text{ grasses}$$

4. A farmer set up a trap and collected 36 birds over one night in his garden, marked the and released. One week later he placed the same trap over the same location and collected 20 birds, of them 8 had paint marking. Estimate the number of population of birds in the garden?

Answer: C

A. 90 B. 180 C. 45 D. 60

Given:

First marked (A) = 36, second captured (n) = 20, number of marked and recaptured (x) = 8

$$N = \frac{A \times n}{x} = \frac{36 \times 20}{8} = 90 \text{ birds}$$

Answer: A

6.1.2 Population Density and Dispersal

Population density is the number individuals of the same species within a specific area or volume at a given time.

Density is a dynamic characteristic that changes over time as individual are added to or removed from the population.

- birth and immigration can increase a population's density.
- death and emigration can decrease a population's density

Population dispersal refers to how individuals in the population often exhibit characteristics pattern of dispersion.

- Spacing between individuals in the population (relative to each other).
- Provide more information on interaction than population density
- There are three dispersal patterns of species. These are:

1. **Uniform dispersion:** occurs when individuals are more evenly spaced

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Example, a nesting colony of sea birds

2. **Random dispersion:** occurs when individuals in the population are spaced unpredictably.
   - Very rare and hardest to observed
3. **Clumped dispersion:** occurs when individuals are concentrated in specific parts of a habitat.
   - Common among social animals.
   - Example, many fish species, lives in pack.

Studying the Human Population

The human population can be studied using a census. In Ethiopia a population census is conducted once every 10 years at least.

- **A census** is a systemic collecting of information from all units or individuals in the population
- Census involves the official counting of a population in the territory of a country.
- Focus on selected demographic and socioeconomic characteristics of an individuals, such as age sex, ethnic group religion, marital status occupation economic activity, house hold size:

There are two methods of conducting a census. These are:

1. De Facto Method of Census:

The government fixes one date for conducting the census throughout the country. People are counted where they are found in the census, regardless of where they usually reside.

It is usually done on a full - moon night, because

- it is believed that all households are present at their residences during that time.

Advantage of De facto Method of Census:

- It takes vary short period of time

Disadvantage of De Facto Method of Census:

- Peoples outside of the house or in transportation are not counted.
- People may not cooperate to give data at sleep time.
- It requires a large number of trained enumerator (counter)

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2. De Jure Method of Census:

The government fixes the counting period (2-3 weeks). Peoples are counted where they usually reside, regardless of where they are in the census day. The data are collected from households by visiting them frequently to fill in the required information.

Disadvantage of De Jure Method of Census:

- Temporary residents at a place are not counted

Advantage De Jure Method of Census:

- It enables to collect more demographic and socioeconomic data.
- The result of this census is usually preferred by the policy makers.
- It is a kind of census usually practiced in Ethiopia

6. 1. 3 Population Growth

Population growth rate is the percentage rate of change within a specified number of individuals in a population. It measures how fast the size of the population is changing.

There are two ways of describing population growth. These are: logistic growth and exponential growth.

i. Exponential growth

- Describe populations increase in numbers without any limits to their growth.
- Growth of a population without environmental constraints (food, space, etc.)
- The population graph of this model shows a J-shaped curve.
- It shows the growth of a population in an ideal environment with unlimited resources to use.
- Growth rate is determined by subtracting the death rate (D) from the birth rate.
- Considering death and birth rate as single factor (r), growth rate can be expressed in a simple equation as:

  population growth rate = $dN/dt = rN$

  Where d = change, N = population size, t = time, and r = the capita (per individual) rate of increase

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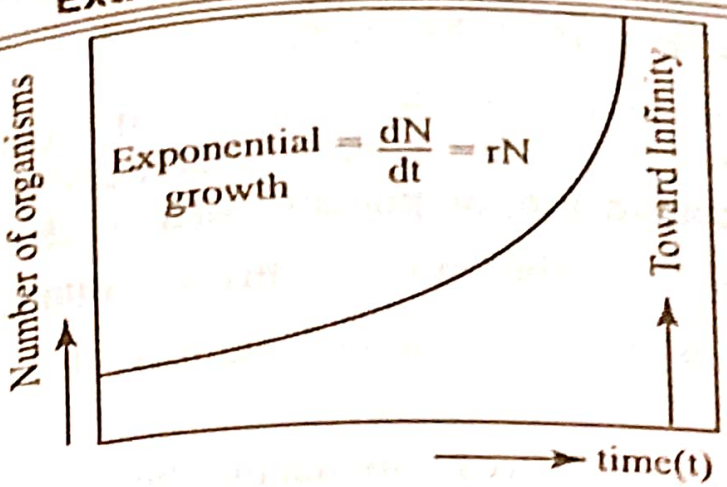
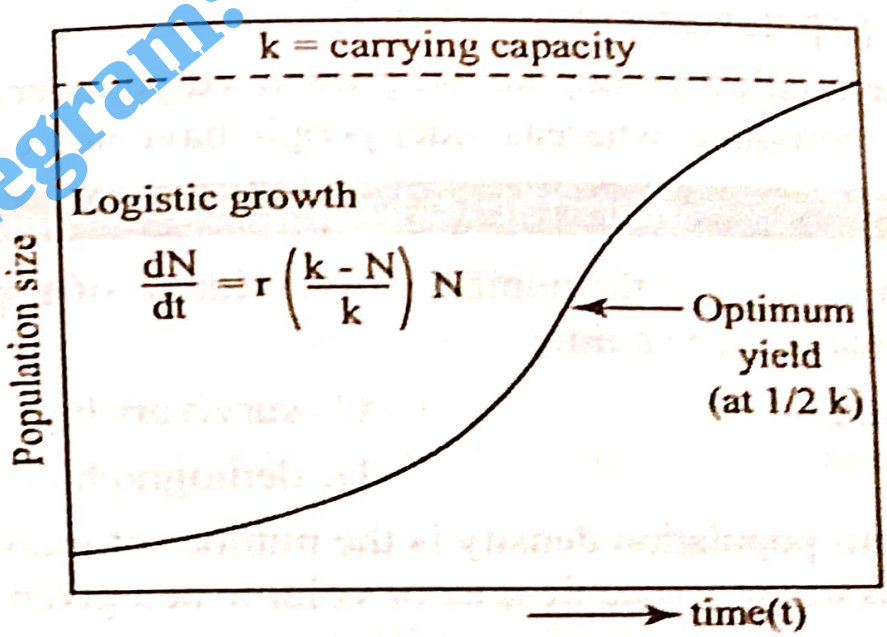


ii. Logistic growth model

- Population growth can be affected by the available of resources.
- Population growth levels off from exponential growth due to limited resources
- It shows a real population growth
- It shows an **S - shaped** curve graph models of population.
- Expressed in terms of **carrying capacity (K)**, which means the maximum population size that the environment can support.
- When the size of the population reaches its carrying capacity, population growth levels off.

Hence, the equation for the logistic model

$$\text{Population growth rate} = dN/dt = rN\,\frac{(k - N)}{K}$$



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6.1.4 Demographic Structure

Demography is the scientific (statistical) study of human populations especially with reference to size and structure. Demography describes the dynamics of a population, or parameters, such as:

Crude birth rate - is the number of live births per 1000 people in a year

Crude death rate - is the yearly annual number of deaths per 1,000 people.

Infant mortality rates - refer to the annual number of deaths of children aged less than a year old per 1000 of those born alive.

Life expectancy - It refers to the number of years that a human being can live in relation to the current mortality rates.

Fertility rate - is a measure of the number of children born.

Mortality rate - is a measure of the number of people who die.

Migration - means the movement of people from one area to another area. It could be:

- Immigration - movement of people into an area to take up permanent residence
- Emigration - refers to the movement of people out of an area to another place of permanent residence.

Note : Why the age and sex are the most commonly study elements of population composition?

This is because it influence population growth. At younger age people are more likely to reproduce, whereas older people have higher rates of dying.

Illustrative Example

1. Population _____ is the number of individuals of a species per unit of habitat area at a given time

A. Dispersion
B. density
C. survivors hip
D. demographics

Explanation:- population density is the number of individuals of the same species within a spacific area or volume at a given time

Answer: B

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2. The type of dispersal that occur when individuals are concentrated in specific parts of a habitat.

A. Clumped
B. Uniform
C. Random
D. Similar

Explanation: Clumped dispersion - occurs when individuals are concentrated in specific parts of a habitat.

Answer: A

3. Which of the following factor increases the density of a population

A. Death
B. Emigration
C. Immigration
D. Extinction

Explanation:- immigration and birth increases the size of population, while emigration and death decrease it

4. Which of the following is not true about De facto method of census?

A. Conducted in one full moon night
B. People on the street/transportation are considered
C. It require large number of enumerator
D. Temporary residents are counted

Explanation:- De facto method of census is carried out within one day at night, it requires large number of data collector (enumerator), and every citizen is counted if available in the house.

Answer: B

5. The main advantage of De Jure method of census is

A. Takes shorter period for counting
B. able to collect more demographic data
C. require large number of enumerator
D. temporary residents are counted

Explanation: De Jure method of census

It enables to collect more demographic and socioeconomic data of population. Therefore, the result of this census is usually preferred by the policy makers.

Answer: B

6. What is the best term that refers to the maximum number of individuals

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that a given environment can support over a long period of time without degradation?

A. Density
B. Climax community
C. Carrying capacity
D. Limiting factor

Answer: C

7. Logistic population growth differ from exponential growth in that exponential growth of population is

A. Is expressed in interms of carrying capacity
B. It is a real type of population growth
C. Affected by resources availability
D. It has no environmental factors

Explanation

Logistic population growth	**Exponential population growth**
Population growth are affected by the availability of resources	Population gowns without environmental constraints (food, space)
It shows an S - shaped curve graph models of population	this model shows a J-shaped curve
it shows a real population growth	It shows an ideal population

Answer: D

8. Assume that you have 100 rabbits, and have the per pita rate of increase is 0.2 year. What will be the growth rate?

A. 40 B. 20 C. 30 D. 60

Given N = 100, r = 0.2

Population growth rate = dN/dt = rmax N

$0.2 \times 100 = 20$ rabbit per year.

Answer: B

9. Assume that you have 100 rabbits in the garden. The garden has a resources that can support 200 rabbits. The percapita rate of increase is 0.2 per year. What will be the growth rate?

A. 20 B. 10 C. 40 D. 50

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Given N = 100, r = 0.2, K = 200

Population growth rate = dN/dt

$$= r_{max} N \frac{(K-N)}{K} = 0.2 \times 100 \frac{(200 - 100)}{200} = 10$$

Answer: B

10. If five cars are in the pattern: car, empty space, car, empty spaces, etc. What type of dispersion is takes place?

A. Random
B. Uniform
C. Clumped
D. Clumped and random

Answer: the car is dispersed evenly/ uniform everywhere.

Answer: B

11. List the type of population dispersion?

Explanation: Random dispersion, uniform dispersion, and clumped dispersion.

6.1.5 Population Pyramid

Population pyramid is a diagrammatically representation of the sex composition of a specific population

- A population pyramid is also known as an "age pyramid" or "age and sex pyramid".
- Used to evaluate the extent of development for a given population in a country.
- It helps to make predictions about services that population needs (schools, hospitals, homes, etc)

There are three types of population pyramids

i. Expansive Population pyramid

- The shape is a broad based and a narrow top.
- Are used to describes a young and growing population.
- Shows a larger percentage of a younger age group.
- Populations of hight fertility rates with lower life expectanices.
- Represents a population of third world countries, such as Ethiopia

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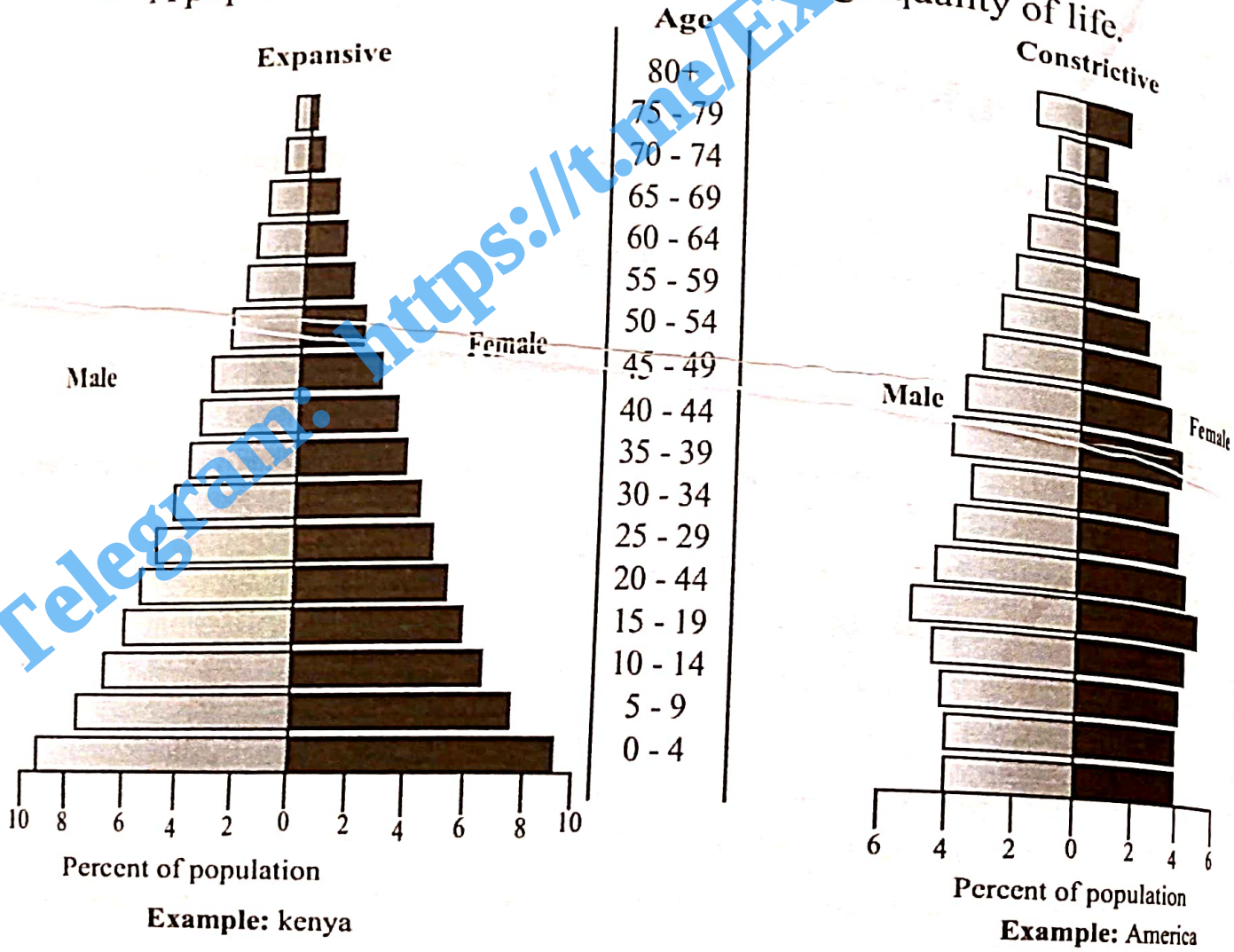
ii. Constrictive population pyramids

- Represents populations that are elderly and shrinking.
- Have smaller percentages of people in the younger age group.
- Represents a population of developed countries
- These populations have access to quality education and health care

iii. Stationary population pyramids

- Are used to describe populations that are not growing.
- It is not a decrease or increases population, it is stable.
- It shows an almost equal proportion of the population at each age group
- It is tapers off at the top.
- Represent developed countries.
- A population of a low birth rates and high quality of life.



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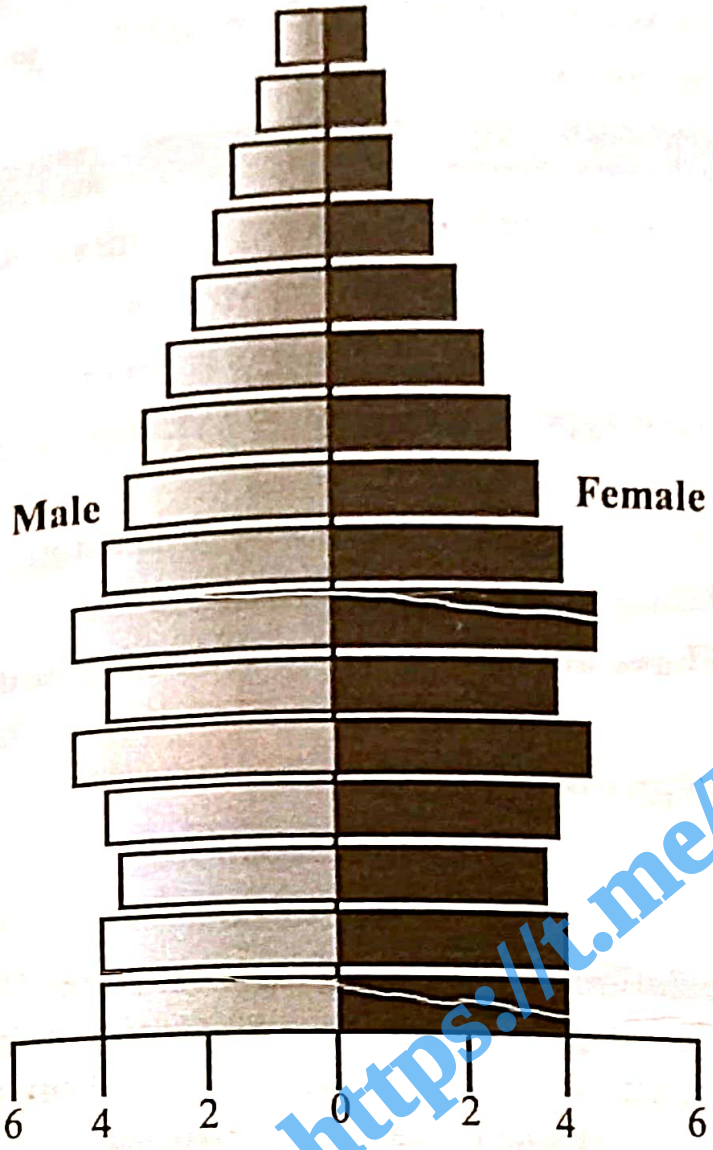

Example: Italy

6.1.6 Population of Ethiopia

The population census in Ethiopia is conducted in every 10 years. The data showed that it has been growing substantially. The following data shows the census of Ethiopia from 1984 - 2012.

- 42.6 million in 1984,
- 53.5 million 1994,
- 73.5 million in 2007 and
- 83.7 million in 2012 and is
- Currently it is estimated to be more than 110 million.

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- The 2007 census showed that Ethiopia has a younger population
- 45% is the age of 15 years and only 4.8% is aged 60 years or older
- 23% is women of the age's b/n 15 and 49.

Illustrative Example

12. A population pyramid that represent shrinking and more elder are

A. Stationary
B. Expansive
C. Constrictive
D. All is answer

Explanation: constrictive type of population pyramid is characterised by:

- Represents populations that are elderly and shrinking.
- Represents a population of developed countries
- Theses populations have access to quality education and health care.

Answer: C

13. The trend of population growth of Ethiopia is

A. Decreasing
B. Shrinking
C. Expanding
D. Stable

Explanation: the population census of Ethiopia from 1984 to 2012 shows that the population is growing faster (42.6 million in 1984, 53.5 million in 1994, 73.5 million in 2007 and 83.7 million in 2012 and Currently it is estimated to be more than 110 million).

Answer: C

14. If the age pyramid of a certain country is narrowing at the base, what does this tells about the trend of the population size of the country? The population size is

A. Growing fast
B. declining
C. Increasing slowly
D. Stable

Explanation:- declining population pyramid is represented by narrower at the base, which means it has low birth rate and has more elder population.

Answer: B

15. Which of the following nations has stable population?

A. Ethiopia B. USA C. Egypt D. Nigeria

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Explanation: population pyramid of developed countries are either stationary or constrictive.

Answer: B

6.1.7 Survivor ship Curves

Survivor ship is the probability that a given individual in a population will survive to a particular age. There are three general types of survivor ship curve

i. Type I - survivor ship curve

- Reflects high survivor ship until old age in life, then a lager increase in death.
- The probability of survivor decreases more rapidly with increasing age.
- Represents large mammals that bear one or few offspring and extended parental care.
- Mortality is concentrated at later age (old age)

Example: elephants, human

ii. Type II - survivor ship curve

- Reflects a fairly constant death rate at all ages.
- The probability of death is equally likely across all age groups.
- The probability of survival doesn't change with age
- The cause of death is due to a random event (such as disease) at any age.
- Example, large birds, lizard.

iii. Type III - survivor ship curve

- The probability of mortality is greatest at child hood or early life
- The probability of survival increases with increasing age.
- They produce many offspring and do little or no parental care.
- Individual which avoid early death have probability of survival

Example, invertebrates, insects, fish, plants and fungi

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Illustrative Example

16. At the begining of the summer farmers used to plant many seedlings around their farm land. However, the number of plant seedling that reaches to maturity is very few. Such type of survivor ship curve is ____ ?

A. Type II B. Type II C. Type III D. None of this

Explanation: type III survivor ship curve is represented by high birth rate, no/little care, the number of elders are very few. Examples: plants, insects, fish, etc.

Answer: C

17. A population of Birds and lizard are classified into ____ type of survivor ship curve.

A. Type I B. Type II C. Type III D. Type I and II

Explanation: population of birds and lizard are grouped under survivor ship curve type II, the probability of survivor doesn't change with age.

Answer: B

18. The population of large mammals such as elephants and human are grouped under the type of survivor ship curve ____

A. Type I B. Type II C. Type III D. None of this

Explanation: An example of survivor ship curves of type I are animals bear small number of offspring such as elephants, human.

Answer: A

19. A few elephants bear a single offspring every two to four years. Based on this information, which survivor ship curve do you thinking is representative of elephant? Explain your answer.

Explanation: type I survivor curve, have a better family care, low death rate at early age, the probability of survivor decreases more rapidly with increasing age, it represents large mammals that bear one or few offspring and extended parental care.

20. Do all survivor ship cures neatly fit the type I, II or II models?

Explanation: The three survivor ship curve are generalization, and only few population exactly fit one of the three. Some species have one type of survivor ship at early stage and another type at adult.

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6.1.8 Population Regulation

Population regulation is a density - dependent process, meaning that population growth rates are regulated by the density of a population

There are two resistance to population growth. These are: **density - dependent** and **density - independent factors.**

A. Density - dependent factors - regulate population growth by affecting a larger proportion of the population as population density rises. The more organism in the ecosystem, the larger the effect.

- It is biotic factors that affect a population
- It includes **predation, competition** (inter specific and intra specific) , accumulation of waste, and **diseases.**
- Intra specific - competition between the same species for a common resources
- Inter specific - competition between different species for a common resources.

Usually, the denser a population is the greater its mortality rate will be.

For example

1. During competition, the reproductive rate becomes lower, reducing the growth of population.
2. When there is a low prey density, it results in increasing the mortality of its predators. This is due to difficulty of ensuring their food source.
3. The diseases in the area regulate population growth. For instance, communicable disease spread quickly and increase mortality in a dense population.

B. Density - independent factor - limits population growth but is not influenced by changes in population density.

- It is non - living abiotic factors that regulate population growth rate regardless of population density
- Factors include weather, natural disasters, and pollution.

Example:

An individual may be killed in a catastrophic earthquake regardless of the number of individuals present in that area.

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6.2 Natural resources

The materials from the earth that are used to support life and meet people's need, or any natural substances that human use can be considered as natural resources. It can be of two types: renewable and non-renewable.

I. Renewable

Renewable natural resources are resources that can be replaced after utilization.

Resources that cannot be exhausted even after continuous utilization are termed renewable resources. Examples of renewable resources include sunlight, air, trees, water, wind, tidal energy, solar and wind energy, biomass energy, and hydro power.

They are available continuously and their quantity is not noticeably affected by human consumption.

However, renewable resources do not have a rapid recovery rate and are susceptible to depleting if they are overused.

They can also be depleted if not properly managed or conserved.

II. Non - renewable

Non - renewable natural resources are resources that do not naturally replenish at the same speed at which they are used to meet the growing demands.

These resources may take millions of years to form and replenish.

A resources is considered non - renewable when its rate of consumption exceeds its rate of recovery. Example of non - renewable natural resources are minerals, fossil fuels, coal, and natural gas.

Illustrative Example

21. Density - independent factor of population could be due to

A. Competition
B. Predation
C. Pollution
D. disease

Explanation: density - independent factor of population

- It is non - living (abiotic) factors that regulate the population growth rate regardless of population density
- Factors include weather, natural disasters, and pollution.

Answer: C

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22. Which of the following is not true about conservation of natural resources?

A. Is the wise use of natural resources

B. Is the use of natural resource on the basis of sustainability

C. Stop using natural resources

D. Reserving resources for future generation

Explanation: the principle of conservation doesn't preach not to use natural resources, It says resources are used on the basis of sustainability by reserving the resources for future generation.

Answer: C

23. Compare renewable and non - renewable natural resources?

Explanation

Renewable natural resources:	**Non - renewable natural resources:**
Resource that have rapid recovery	Resources that don't have rapid recovery
Examples, solar energy, wind, water, plant and animal	Examples, mineral, fossil fuel, coal, natural gas, etc

6.3 Conservation of Natural Resources in Ethiopia

Natural resources are always in short supplies due to a growing demand of population, rapid industrialization, and urbanization. Therefore, there must be a need to use existing resources wisely so they will be available for the next generation.

- **Conservation:** is the care, protection, and wise use of natural resources so that the resources can be used for future generative
- Conservation aims to manage human use of natural resources for the current public benefit and sustainable social and economic utilization.
- Ethiopia has a wide range of natural resources such as plants, water, and soil. This is because of topography and diverse climatic conditions.

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6.3.1 Wild life in Ethiopia

Wildlife refers to living things that live in the natural environment and are not yet domesticated

Importance:

- Used to maintain natural ecological process, store genetic resources for the future, secure wildlife tourism, and contribute to the national economy, and manufacture goods.

Examples of wildlife in Ethiopia are: Lion Statesman's Bush crow, Walia Ibes, and the Ethiopian Mountain Chameleon.

The causes of loss/decline natural resources: are due to natural and human activities.

- Natural disasters, such as: earthquakes, floods, droughts, tornadoes and fire.
- Human activities, such as: agricultural activities, development activities, deforestation, poaching, illegal wildlife trade, habitat change, climate change, overexploitation, and pollution

Conservation mechanisms:

There are two strategies practised for the preservation of a variety of living species globally:

1. **In - suit conservation** - is the process of protecting an endangered species its natural habitat
2. **Ex - situ conservation** - is the relocation of endangered or rare species from their habitat to protected areas equipped for their protection and preservation.

Ethiopia has established many national park, wild life reserves, community conservation area and more controlled hunting areas as a conservation mechanism.

I. National Park of Ethiopia

1. Abijata Shala Lakes
2. Alatish
3. Awash
4. Bale Mountains
5. Gambella
6. Geralle Somail
7. Kafta Shiraro
8. Nechsar
9. Omo
10. Simien Mountains
11. Yangudi Rassa
12. Denkoro Chaka

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13. Mago Yabello 14. Yebello Oromia

II. Wildlife Sanctuaries

1. Babile Elephant Sanctury
2. Senkele Swaye'n harte beets sanctuary

III. Wildlife reserves

1. Alledeghi 2. Awash 3. Bale

IV. Community Conservation Areas

1. Abune Yosef 2. Guassa Meze 3. Tama

6.3.2 Plants in Ethiopia

Ethiopia is conserved as a center for crop genetic diversity

- ☑ This is due to the topographical and diversity of climatic condition.
- ☑ Plant species in Ethiopia are divers and have a rich indigenous element.
  - ➥ Endemic species - are native to a particular country or prevalent to particular environment.
  - ➥ Indigenous species - is produced, growing, living naturally in a particular region
- ☑ Examples of plants in Ethiopia: Mekmeko **(Rumex Abyssinian)**, African juniper **(Juniperus procera)**, Cabnage tree **(Moringa stenopetala)** and Kosso **(Hagenia Abyssinia)**.

Cause of loss/decline of plant species in Ethiopia are: due to natural disaster and **human activities.**

1. Human activities

Such as deforestation, climate change, invasive species, over exploitation, pollution and land degradation.

2. Natural disaster

Such as: earthquakes, floods, droughts, tornadoes and wild fire.

Conservation mechanisms: include **indigenous** and **modern practices.**

I. Indigenous practices:

- ➥ Long - standing traditions and ways of life of specific communities or localities.

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- It relies on taboos or forbidden practice for using or consuming some plant species; domestication, reserves, selective harvesting sacred groves etc.
- Sacred place and place of worship, such as church and mosques, graveyards, and monasteries are reserved areas for indigenous plants.
- Traditional farming systems to maintain a rich genetic diversity of vegetables and other crops through selection, transfer, exchanges, and conservation.

II. Modern practice: such as

- Establishing botanical gardens
- Controlling invasive species
- Preserving genes/seed in a gene bank.

Note: invasive species are alien species which threatens ecosystem, habitat or species. Examples of invasive species in Ethiopia are: julifora, striga species, water hyacinth, L. camera, etc.

6.3.3 Soil and Water in Ethiopia

Ethiopia has a huge soil and water resources. This is due to a wide range of climate, topography, parent material, and land use. water resources include lakes, river basins, and ground water.

The country's water resources have a huge potential for:

- Hydro power generation. For example, the Grand Ethiopian Renaissance Dam (GERD)
- Irrigation
- Fishing

The cause of soil erosion and the decline in water quantity and quality are: Human activities such as

- Deforestation,
- Poor land husbandry practices
- Wasters emitted from industries and homes.

Conservation measure of soil and water involves: mechanical and biological measure.

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1. Mechanical measures:

It includes constructing bunds, terraces, diversion ditches, check dams, micro - basins, and hillside terraces:

2. Biological measures:

- It includes enclosing degraded land free from human and animal interferences
- Planting of tree on farmlands (agrofroestry)
- Afforestation (establishing a forest in areas where there was no recent tree cover).
- Tree plantations around homesteads.

For example. The Green legacy initiative that was launched in June 2019; with a vision of building a green and climate resilient Ethiopia. It can reduces environmental problems such as soil erosion and loss of water.

6.4 Impact of Accident on Wild and Domestic Animals

Road traffic accidents have become a huge global public health and development problem, killing and injuring many people and destroying properties every year. Developing countries, including Ethiopia, have accounted from a large number of global road traffic deaths. Car accident causes:

- The lose of a human being
- Death for both wild and domestic animals
- Death of wild animals in the conservation areas when crossing highways.

For example, 29% of the wildlife mortality was due to car accidents in Geralle National park (Addisu et al, 2017).

Methods to reduce traffic accidents includes:

- Road designs that allow safe passage for animals and wildlife,
- Constructing road in the way promoting habitat connectivity
- Considering natural wildlife habitats when roads are constructed
- Deriving slowly.

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Illustrative Example

24. Which of that following plant conservation method is not a modern practice?

A. Establishing botanical garden
B. Establishing gene bank
C. Controlling invasive species
D. Preserving plants in the monasteries

Explanation:- Conservation of plants in religion areas such as church, mosque and monasteries are an indigenous activity.

Answer: D

25. In - situ conservation differ from ex - situ conservation in that in - situ conservation is

A. Used for conservation of endangered species in their nature environment
B. Used for conservation of endangered species out - side the conservation areas
C. Are strategies to exploit wild life in their natural environment
D. Are strategies to exploit wild life outside of their natural environment

Explanation

In - situ conservation	Ex - situ conservation
Conservation of endangered species in natural environment	Conservation of endangered species out - side the conservation areas

Answer: A

26. Which one of the following conservation method is more appropriate to conserve very rare and endangered species?

A. National parks
B. Sanctuaries
C. Wild life reserved
D. Community conservation areas

Explanation: Sanctuaries are designated to protect and reserved very rare or endangered species.

Answer: B

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27. Ethiopia has a huge potential water sources, which is used for all, except?

A. Hydro power B. irrigation C. fishing D. navigation

Explanation: Ethiopia has a huge water potential, hydro power and fishing

Answer: D

28. Which one of the following human activities doesn't cause the decline of water and soil conservation?

A. Deforestation C. Poor land husbandry

B. Waste emission D. Afforestation

Explanation:- Afforestation is establishing a forest in area where there was no recent tree cover.

Answer: D

29. A species that are native to a particular country or environment is called _____ ?

A. Indigenous C. Wild life

B. endemic D. Exotic

Explanation: Endemic species- are native to a particular country or prevalent in a particular field or environment.

Indigenous species - is produced, growing, living naturally in a particular region

Answer: B

30. The biological measure that used to conserve soil and water is

A. Building check dam C. tree plantation

B. terracing D. construction of micro - basins

Explanation: the biological and mechanical soil and water conservation mechanisms are:

Biological means of soil and water conservation	Mechanical means of soil conservation
Tree plantation, agro forestry, are enclosure in degraded land	Building check dam, terracing, tree plantation, and construction of micro - basins, terracing

Answer: C

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31. What was the vision of green legacy initiative that was launched in 2019 in Ethiopia?

Explanation: building a green and climate resilient Ethiopia

32. What are the methods to reduce road traffic accidents on wild life in Ethiopia?

Explanation: Road design that allow safe passage for wild and domestic animal, road construction should be designed in a way to stop habits fragmentation: not driving faster.

6.5 Impact of Human Activities on the Environment

Human beings are obtaining all the materials needed for survival and development from the environment we live in. However, we are exploring resources in destructive ways that destroys the entire ecosystem and our lives. This has an effect on the present and future generation.

The human activities that affect the environment include:

- Burning of coal, natural gas
- Using refrigerants and coolants
- Applying pesticides
- Waste incineration and waste disposal
- Using various volatile chemical
- Agricultural activities
- Industrial and other sewage discharges
- Construction of infrastructure and transportation
- Urban development
- Mining activities

What are the consequences of the human activities on the environment?

- Environmental pollution
- Climate change
- Global warming
- Acid rain
- Ozone layer depletion
- loss of biodiversity
- Toxic bio accumulation
- Resource depletion

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6.5.1 Environmental Pollution

Pollution refers to the introduction of contaminants (in the form of substance or energy) in to natural environment at a rate faster than they can be dispersed, diluted, decomposed, recycled, or stored in some harmless form.

Environmental pollution are caused by:

- Natural events such as forest fire and active volcanoes
- Human activities, such as: rapid human population growth and industrialization

Types of environmental pollution

- Air pollution
- Water pollution
- Land pollution
- Noise pollution
- Light pollution

Note: pollution of all kinds can have negative effects on the environment and wildlife which in turn have impacts on the health and well being of human beings.

A. Air Pollution and its effects

Air pollution is the contamination of the air in the atmosphere by harmful gases, dust, and smoke that affect plants, animal, and human drastically.

Pollution that cause air pollution are:

- Carbon monoxide,
- nitrogen dioxide,
- Ozone
- lead,
- sulphur dioxide,
- Smog and particles

Air pollutants can be released from human activities such as

- Transportation vehicles
- Industrial activities
- Burning of coal and natural
- Chemical from refrigerants
- Volatile chemical

What are the effects of air pollution?

- It cause depletion of the ozone layer
- It cause respiratory and cardiovascular disorders in human beings
- Changes the pH levels of the soil
- cause acidic rain

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➥ Infects plants and other forms of vegetation (harmful and toxic for

B. Water Pollution and its effect

Water pollution - is due to the release of toxins, organic molecules, he or microbes into water that make water unsafe for human use and disrup aquatic ecosystems.

The causes of water pollutions are:

- ➥ Chemicals from factories and industries,
- ➥ Agriculture wastes
- ➥ Sewage
- ➥ Transportation of oil,
- ➥ Chemical fertilizers, insecticides and herbicides
- ➥ pathogenic micro organisms,
- ➥ Water hyacinth (Emboch)

The effects of water pollution:

- ➥ Depletion of fish populations
- ➥ Destruction of biodiversity
- ➥ Water borne diseases that lead to death.

6. 5. 2 Climate Change

Climate change is generally defined as a significant variation of average weather conditions or a change in the usual patterns of weather. This could be a change in the amount of rain, temperature, etc. for a month or season.

Cause of climate change:

Global climate changes naturally over time scales is due to:

1. Internal fluctuations such as exchange of energy, water and carbon between the atmosphere, oceans, land and ice,
2. External influences such as variations in the amount of energy received from the sun, variation in the earth's orbit around the sun and the effects of volcanic eruptions.
3. Human activity. For example, the combustion of fossil files.

➥ The release of greenhouse gases in the atmosphere, such as carbon dioxide (CO_2), methane, nitrous oxide, and fluorinated gases. Many of these greenhouse gases occur naturally, but human activates are

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increasing the concentrations.

6.5.3 What are the effects of greenhouse gases?

The main effect of increasing green house gases in the atmosphere is global warming. This is done by trapping the heat of the sun and preventing it from returning into spaces. This results in an increase in the temperature of the earth's atmosphere, the ocean, and land surface. Global warming is an average rapid rise in the earth's temperature.

Global warming has an effect on living things, such as (extinction) of species, health problems, high heat stress and health risk, poverty and displacement.

Some of the effects include flooding, melting of ice caps, rising oceans/sea level, drought, heavy rainfall

6.5.4 Ozone Layer Depletion

Ozone is an inorganic molecule with the chemical formula (O3), made of three oxygen atoms, and it is toxic gas.

The atmosphere has four layers. These are:

- Troposphere
- Stratosphere
- Mesosphere
- Atmosphere

☑ Ozone is found in the stratosphere of the atmosphere.

☑ It protects the earth from harmful radiations (UV rays).

☑ When the ozone layer is depleted, the sun release UV rays onto the earth, causing skin cancer in humans. Currently, this protective layer (ozone) is getting depleted, creasting ozone hole.

What are the main causes of ozone layer depletion?

Some industrial products release gases such as chlorofluorocarbons (CFCs), hydro fluorocarbons (HCFCs), and halon in to the atmosphere, which plays a role in ozone depletion.

For instance, how do Chlorofluorocarbons (CFCs) affect the ozone?

- CFCs are gases, released from refrigerators and fire extinguishers.
- Causes chemical reaction that break down ozone molecules, reducing the ozone's ultraviolet radiation absorbing capacity.
- Chlorine reacts with ozone and forms oxygen (O2) and chlorine

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monoxide (ClO).

$$Cl + O_3 \longrightarrow ClO + O_2$$

When the molecule of chlorine monoxide (ClO) meets with another molecule of oxygen (O), it breaks up, releasing chlorine (Cl), which can react with another molecule of ozone (O_3), creating the catalytic cycle of chlorine.

$$ClO + O \longrightarrow Cl + O_2$$

What is the effect ozone layer depletion?

Ozone layer depleting exposes the to UV rays that can lead to:

- Skin cancer and cataracts and weaken the immune system response.
- Ageing of the skin, making one look much older than he/she actually is
- Cause respiratory diseases such as chest pain, difficulty in breathing or even throat irritation.
- It also increases the formation of ground - level ozone (smog)

Illustrative Example

33. Which one of the following environmental pollution declines the quality and quantity of aquatic organism?

A. Air pollution
B. Noise pollution
C. Water pollution
D. Light pollution

Explanation: Water pollution directly affects aquatic animals (animals living in the water).

34. Which one of the following gas is most responsible to deplete the ozone layer?

A. CFC B. NO_2 C. CO_2 D. SO_2

Explanation: ozone layer depletion is occur due to the release of CFC, HCFC, and Halons.

Answer: A

35. List some examples of air pollutants?

Explanation: Air pollutants are gases or chemicals that contaminate the atmosphere. These are: CO, NO_2, Pb, SO_2, O_3, and smog

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36. Write some examples of the effects of water pollution ?

Answer: Depletion of fish,destitution of biodiversity, water born disease

37. Explain how the greenhouse gases increases global warming?

Explanation: the accumulation of greenhouse gases into the atmosphere cause global warming. Global warming is occurring by trapping the heat of the sun and preventing it from returning into space. This results in an increase in the temperature if the earth's atmosphere, the ocean, and surface.

38. Discuss briefly how the depletion of ozone layer affects living organisms on earth?

Explanation: depletion of ozone layer means UV - rays reach to earth, which causes problem such as: skin cancer, cataract, weaken immune system, ageing skin, respiratory diseases, some formation, etc.

6.5.5 Acidic Rain

Acid rain is rain or any other form of precipitation that is unusually acidic (lower pH value usually from 4 - 5). Acid rain formed when acid - forming pollution dissolve in water vapour in the atmosphere.

The major causes of acid rain are:

The burning of fossil fuels, release gases such as sulphur dioxide and nitrogen oxides into the air, react with water to form solutions of sulphuric acid (H_2SO_4), and nitric acid (HNO_3) respectively.

- $2SO_2(g) + O_2(g) + 2H_2O(1) \rightarrow 2H_2SO_4$ (aq)
- $4NO_2(g) + O_2(g) + 2H_2O(10 \rightarrow 4HNO_3$ (aq)

Sulphur and nitrogen gases, released from combustion of coal/oil, can also combine with oxygen to form sulfur dioxide (SO_2), and nitrogen dioxide (NO_2), which contribute to the acidity of rain.

Carbon dioxide (CO_2) in the atmosphere and water combine to form carbonic acid, commonly known as carbonated water.

What are the effects of acid rain?

Acid rain damage plants, soil, and water

- Affects many living organisms that depend on theses resources for their survival.

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Acid rain damages the waxy cuticle of plant leaves

This results in leaching of nutrients such as calcium, magnesium, and potassium out of leaf tissue and drying.

Acid rain can stop seed from germination.

Degrade available nutrients in the soil, which stops the growth of plants.

For example, it dissolves aluminium in the soil, which stops the growth of plants.

Acid rain affect soil micro organisms, aquatic animals such as fish and amphibians

It affects the quality of the water that people use for drinking and cleaning

- This results in acute toxicity or chronic health problems due to dissolved trace metals.
- Corrosion of water pipes, which further results in leaching of heavy metals such as iron, lead, and sipper into drinking water.

It also damages the building and monument made up of stones and metals.

6.5.6 Loss of Biodiversity

Biodiversity refers to the variety of life on earth in terms of genes, species, individual organisms within a given species, and biological communities from an ecosystem to the global biosphere.

What are the major causes of loss of biodiversity?

Rapid growth of human population and increased demand .This results in rapid and permanent change in the ecology, ecosystem, landscapes, and the global biosphere.

Loss of habitat and degradation. The fragmentation and destruction of habitat reduces the food resource and living space for most species. This results in the loss of biodiversity and extinction of species.

Overexploitation: exploitation of animals or other organisms beyond the capacity for surviving populations result in biodiversity loss.

Pollution: Pollution results in killing of organisms or creating reproductive problems that threaten the species' survival.

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Climate change: Heating of the earth's surface affect biodiversity, because it endangers all the species, especially those species that are adapted to the cold due to latitude or altitude.

Invasive species: The introduction of non - native species to the country, disrupt the ecosystem they colonize, my out compete native species for food and habitat, which results population declines in the native species.

What are the effects of biodiversity loss?

- Material scarcity such as food,, water, fertile soils, and medicine.
- Declining of an ecosystem's productivity and service.
- It affects the economic system and human society. For example: food, building materials, and medicines.
- Threatens the global food security
- Threatens the development of new pharmaceuticals to deal with future disease, which are derived from the chemicals in plants and animals

6.5.7 Toxic Bio Accumulation

Bio accumulation is the gradual accumulation of substances in an organism. Persistent bio accumulative toxic substance (PBTs) are chemicals that do not degrade easily in the environment. PBTs are typically accumulated in fatty tissues and are slowly metabolize, often increasing concentration within the food chain.

Sources of toxic boi - accumulates:

The major bio - accumulates are organic compounds and metals. They include synthetic chemicals:

- halogen atoms (fluorine, chlorine, or bromine)
- DDT
- Metals, such as lead and mercury.

The process of bio accumulation occurs as follows:

- Plants absorb toxic substance (pesticides or pollutants).
- These plants are eaten by primary consumers.
- The primary consumers are eaten by secondary consumer then to higher - level consumers.
- At each tropic level of the food chain, the toxins remain in the tissues of the animals.

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Note: The concentration of toxins is higher in the animals which are at the top of the food chain.

What are the effects of toxic bio - accumulates?

Human, domestic animals, and wildlife are more likely to be exposed to these chemicals in the environment. Bio - accumulates is not degraded by natural means, such as: sunlight, reaction with substance, bacteria. Some of the effects of bio - accumulate are:

- they have toxic properties, which results in health effects such as:
- Mutation on DNA
- Cancer
- Neurological toxicity
- Reproductive toxicity
- Developmental toxicity
- Immunise system damage.

Example

Lead contamination on air, soil, or drinking water results

Result in impaired brain development in foetuses, infants, and children.

When mercury is consumed by fish and plankton, it passes in the food chain and

Affects nervous systems and the reproductive systems of mammals, including humans.

DDT affects the population of birds.

When the bird consumes prey, which are affected by DDT in the food chain, it makes the shells of their eggs very thin and break easily when the birds try to incubate them/

6.5.8 Resources depletion

Natural resources have been depleted primarily because of human activities, such as increasing global population. It occurs when the consumption of natural resources are faster than that are recovered.

The major causes of the depletion of natural resources are:

i. Overpopulation.

As long as global population is increasing, the demand for natural resources

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also increases, which lead to resource depletion.

ii. Poor farming practices, such as:

Poor irrigation practices, poor soil management practices, the use of heavy machinery and farming equipment.

For example, destroying the soil structure makes it unsuitable for plants growth, excessive use of pesticides, fungicides, and herbicides kills important soil micro organisms that are essential in replenishing nutrients in the soil.

iii. Over consumption of natural resources

large - scale mineral and oil exploration, leading to more and more natural oil and mineral depletion.

iv. Industrial and Technological Development

Industrial wastes contain toxins and chemical by - products, affecting natural resources which lead to increased natural resource depletion.

6.6 Indigenous Conservation Practices in Ethiopia

The people of Ethiopia have indigenous knowledge about natural phenomena, natural resources and their environment. Indigenous knowledge is a body of knowledge built over generations by a group of local people living in a particular environment.

For instance, they have indigenous knowledge of natural phenomena such as:

- Astronomy,
- The proper use of natural resources, including the conservation
- Prediction of seasonal conditions such as rainy season, and drought season, productivity of the season, etc.

Ethiopia has a known indigenous culture, tradition, religion, and knowledge. This indigenous knowledge was developed by observation of the sun, moon and stars, behaviour of animals, winds, and the level of temperatures.

The indigenous knowledge helped us to have our own calendar, predict the weather conditions, treat different types of diseases, and produce crops to stain our lives, protect our environment and conserve natural resources.

Ethiopians have indigenous knowledge in the fields of conservation of

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natural resources (soil, water, crop seeds, forest, medicinal plants, animals) etc.

- For example, the well - known and internationally recognized indigenous soil and water conservation practices of the Konso people, registered by UNESCO as a world heritage.

The conservation practices among the Konso people are making terraces, contour ploughing, crop rotation, mixed cropping, surface mulching and agro forestry.

i. Making Terraces:

Terraces are structures, built mainly in hilly areas to intercept runoff water to reduces soil erosion and for soil and water conservation.

- The terrace can retain the soil from erosion
- The terraces can collect maximum water and discharge the excess,
- The terraces can creates saddles, used for agriculture.
- The best example of terrace in Ethiopia is the Konso Cultural Landscape

ii. Contour ploughing

Contour ploughing is the act of farming on a hill to reduce runoff of water and prevent soil erosion

iii. Crop Rotation:

Crop rotation is planting different crops sequentially on the same plots of land to improve soil fertility and reduce the effects of pests and weeds.

- Improve soil fertility by alternating high residue - producing crops with low residue - producing crops.

iv. Mixed Cropping

Mixed cropping is the growth of two more species on the same field at the same time.

- The best example is a mix of maize and groundnut, maintain soil fertility and are cost - effective.

v. Surface Mulching:

Surface mulching is applying a layer of materials to the surface of the soil for conservation of soil moisture to improve the fertility of the soil and reduces weed growth.

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- Most farmers use surface mulch on their fields by using crop residue and branches.

vi. Agro - forestry:

Agro - forestry is the widely used activity to conserve soil, small and big trees inside and just outside the farm land. These indigenous conservation practices are observed among peoples in different parts of Ethiopia, such as in Konso.

- ☑ The best example of plant used is *Moringa stenopetala*.
- ☑ There are areas protected by the community for conservation purposes.

For example: the Guassa community conservation area of Menz

- It is protected by communal management system known as Qero.
- They can close the Guassa area from any use for about 3 - 5 years.

- ☑ The community elects headmen who determine when to open the area, for how long local people can harvest the grass, graze their livestock, and close it again.

The opening of the area for use depends on:

- The growth and recovery of the grass
- Community requirements for resources
- The success of the local crop harvest
- The frequency of drought in the Guassa area.

Illustrative Example

39. Which of the following substances is not bio - accumulates?

A. DDT
B. halogen atom
C. lead
D. Magnesium

Explanation: Bioaccumulation is the gradual accumulation of substance in an organism. Examples:

- halogen atoms (fluorine, chlorine, or bromine)
- DDT
- Have metals, such as lead

Answer: D

40. Which of the following tropic level of organism has more bio

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accumulation than the other?

A. Primary producer
B. Primary consumer
C. Secondary consumer
D. Tertiary consumer

Explanation: the concentration of bioaccumulation is the highest at top tropic level.

Answer: D

41. Which part of the Ethiopia got has an indigenous knowledge of conservation of water and soil, and got UNESCO recognition?

A. Konso B. Welayita C. Tigray D. Jinka

Explanation: Konso area is best known by soil and water conservation and has got recognition from UNESCO

Answer: A

42. A water and soil consecration methods that apply a layer of material to the surface of soil to conservation soil moisture and improve soil fertility is

A. Mixed cropping
B. Surface mulching
C. Crop proration
D. Contour ploughing

Explanation: surface mulching is applying a layer of materials to the surface of the soil for conservation of soil moisture to improve the fertility of the soil and reduce weed growth.

Answer: B

43. A soil conservation methods that improve soil fertility because of the residue left in the soil

A. Surface mulching
B. Crop rotation
C. Contour ploughing
D. Terracing

Explanation: Crop rotation is planting different crops sequentially on the same plots of land to improve soil fertility and reduces the effects of pests and weeds. Improve soil fertility by alternation high residue - producing crops with low residue - producing crops.

Answer: B

44. The most commonly used example of plant used in agroforestry in Ethiopia is

A. Zigba B. Shola C. Moringa D. Acacia

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Explanation: the most commonly used example of plant used in agroforsty is **Moringa stenopetala.**

Answer: C

45. Which of the following is not true about activities of terracing?

A. Retain soil from erosion
B. Create saddle for agriculture
C. Collect maximum water
D. usually done on plain field

Explanation: Terraces are structures, built manly in hilly/sloppy areas to intercept runoff water to reduce soil erosion, improve soil and water conservation.

- retain the soil from erosion
- collect maximum water and discharge the excess,
- Creates saddles, used discharge the excess,

Answer: D

46. The communal management system that is elected to conserve the Guassa in Menz, Ethiopia is called _____ ?

A. Wild life reserves
B. Sanctuaries
C. Monasteries
D. Qero

Explanation: Guassa, the community reserved area is protected by communal management system known as Qero

Answer: D

47. Discuss the effects of bio - accumulation in living organism?

Explanation: mutation, cancer, neurological toxicity, immune system damage, respiratory toxicity.

48. What is the difference between terrace farming and contour ploughing?

Terrace Farming	Contour Ploughing
terrace farming shifts the slop's structure to create flat areas that provide water catchment On the steep slopes, wide flat terraces are designed so that flat area are usable for cultivation, thus minimization surface run off and soil erosion	Contourr ploughing suits the slops natural shape without change it. Ploughing parallel to the contours of a hill slope to create s natural obstacle for fluid to drain down the slop.

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Review Exercise

I. Match each term with its most suitable description

A	B
1. Carrying capacity	A. Population growth plots out as a J. shaped curve
2. Exponential growth	B. Maximum number of individuals sustainable by the resources
3. Logistic growth	C. Population growth plots out as a S. shaped curve
4. Survivorship curve	D. a variety of living organisms in area
5. Population dispersion	E. The gradual accumulation of substance in organisms
6. Population density	F. An alien species which threatens ecosystem, habitat or species
7. Bio - accumulation	G. A group of similar organisms in area
8. Invasive species	H. Spacing between individual in the population
9. Population	I. Is the probability that a given individual in a population will survive to a particular age
10. Biodivesity	J. The number of individuals of the same species within a specific area in a population

II. Multiple choice

11. The rate at which a population grows or declines depends on the rate of _____ ?

A. Births
B. Death
C. Immigration & emigration
D. All of the above

12. Resource competition, disease, and predation are examples of ___ population growth factor.

A. Density - independent
B. Exponential
C. Dispersal
D. Density - dependent

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13. Population _____ is the number of individuals of a species per unit of habitat area at a given time
A. Dispersion
B. Density
C. Survive ship
D. Demographics

14. The type of dispersal that occur when individuals are concentrated in a certain parts of the habitat
A. Clumped
B. Uniform
C. Random
D. Similar

15. The largest population that can be maintained by a particular environment for an indefinite period is known as
A. Carrying capacity
B. declining phase
C. log phase
D. Exponential growth

16. A few elephants bear a single offspring every two to four years. Based on this information, which survivorship curve do you think is representative of elephant?
A. Survivorship type I
B. Survivorship type II
C. Survivorship type III
D. Have no curve survivor

17. Which of the following is not true about quadrant method of determining population size?
A. Used to measure small, slow moving organisms
B. It can be a square or rectangle in shape
C. It is a sampling area
D. Used to measure large sized animals

18. The chance of survival of organisms in a large population is higher than small population. This is due to
A. More genetic variation
B. Easily suffer from disease
C. Least natural selection
D. Less genetic variation

19. Expansive population pyramid is characterized by
A. Low birth and death rate
B. Have a better health care system
C. Have more older age population
D. It represents a third world population

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20. Which of the following is not true about renewable natural resources?
   A. Replaced after utilization
   B. Not exhausted
   C. Not susceptible to depletion
   D. Examples are solar energy and mineral

21. The human activities that contributes to the loss of wild life is
   A. Wild fire
   B. Earth quake
   C. Pollution
   D. Floods

22. Churches, mosques and monasteries are sacred Places in Ethiopia which used to preserve wild life. This activity is
   A. Modern practice
   B. Indigenous practice
   C. Botanical garden
   D. A gene bank

23. Acidic rain affect the quality of water due to
   A. Corrosion of water pipe
   B. Leaching heavy metal from water pipe
   C. Toxicity due to release of trace metals
   D. All of the above

24. Acidic rain affects
   A. Plant and soil
   B. Waxy cuticle of plant leaf
   C. Aquatic animal
   D. All are the answer

25. The main difference between exponential growth and logistic growth is that, exponential growth is _____
   A. Is a S shaped curve
   B. Resources are limited
   C. Ideal population growth
   D. Real type of population growth

26. Which one of the following is not true abut type 1 survivorship curve?
   A. Have great parental care
   B. Death at earlier age is lower
   C. exhibit high death rate at later life

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D. Give birth to many offspring

27. If the age pyramid of a certain country is narrowing at the base, what does this tells about the trend of the population size of the country? The population size is

A. Growing fast
B. Declining
C. Increasing slowly
D. Stable

28. Which of the following nations has stable population?

A. Ethiopia B. USA C. China D. Nigeria

29. Which of the following can be given as a good reason for finding a variety of plant and animal species in Ethiopia?

A. Lack of ecological disturbance
B. Good biodiversity management
C. Climate diversity
D. Environment free from predator

30. What could be the main reason behind the current observed low death rate and stable population growth in the industrialized countries?

A. Poor family planning
B. Improved health care
C. High infant mortality
D. Poor nutrition status

31. The burning of coal and natural gases would have greatest impact on which environmental issue?

A. Depletion of ozone layer
B. Exposure to UV - radiation
C. Production of acid rain
D. Global coldness

32. The total variety of living things on earth is described as

A. An ecosystem
B. Biodiversity
C. A population
D. Competition

33. The rate at which a population grows or declines depends on the rate of _____.

A. Births
B. Death
C. Immigration & emigration
D. All of the above

34. Resources competition, disease, and predation are examples of ____ population growth factor.

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A. Density - independent
B. Dispersal
C. Population sustaining
D. Density - dependent

35. The largest population that can be maintained at a particular environment for an indefinite period is known as
A. Carrying capacity
B. Declining phase
C. Resources are limited
D. Real type of population growth

36. The main difference between exponential growth and logistic growth is that, exponential growth is _____
A. Is a S shaped curve
B. Resources are limited
C. Ideal population growth
D. Real type of population growth

37. Which one of the following is not true about type I survivorship curve?
A. Have great parental care
B. Death at earlier age is lower
C. Exhibit high death rate later life
D. Given birth to many offspring

38. A population of Birds and lizard are classified into ____ type of survivoship curve.
A. Type I B. Type II C. Type III D. Type I and II

39. A fish population of 250 in a pond has a maximum annual rate of increase of 0.8. If the carrying capacity of the pond is 1500 for this species, what is the
A. 167 B. 417 C. 450 D. 500

40. Which of the following plants are not endemic to Ethiopia?
A. Kosso B. Bahirzaf C. Junipers D. Mekmeko

41. The population of insects and fish are grouped under _____ survivorship curve?
A. Type I B. Type II C. Type III D. None of these

42. A population of Birds and lizard are classified into _____ type of survivorship curve.
A. Type I B. Type II C. Type III D. Type I and II

43. In a population of 200 mine if 2000 mices die per year. The death rate will be

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A. 0.1 B. 0.5 C. 0.2 D. 2

44. If five cars are in the pattern: car, empty space, car, empty space, etc. What type of dispersion is takes place

A. Random C. Clumped

B. Uniform D. Clumped and random

45. A fish population of 250 in a pond has a maximum annual rate of increase of 0.8. If the carrying capacity of the pond is 1500 for this species, what is the expected population size after one year?

A. 167 B. 417 C. 450 D. 500

46. A population of 112 deer inhabits a forest ecosystem can support 250 deer. If the population grows to 145 after one year, what is the maximum growth rate (rmax) of the population?

A. 0.1 B. 0.3 C. 0.9 D. 0.5

47. A population 150 rabbit inhabit a forest ecosystem that can support 450 rabbits. If the population growths to 200 after one year, what is the maximum growth rate of the population?

A. 0.5 B. 0.9 C. 0.2 D. 0.6

48. Based on the following data, calculate the density population of species X in all five quadrants from (1 to 5)

Plant species	Number of quadrant ($1m^2$ each)					density
	1	2	3	4	5	
No of X	2	8	1	3	1	

A. 5 B. 0.3 C. 0.5 D. 1

49. Which of the following terms includes all the others?

A. Community C. Population

B. Species D. Ecosystem

50. In an ecosystem the term population is refers to _____

A. A member of different species

B. A member of a single species

C. All the consumer in the ecosystem

D. All the producers in the ecosystem

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51. Developed countries have a demographic transitions, characterized by
   A. High death rate at earlier
   B. High birth rate and low death rate
   C. Low birth and death rate
   D. High birth and death rate

52. Which one of the following is not biotic factor?
   A. Temperature
   B. Predation
   C. Disease
   D. Competition.

53. The demographic characteristic of developing countries are
   A. High birth rate
   B. High death rate
   C. Greater number of young.
   D. All of the above

54. Which of the following factor doesn't affect the global population size?
   A. Natality
   B. Mortality
   C. Migration
   D. None of the above

55. Which one of the following are the indirect practise that affect biodiversity?
   A. Deforestation
   B. Overgrazing
   C. Population growth
   D. Invasive alien species

56. Which one of the following is abiotic factor influencing population growth?
   A. Predation
   B. Completion
   C. Disease
   D. Earth quake

57. Which of the following is not represented at a population pyramid?
   A. Male and female ratio
   B. Number of birth and death
   C. Individuals at each age group
   D. None of the above

58. Which one of the following is not true about ozone?
   A. The chemical formula O3
   B. Located in stratosphere
   C. Protect the earth from UV - rays
   D. Depleted by any gas

59. A gas which is responsible for ozone layer depletion is
   A. Chlorofluorocarbon
   B. Oxygen

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C. Hydrogen D. Nitrogen

60. How does DDT affect the population of birds?

A. Nervous system C. Impair brain

B. The egg shell D. Reproductive system

61. Assume that the parking lot contain cars that are distributed evenly. Such type of dispersion resembles

A. Clumped C. Uniform

B. Random D. None of this

II. Short answer item

62. A group of similar organisms of the same species, which can interbreed together and produce viable offspring within a given area and a given time, is called _____ ?

63. A type of dispersion that occurs when individuals are concentrated in specific parts of a habitat is _____ ?

64. The type of population census that government fixes the counting period to be from 2 - 3 weeks is _____ ?

65. Population growth models that can be affected by the availability of resources are known as _____ ?

66. Maximum population size that the environment can support is _____ ?

67. _____ refers to the number of years that a human being can live in relation to the current mortality rates?

68. A population pyramid that shows a high fertility rates with lower life expectancies is _____ ?

69. The type of survivorship curve that shows higher probability of survival when age is increasing _____ ?

70. Write the two types of strategies that used to regulate population?

71. Natural resources that do not naturally replenish at the same speed at which they are used up to meet the growing demands are knows as __?

72. _____ is an inorganic molecule with the chemical formula made of three oxygen atoms that cover the stratosphere of the earth?

73. Write the two strategies used to conserve wild animals?

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74. Mention some modern practices used to conserve plants?

Essay type

75. Discuss the effects of acidic rain on the growth of plants?

76. Explain the main reason how DDT affect the population of birds?

77. Explain how the indigenous knowledge of people is important in conserving plant in Ethiopia?

78. A population of 112 deer inhabits a forest ecosystem that can support 250 deer. If the population grows to 145 after on year, what is the maximum growth rate (rmax) of the population?

79. List the type of population dispersion?

80. List some example if invasive species Ethiopia?

81. Discuss briefly how the depletion of ozone layer affects living organisms on earth?

82. How does an exponential population growth differ from a logistic population growth?

83. List some mechanical conservation for soil and water?

Answer to Review Exercise

I. Matching

1. B	3. C	5. H	7. E	9. G
2. A	4. I	6. J	8. F	10. D

II. Multiple Choice

11. D	19. D	27. B	35. A	43. A	51. C	59. A
12. D	20. C	28. B	36. C	44. B	52. A	60. B
13. B	21. C	29. C	37. D	45. A	53. D	61. C
14. A	22. B	30. B	38. B	46. D	54. C	
15. A	23. D	31. C	39. A	47. A	55. C	
16. A	24. D	32. B	40. B	48. D	56. D	
17. D	25. C	33. D	41. C	49. D	57. B	
18. A	26. D	34. D	42. B	50. B	58. D	

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62. Population
63. Clumped
64. De - Jure
65. Logistic growth
66. Carrying capacity
67. Life expectancy
68. Expanding
69. Type II
70. Density - dependent and density independent
71. None renewable
72. Ozone
73. In - situ and ex - situ
74. Establishing botanical garden, control invasive species, set up seed bank

III. Essay type

75. Acidic rain damages waxy cuticle of plant leaf, so that important nutrients such as Ca, Mg and K are leaches out of the leaf tissues and dying

76. Birds consume DDT containing prey in the food chain: this makes the egg shell thin, so that it breaks up easily during incubation

77. Taboos and forbidden practices of using some plant species, religious sites are used as preservation of plants, and traditional farming system is used as a preservation for some vegetables and crops.

78. given

$N = 112$, $r_{max} = ?$ $K = 200$

Population growth rate $= DN/dt = r_{max} N \dfrac{(K - N)}{K}$

$$145 - 112 = ramx \times 100 \frac{(200 - 100)}{200}$$

$$r_{max} = \frac{200 \times 33}{100 \times 100} = 0.66$$

79. Random dispersion, uniform dispersion, and clumped dispersion

80. L. camara, juliphora, striga species, water hycinth

81. Depletion of ozone layer let UV - rays reach to earth, which causes problem such as: skin cancer, cataract, weaken immune system, ageing skin, respiratory disease, smog formation, etc.

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82.

Exponential population growth	Logistic population growth
Growth is unlimited by resources	Growth is limited by resources
Have J - shaped curve	Have S - shaped curve
Shows an ideal growth of population	Shows a real growth of population

83. The mechanical conservation of soil and water are: terrace, bitches, check dam, micro basin and bunds.

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UNIT 7

7. Application of Biology

Biology:- is the science that studies about living things, Biological science can be applied and used in different aspects, such as:

- To determine the origin of diseases and pests, and factor that cause disease.
- To study the functions of living organisms
- To study the evolution of species and rate of extinction
- For the discovery of new drugs, such as antibiotics
- To explore topics such as genetic engineering, mother cells, and global warming
- To study about conservation on of natural resources.
- To study how living things are interact in life

It can be used in practical applications at home level, such as:

- Hand washing with soap to kill pathogens
- Domestication of animal and plants for human use (medicinal plants, farm animals, etc.)
- Traditional fermentation using microbs to make injera, tella, etc.

7.1 A practical Application of Biology

The science of biology can be applied in different sectors to maximize the benefits of human. It includes:

1. Conservation of natural resources
2. Food and nutrition security

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3. Creating conscious citizen
4. Biotechnology and genetic engineering
5. Bio-mining
6. Environment
7. Industries
8. Agriculture
9. Biological warfare

7.1.1 Application of Biology in Conservation of Natural Resources

Natural resources are any materials from the earth that are used to support life and meet people need, or any natural substance that human use.

It can be of two types: renewable and non - renewable

i. Renewable Natural Resource:

Renewable natural resources are resources that can be replaced after utilization.

Example: Soil, water air, plants, animals, and energy

Resources can be lost (extinct) if we do not manage them carefully.

ii. Non-renewable Natural Resource

A resources is considered non-renewable when its rate of consumption exceeds its rate of recovery.

Example: Minerals, fossil fuel coal and natural gas.

Conservation: is the care, protection, and wise use of natural resources so that the resources can be used for future generation.

Conservation biology is a mission - oriented science that focuses on how to protect and restore biodiversity, or the diversity of life on Earth.

Ethiopia has many natural resources, such as minerals, lakes, river, plants and animals that make up rich ecosystem.

Therefore, the application of the knowledge of conservation biology is required as a grater scale.

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For example, what should be done to conserve trees?

- Coppicing
- Planting trees to replace the ones being cut down
- Establish seed bank

Conservation of animals can be done through: establishing zoo, stop destruction of habitats.

7.1.2 Application of Biology in Food and Nutrition Security

The application of the biological senescence is essential to produce high - nutrient crops to combat malnutrition. Produce processed food and drink through biotechnology, which have consistent flavour, and texture to be produced in large quantities.

Food Security is a state in which all the people have physical, social, and economic access to sufficient, safe, and nutritious food that meets their food preference and dietary needs for an active and healthy life at all times, food security is ensured only if enough food is available for all in a county:

- When all individuals have the capacity to buy quality food without any barrier.

Note: If food is not secured, it has an impact on growth, cognitive development and risk of disease.

Nutrition security is a situation whereby individuals have access to sufficient, safe and nutritious foods, safe water and adequate sanitation; the ability to access to health care services, and knowledge of child care, food storage and preparation hygiene.

7.1.3 Creating Conscious Citizen and Ensuring Sustainable Development

Conscious citizen: is a person who places value on being fully human while connecting with a higher purpose.

Example of conscious citizen, is the one who:

- Values human life and a relationship with all living things,
- Transform skills into action through ethical decision to improve the living conditions
- Develop innovative and cost effective bio - based technologies

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➥ Involved in sustainable development through conservation of resources . For example, analysing soil, water and air for chemical pollution, finding ways to clean up pollution

Illustrative Example

1. Which of the following is least studied under the science biology?

   A. Determine origin of disease
   B. Function of living things
   C. Evolution of species
   D. The study of soil and water

   Explanation

   Biology is the study of living things: conservation, origin, function.

   Answer: B

2. The practical application of biology, which is the most familiar at home level is _____ ?

   A. Hand washing with soap
   B. Bread making
   C. Domestication of organisms
   D. All are the answer

   Explanation:- hand washing with soap (kills microbes), bread making (uses yeast calls), and domestication of farm animals and plants are the most familiar application of biology at home level.

   Answer: D

3. Conservation of natural resources preaches all, except _____ ?

   A. Preserving resources for future uses
   B. the wise use of resources on the basis of sustainabilit /
   C. extensive use of resources
   D. protection of natural resources

   Explanation: Conservation is the protection, preservatio n and wise use of natural resources.

   Answer: C

4. Which one of the following is the method of conserving plants?

   A. Establishing zoo
   B. Coppicing
   C. Pruning
   D. Grafting

   Explanation:- Coppicing involves cutting back the shoots of the plant to about 10 - 15 cm above ground and allowing the stump to regrow and produce more branches.

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Answer: B

5. A condition in which all people have an access of sufficient, safe and nutritious food to physical and social well being is _____ ?

A. nutrition security
B. shelter security
C. body security
D. food security

Explanation:- Food security is a state in which individuals have physical, social, and economic access to sufficient, safe, and nutrition food that meets their food preference and dietary needs for an active and healthy life at all times.

Answer: D

6. Conscious citizen is individual who values all, except _____ ?

A. Ethical decisions in transforming skill
B. improving the life of people
C. Innovative and cost effective technologies
D. self - interest

Explanation:- conscious citizen don't value self - interest, rather works for the mass of people.

Answer: D

7.1.4 Application in Biotechnology and Genetic Engineering

Biotechnology is the integration of natural and engineering sciences to achieve the application of organisms, cells parts and molecules to make product and services that benefit humans.

I. Food processing, and foods productivity using micro organisms.

For example: single cell protein (SCP), vitamins, dairy products, fermented meat, production of alcohol

Single - cell protein (SCP) refers to dead and dry cells of micro organisms like yeast, bacteria, fungi and algae, which serves as a food or animal feed supplements.

- It includes: mycoprotein, pruteen, and supplements.
- Microbes for SCP are growth on waste materials such as molasses, petroleum by - products, and agriculture wastes.

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- It can be used as a substitute for protein food, a potential protein source for developing nation

Mycoprotein is a fungal protein, which is a form of SCP, derived from fungi known as *Fusarium venenatum* for human consumption or meat substitutes.

Pruteen is one of the first product of SCP, which is produced by a bacterium called *Methylophilus methylotrophus*, for additive of animal feed

Spirulina is a mass of cyanobacteria (blue green algae), that is rich in protein (62%) , vitamins, mineral and antioxidants.

- It is harvested from the surface of lake and ponds.
- It is served as a dietary supplement.
- It can be available in the form of **pill** or **powder** from.

Production of Vitamins

- Vitamin C was the first vitamin to be produced during a fermentation process by using bacteria.
- Previously, Vitamin B12, B2 or Riboflavin was obtained from animal liver extract. But, nowadays it is production by propionic bacteria.
- In nature vitamin B is found in cereals, vegetables and yeast but the yield can be enhanced hundred to three hundred fold by using microbes.

Dairy products

Dairy products are made from milk, different bacteria are involved in the process. Examples:

- Yoghurt is made by adding *Streptococcus thermophiles* and *Lactobacillus bulgaricus*
- Cultured butter milk is made by adding *streptococcus cremoris*
- Sour cream is made by adding *Streptococcus lactis* or *Leuconostoc citrovorym*

Note: fermented milk can vary from each other in their acidity and alcohol content, which is on the type of bacteria, is used. For example, Bulgaria milk is more acidic, which is made by adding Lactobacillus bulgaricus

Fermented meat

- Meats such as salami, summer sausage and Lebanon bologna have got their characteristic flavour by microbial fermentation, such as

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Lactobaillus plantarum, Pediococcus cerevisiae

- Fermented country ham (sheep meat) has got its flavour from a fungus called penicillin and asperigillus

II. Production of alcohol:

Alcohol drinks such as beer, wine spirits are produced through microbial fermentation process. A fungus, known as saccharomyces are used as a source of yeast and flavouring during fermentation of any alcohol.

How to make beer ?

Material needed and procedure:

a) Crushed malted barley (a source of sugar or energy for yeast) is mixed with hot water (65°c)
b) liquid extract (worts) is separated from the mix
c) Hops are added for flavouring
d) The mix boiled to stop enzyme action and to precipitate protein
e) Saccharomyces are added for flavoring
f) Ethyl alcohol is produced with other substance such as acetic acid, butric acids, CO_2, amyl, and asoamyl alcohol
g) Finally the yeast removed and beer is filtered.
h) The alcoholic content of beer is about 6%

How to make wine ?

Material needed and procedure:

a) fruit such as grapes, is the main source of sugar
b) The fruit are treated with SO2 to kill wild yeasts from the surface
c) Sugar and saccharomyces are added into the fruit juice and fermented
d) Ethyl alcohol is produced with other chemicals similar to those in beer
e) Wine is separated from yeast sediment, aged in cool places and bottled.

Note: The type of fruit juice and yeast determine the flavour of wine. The alcoholic content of wine is about 12%

How to make sprits?

Sprit includes whiskey, gin and rum. They are made by microbial fermentation and then distillation is followed. The alcoholic content of sprit ranges from 40% - 50% following distillation.

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- Scotch Whiskey is made of fermented barley
- Vodka is made from potato
- Brandy is made from fruit juices
- Rye whiskey and gin are made of rye
- Rum is made from sugar cane molasses

Bread making

Procedure:

a) Mix the flour and water
b) Add baker's yeast (*Saccharomces cerevisae*)
c) Flour contain maltose, which used as energy source for yeast
d) The yeast respire aerobically to produce CO_2 and water
e) The CO_2 is used to raise the dough, this process is called leavening. These make the bread spongy, porous and good flavor.
f) Enzyme from microbes in the dough break down flour protein (glutten) and create elastic dough

Note: beside CO_2, other organic acids and alcohol produced during fermentation are essential for flavoring and aromas on the bread. Micro organism in the dough is essential for leavening, flavor and odor, and conditioning dough

Illustrative Example

7. A type of single cell protein (SCP), made of fungus called *Fusarium venenatum* for human consumption as a meat substitute is ____?

A. Spirulina
B. Mycoprotein
C. Pruteen
D. Fermented meat

Explanation: Mycoprotein is a type single cell protein (SCP), made of fungus called *Fusarium venenatum*.

Answer: B

8. Which of the following is not true about spirolina?

A. Is a type of single cell protein
B. Can be available in the from green powder or pills
C. Is made of cyanobacteria (blue green algae)
D. Produced in dry land

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Explanation: Spirulina is a mass of cyanobacteria (blue green algae), that is rich in protein (62%), vitamins, mineral and antioxidants.

- It is harvested from the surface of lake and ponds
- It is served as a dietary supplement.
- It can be available in the from of pill or powder form.

Answer: D

9. The first vitamin produced by bacterial fermentation is _____?

A. Vitamin B12 B. Vitamin C C. Vitamin B D. Vitamin D

Explanation: Vitamin C was the first vitamin to be produced by a fermentation process of bacteria.

Answer: B

10. Fermented milk can vary from each other in their _____ contents?

A. Acidity B. Protein C. Alcohol D. A&C

Explanation: Fermented milk can vary from each other in their acidity and alcohol content, which is based in the type of bacteria is used

Answer: D

11. The type of fungus that used in the fermentation of any alcohol is ?

A. Fusarium
B. Candida
C. Mould
D. Saccharomyces

Explanation: A fungus, known as Saccharomyeces are used as a sources of yeast and flavouring during fermentation of any alcohol.

Answer: D

12. One of the following is not true about wine preparation?

A. The main source of sugar is grapes
B. The flavor is determined by the type of yeast used
C. It can be produced from fruits
D. Its alcoholic content is less than beer

Explanation: the alcoholic content of wine is about 12%

Answer: D

13. The type of alcoholic spirit, made by fermentation of potato followed by distillation is _____?

A. Whiskey B. Run C. Vodka D. Brandy

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Explanation: Scotch Whiskey is made of fermented barley, Vodka is made from potato, Brandy is made from fruit juices, Rye whiskey and gin are made of rye, and Rum is made from sugar cane molasses.

Answer: C

14. The type of yeast used for making bread known as _____?

A. Any form of yeast
B. *Candid albican*
C. *Fusarium venatum*
D. *Saccharomyces cerevisae*

Explanation: Saccharomyces cerevisae type yeast used to ferment dough in bread making

Answer: D

III. What is Tissue Culture

- Tissue culture is a technique in which cells are isolated from plant and grown in nutrient medium.
- It is used to propagate large quantity of plants quickly:
- Produces high quality and disease free plants
- Plants that are produced in tissues culture frequently are palm, orchids, banana and carrot

The steps of propagation of plants in tissue culture involves

a) Tissue are taken from parent plants
b) Tissue are placed in agar (nutrient medium)
c) It develops tiny plantlets (seedling), which are identical to parent plants.

IV. What is Biosensor Technology

Biosensor is an electronic monitoring devices, which uses a biological material, such as cells, enzymes or antibodies, to detect or measure a chemical compounds.

- They are specific and pick out particular molecule from complex molecule
- They can be used to measure specific components in the beer, monitor pollutants, detect flavor from food.
- Detect glucose, acetic acid, glutamic acid, ethanol, biochemical oxygen from industrially produced food, pharmaceutical products
- Recently discovered biosensor can detect pathogens, herbicides, toxins, protein and DNA. Example hand held aflatoxin is used to monitor

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quality of food

- The main advantage of biosensor is: they are rapid in response, safe to use, accurate, produced in mass
- The main problem of using biosensor is they are not stable, not sterilized and need careful handing.

V. What is Forensic science

Forensic science is the application of scientific knowledge to questions of civil and criminal law such as paternity case of child, crime investigation, identifying human cancer cells, to study genetic ancestors of human population.

- Genetic finger printing is a technique to compare DNA from different people
- It is based on DNA analysis, DNA extracted from individual is unique to that individual, which is called DNA fingerprinting.
- Two random individual would have only a chance of 1 in 10 billion similar genetic patterns.
- Sample of DNA can be taken from blood cells, saliva and hair
- DNA analysis is made by gel electrophoresis in PCR machine.
- Gel electrophoresis is a technique that use a thick jel to act as a molecular sieve. This result can be used for different purpose.

For example, in crime inverstigation, we compare weather the DNA take from suspect and crime scene is similar or not and report to the court.

- Bioinformatics is widely used in the field of forensic science.
- Bioinformatics is the application of information technology in genetics.

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Illustrative Example

15. Which of the following is not true about tissue culture?

A. It is asexual reproduction
B. Produce healthier plants
C. Produce identical plants
D. Carried out by natural means

Explanation: Tissue culture is an artificial technique in which cells are isolated from plant and grown in nutrient medium. It is an asexual reproduction, produces identical plants.

Answer: D

16. The main advantage of using biosensor is all, except _____?

A. they are rapid in response
B. Safe to use
C. Accurate
D. Can's be sterilized

Explanation: The main advantage of biosensor is: they are rapid in response, safe to use, accurate, produced in mass

Answer: D

17. The recently discovered biosensor can detect _____ in the mixture of chemicals?

A. Pathogens and DNA
B. Biochemical oxygen and lactic acid
C. Glucose and alcohol
D. Lactic acid and glutamic acid

Explanation: Recently discovered biosensor can detect pathogens, herbicides, toxins, protein and DNA. Example hand held afloxin is used to monitor quality of food.

Answer: A

18. In forensic investigation the sample of DNA is take from _____?

A. Blood cell
B. Hair
C. Saliva
D. All are answer

Explanation: Sample of DNA can be taken form blood cells, saliva and hair.

Answer: D

19. Which one is not correct about genetic finger printing?

A. Is unique to individual
B. Is used for criminal investigation
C. Is made by gel electrophoresis in the PCR machine
D. Is based on RNA analysis

Explanation: genetic finger printing is a technique used to compare DNA from different people. Separation of DNA is made by gel electrophoresis. Gel electrophoresis is a technique that uses a thick jel to act as a molecular sieve. Everybody has a unique DNA fingerprinting.

Answer: D

20. Which of the following is not true about tissue culture?

A. It is asexual reproduction
B. Produce healthier plants

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C. Produces identical plants D. Carried out by natural means

Explanation: Tissue culture is an artificial technique in which cells are isolated from plants and grown in nutrient medium. It is an asexual reproduction, used to produce genetically similar and healthier plants.

Answer: D

21. The main advantage of using biosensor is all, except _____ ?

A. They are rapid in response C. Accurate

B. Safe to use D. Non sterilizable

Explanation: Biosensor is an electronic monitoring device, which use a biological material, such as cells, enzymes or antibodies, to detect or measure a chemical compounds. The main advantage of biosensor is they are rapid in response, safe to use, accurate, produced in mass.

Answer: D

22. The recently discovered biosennsor can detect ___ in the mixtures of chemicals?

A. Pathogens and proteins

B. Glucose and alcohol

C. Biochemical oxygen and lactic acid

D. Lactic acid and glutamic acid

Explanation: Recently discovered biosensor can detect pathogens, herbicides, toxins, protein and DNA. Example hand held aflatoxin is used to monitor quality of food.

Answer: A

23. In forensic investigation the sample of DNA is taken from _____ ?

A. Blood cell C. Saliva

B. Hair D. All are answer

Explanation: Sample of DNA can be taken from blood cells, saliva and hair.

Answer: D

VI. What is Genetic Engineering

It is the process of transferring genetic material from one organism to another for the purpose of creating **transgenic organism.**

An organism that received genetic material from another organism is

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known as transgenic organism or **genetically modified organism (GMO)**.

Genetic engineering can be used in different sectors to increase productivity. For example:

- Treatment of diseases, gene therapy
- Agriculture
- Industrial processing
- Energy production
- Pharmaceutical
- Food and drink production
- To produce transgenic plant and animals (GMOs)

The importance of genetically modified organisms (GMO) are:

- To produce crop or livestock with increased yields, growth rate, drought and disease resistant
- To produce high quality nutrition food
- For the treatment of waste or sewage
- For diagnosis and treatment of disease

☑ **Genetic engineering can be used inhuman health and well being**

Biological science can be applied in the production of drugs and hormones to ensure human health and well being. Genetically modified bacteria can be used to produce:

- Insulin, a hormone used to treat diabetes.
- Human growth hormone, which is used to increase size for dwarf
- Plasminogen activator, used to treat heart attack
- Antibiotic production

☑ Antibiotics are drugs that attack bacteria in different ways:

- Disrupt production of cell wall
- Durst and open the cell
- Interrupt bacterial protein synthesis

☑ Antibiotics that stops bacterial reproduction is known as **bacteriostatic**

☑ Antibiotics that kill bacteria is called **bacterioicidal**

☑ Antibiotics don't harm or kill cell, because animal cells have no cell wall, and also the structure of cell in protein synthesis is different from

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those in bacteria.

☑ The first antibiotic was made from a fungus called penicillun, which was discovered by Alexander Fleming, in 19 28.

☑ Nowadays, antibiotics are produced from genetically modified bacteria, such as: Actinomycests, to produce streptomycin.

☑ Antibiotics such as: ampicillin, methicillin and oxacillin are produced from different strain of fungi.

Genetically modified bacteria are grown in the fermenter in larger scale. **Fermenter** is a large vessel, which used to culture and grows bacteria in large quantity under controlled condition in the laboratory.

7.1.5 Application of Biology in Biomining

Biomining is a process of mining or leaching of minerals from its ore by the aid of micro organisms.

➥ **Bio hydro metallurgy** is the use of microbes to extract metals from ores.

For example: copper is leached from its ore (sulfide) by the action of bacteria called *Thiobacillus ferrooxidans*. The bacteria oxidize the sulfur that binds on iron, copper, zinc, lead and uranium into sulfate to produce pure metal

7.1.6 Application of Biology in the Environment

The knowledge of biological science is important in cleaning waste that pollute the environment by using micro organisms, such as bacteria and fungi.

I. Solid waste treatment and composting

West (refuse) removed from house hold or industries can be of two type: biodegradable and non - biodegradable:

➥ Non - biodegradable waste are wastes that can't be degrade or broken down by micro organisms. Examples: metals, plastic, rubble

➥ Biodegradable wastes are waste that can be decomposed or degraded by natural decomposers (bacteria and fungi) through aerobic or anaerobic respiration. It includes food wastes, plant and animal materials, textile. It can be used to make compost.

How can compost is prepared?

Composting is a conversion of solid organic waste, such as plant, manure

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into stable humus - like material through aerobic process.

- It is used for soil fertility.
- Substrate to make mushroom (*Agaricus bisporus*)

Compost can be produced in large scale in the container containing microbes on stone bed, having a system of aeration for oxygen.

Then organic molecules are oxidized by micro organisms into smaller molecules, CO. H_2O and small organic substance.

Compost is not a strict fertilizer, because its nitrogen content is very low, but it still provides nutrients to the soil.

II. Waste water treatment:

Sewage is a waste water come from home or industries, containing toxic, organic molecules and pathogenic micro organisms.

- Sewage treatment is the process that makes water safer use and joins streams.
- Effectiveness of water treatment can be measured by biochemical oxygen demand (BOD)
- BOD measures the a mount of oxygen needed by micro organism to oxidize its organic content
- The higher BOD is the higher pollution of water.

The treatment can be performed into two steps:

- Primary treatment - is a physical process, which involve sedimentation and removal of floating objects from the water. It reduces 1/3 of BOD
- Secondary treatment is a biological process, which involve micro organisms to oxidize organic pollutants. It reduces 95% of BOD. Secondary treatment can be done into two forms:

1. Tricking filter

- Waste water is filled in the large settlement tank (contaniner), containing micro organisms (bacteria, fungi, protozoa, and algae) on bed of stone to oxidize the organic pollutants.
- 80 - 85% of BOD is reduced

2. Activated sludge method

- The sludge (seeds of micro organism, such as zoogloea) is added to the waste water filled in settlement tank, have aeration system to blow oxygen into the system to oxidize organic molecules.

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- The bacteria produce slim (flock), which is used to attach other micro organisms in the water
- The sludge is recycled and filtered out of the settlement tank
- Now the water is recycled: and safe to join river, watering garden, but can't be used for drinking because it has many pathogenic micro organisms.
- Settlement of sludge is decreased by the action of filamentous bacteria, such as *spherotillus natans*. The process is known as **bulking**
- Anaerobic oxidation of organic molecules in the waste water can produce methane, which can be used as a sources of energy.

3. Bioremediation

Bioremediation is the biological treatment of hazardous waste from environment into harmless and non toxic compounds by natural genetically modified micro organisms.

- It is based on the enzymes produced by microorganisms.
- The enzyme changes the nature of chemicals wastes, which is known as bio - transformation

For example, the redox reaction carried out on heavy metal by micro organism reduces its toxicity.

4. Biofuels

Biofuels are fuel, made from plant materials, containing cellulose and hemicellulose by anaerobic fermentation of micro - organisms. Example of bio fuel includes ethanol

- For example, anaerobic fermentation of molasses by *Saccharomyces Cervisae* produces ethanol, CO2 and water.
- Genetically modified E. coil strain can ferment any form of sugar from plant material, which increase the bio fuel production.

5. Biogas production

Biogas is a flammable mixture of gases produced by anaerobic break down of organic molecule, such as plants, animal by micro - organism.

- The main components of biogas is methane: other gases such CO_2, water vapor, and hydrogen sulfide
- A mixture of microorganisms, archeabacteria and Eubacteria are involved in the process

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- Some of the Eubacteria converts organic acid, hydrogen, and alcohol into acetate: the other Eubacteria are involved in the conservasion of organic acid and alcohol into acetate
- Archea bacteria such as methanol bacteria are involved in methane production

The main advantage of using biogas is

- relative cheaper than petroleum gases
- It can be used as heat, light and refrigeration
- The left over (slurry) is used for fertilizer

Illustrative Example

24. The application of genetic engineering on health directly explained in terms of _____ ?

A. Antibiotic production
B. Insulin production
C. Crop productivity
D. A and B

Explanation: One of the applications of genetic engineering is health. For example: the production of antibiotic, insulin, and gene therapy, etc.

Answer: D

25. Copper is leached from its ore (sulfide) by the action of bacteria called _____ ?

A. *Thiobacillus ferrooxidans*
B. *Strtoptoccus cremoris*
C. *Streptococcus thermophiles*
D. *Streptococcus lactis*

Explanation: Bio hydro metallurgy is the use of microbes to extract metal from ores. For example: copper is leached from its ore (sulfide) by the action of bacteria called *Thiobacillus ferrooxidans*.

Answer: A

26. Which of the following solid west is grouped under non - biodegradable?

A. Food waste
B. Plant material
C. Plastic
D. Animal waste

Explanation: non - biodegradable waste are waster that can't be degrader or broken down by microorganisms. Examples plastic

Answer: C

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27. Secondary treatment of waste water is different from primary treatment in that secondary treatment _____ ?

A. Reduces BOD by 30%
B. It is a physical treatment
C. It uses micro organism for treatment
D. It increases BOD by 95%

Explanation: Secondary treatment of sewage is a biological process, which involves micro organisms to oxidize organic pollutants. It reduces 95% if BOD.

Answer: C

28. What is the importance of using genetically modified E. coli strains than micro organism in the fermentation process to produce biofuel?

A. Because it can digest all types of sugar
B. Because of the lack of other microbes
C. Because it is cheap
D. Because it can digest proteins

Explanation: E. coli can ferment any form of sugar from plant material, which increases the biofuel production

Answer: A

29. Which of the following is not true about Biogas production?

A. The main components of biogas is methane
B. Both archeabacteria and Eubacteria are involved in the process
C. Is is more expensive than petroleum gases
D. Produced by anaerobic process

Explanation: Biogas is a flammable mixture of gases produced by anaerobic break down of organic molecule. The main components of biogas are methane, other gases such as CO2, water vapor, and hydrogen sulfide. A mixture of microorganisms, archeabacteria and eubacteria are involved in the process. It is cheaper than petroleum gas.

Answer: C

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7.1.7 Application of Biology in Industries

The science of biology can be applied in the industrial process through the use of enzymes extracted from microorganism

What is the importance of using enzymes in the industrial process?

- ☑ It reduces the amount of fuel (energy) used
- ☑ Catalyse a vast range of chemical reaction

The commonest microorganisms used for enzyme producing are: Aspergillus (fungus), and Bacillus (bacteria).

- ➥ Because they are non - pathogenic, and don't produce antibiotics

☑ Enzyme can be of two type: intracellular and extracellular

How enzymes are separated from the microbial cells?

- ➥ If an enzyme is intracellular, cells are separated from the liquid in the fermenter, then broken open by various means to release their content. Most enzymes are intracellular.
- ➥ If an enzyme is extracellular, the liquid content containing the enzyme is filtered from the cell

What are the commercials applications of enzymes?

i. Biological washing powder

Why enzymes are used to prepare washing powder?

Because enzymes reduces the amount of energy used to remove dirt from clothes, it washes clothes at lower temperature.

What are the examples of enzymes used to prepare washing powder?

- ➥ Protease: removes stain of biological origin containing protein, such as blood, milk, egg.
- ➥ Lipase - digest fat and oil, grease from clothes

ii. Textile production

- ➥ Cellulase - digest cellulose from cotton fibres, make it soft and brighter
- ➥ Amylase - digest starch from clothes

iii. Fruit juice production

- ➥ Cellulase and hemicellulse - break down cell wall of fruits

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- Pectinase is used for clarification of fruit juices, it removes pectin, which is sticky compound, suspended in the juices and causes jam

iv. **Meat tendering**

Papayin and protease - break down the tough, non - elastic muscle protein (collagen) from meat to make easier to chew

v. **Leather industries**

- Protease - removes hair from animal hides

vi. **Diary industries**

- Rennin - curding milk to make cheese
- Biochymosine - making cheese

7.1.8 Application of Biology in Agriculture

1. Biopesticides

Biopesticides are biological agents, such as toxin, protein products extracted from bacteria, fungi and viruses, which are used to kill insect.

For example:

The soil bacterium *Bacillus thuringiensis* produce power toxin, which is used to kill insect pests (80,000 times powerful than organophosphate (chemical pesticides)

- The toxin binds in insects' gut and damage epithelial cells, that it can't absorb digested food and starve to death.
- This bacteria is harmless to human

A bacterium called *Pseudomonas fluorescence*, contain toxic producing gene, which is taken from *B. thuringiensis*, is used in maize to kill **black cut worm**

The three viruses that can be used to produce bio pesticide are:

- Nuclear polyhdrosis virus (NPV)
- granulosis virus (GV)
- cytoplasmic polyhedrosis virus (CPV)

Four major genera of fungi are used to produce biopesticides.

Example

- *Beaveria bassianais* is used to control colorado potato beetles
- *Metarhizima anisophle* - is used to control frog hopper in sugar cane

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- *Entomopthora Spp* and *Verticllum lecanii* is used to control aphid in green house and field

II. The use of Ti plasmid as a vector

Plasmid are small circular DNA in bacteria, which are used as a vector to carry genes and transfer into plant cells to produce genetically modified plants through genetic engineering.

Tumor inducing (Ti) plasmid from *Agrobacterium tumifaciens* is used to inoculate (tranasfer) glyphosate resistance gene into soy bean crops.

- Glyphosate is herbicidal chemical that would kill crops.
- The cells having glyphosate resistant gene are cultured to duplicate in mass.
- When these plants are grown it will resist herbicides.

III. Insect resistant crops

Genetically modified plants (carrying foreign gene) are made to protect insect infestation. For example:

- Maize plant is genetically modified to protect corn borer.
- Cotton plant is genetically modified to protect boll weevil.
- Tobacco plant is genetically modified to protect bud worm.

The disadvantage of using generically modified plant against insect infestation is:

- A resistant gene may jump into weeds
- Can kill other non - targeted insects
- The insect may develop resistant genes, which enable them feed GM plants

IV. Transgenic plants and animals

Genetically modified organisms (plant or animals), which contain recombinant DNA are called **transgenic organisms**

These organisms are capable of producing special proteins to fix certain problems.

Transgenic plants are capable of:

- Resistant to disease, herbicide, and pesticide.
- Produce nutritious food

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Several human proteins are produced in the milks of transgenic sheep and goat, eggs of chicken.

Mice have been used extensively to test the effect of recombinant genes and mutations.

Note: Transgenic organisms need close monitory by governmental agencies to ensure they are safe for human use and don't pose danger on other organisms and disturb ecological stability.

V. Herbicide resistant

Transgenic plants that contain special gene, which produces special (enzyme) can break down herbicides such as glyphosate.

- So that spaying herbicides don't have any impact on plants.

Note: glyphosate is the well - known herbicides used by farmers to kill weeds

VI. Animal cloning

Cloning is the production of many genetically identical copies of an individual by asexual reproduction.

- A group of genetically identical individuals is called **clone.**
- The process is carried out artificially in the laboratory.

Advantage of cloning:

- To produce large number of identical organisms those have desired characteristics.
- To produce biosynthetic products such as body organs(skin, kidney, heart, etc)

The first successful cloning was carried out on sheep, known as "**Dolly**" the sheep.

Procedures of cloning to produce Dolly:

- E nucleated egg (nucleus removed egg cell) is taken form one sheep.
- Nucleus from udder (body cell) is taken from another sheep and fused into enucleated egg.
- Cells are cleaved and develop into zygorte and then become embryo.
- Embryo is implanted into the uterus of another female sheep called surrogate.
- Finally the lamp was born successfully and was named Dolly.

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7.2 Biological Warfare

Biological warfare is the use of pathogenic microorganisms (virus, bacteria and fungi) or biological toxin to harm or kill human, animals or plants as an act of war.

Biological weapons - are pathogenic microorganisms, such as virus, bacteria and fungi that can kill or harm human and other organisms.

The major concern of biological weapons are:

- If they are get in the hands of terrorist groups, destruction encountered.
- The long term treat of pollution by these microorganisms

Illustrative Example

30. A pair of microorganisms that are used commonly to produce enzymes for industrial process are _____ ?

A. Thiobacillus and E coli

B. Streptococcus and Saccharomyces

C. Spherotilus natasand italics bisporus

D. Aspergillus and Bacillus

Explanation: Aspergillus and Bacillus is pair of microorgaisms that are used commonly to produce enzymes for industrial process.

Answer: D

31. How do intracellular enzymes from microorganisms are separated after production in the fermenter?

Explanation: cells are separated from the fluid, and then broken open by various means to releases their content.

32. An enzyme that used to clarify fruit juices in the industrial process is _____ ?

A. Protease B. Amylase C. Pectinase D. Lipase

Explanation: pectinase is an enzyme used to clarify fruit juices.

Answer: C

33. An enzyme that used to catalyze tough, non - elastic muscle protein (collagen) from meat to make it easier to chew is _____ ?

A. Papyin B. Amylase C. Pectiase D. Lipase

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Explanation: papyin is enzyme used for meat tendering to make it easier to chew.

Answer: A

34. An enzyme that used to removes hair from animal hide in the leather industry is _____?

A. Papyin B. Amylase C. Protease D. Lipase

Explanation: Protease is an enzyme that used to remove hair from hides in leather industries.

Answer: C

35. A bacterium that produces powerful protein toxin, which is used to kill insect pests is _____?

A. Thiobacillus errooxidans C. Streptococcus themophiles

B. Stretococcus cremoris D. Bacillus thuringiensis

Explanation: Bacillus thuringiensis is asoil bacteria, which produces powerful protein toxin, 80,000 times powerful than organophosphate.

Answer: D

36. List the three viruses that used to produce biopesticides?

Explanation

- Nuclear polyhdrosis virus (NPV)
- granulsis virus (GV)
- Cytoplasmic polyhedrosis virus (CPV)

37. What is the name of fungus that is used to produce biopesticides to control Colorado potato beetles?

A. *Beaveria bassiana* C. *Metarhizium anisophle*

B. *Entomophthor Spp* D. *Verticillum lecanii*

Explanation: the four major genera of fungi are used to produce biopesticides are:

- *Beaveria bassianais* used to control Colorado potato neetles
- *Metarhizium anisophle* - is used to control frog hopper in sugar cane
- *Entomophthora spp* and *verticillum lecanii* is used to control aphids in green house and field

Answer: A

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38. What is the name of insect pest that attack maize plant?

A. Corn borer
B. Boll weevill
C. bud worm
D. mosquito

Explanation: Genetically modified plants (carrying forging gene) are made to protect insect infestation. For example:

➥ Maize plant is genetically modified to protect corn borer.
➥ Cotton plant is genetically modified to protect bud worm.

Answer: A

39. The most popular bacteria that used in agriculture to insert foreign gene into plant cells?

A. *Bacillus thuringiensis*
B. *Agaricus bisporus*
C. *Streptococcus cremoris*
D. *Agrobacterium tumifaciens*

Explanation: *Agrobacteriumt tumifaciens* the most popular bacterial that used in agriculture to insert foreign gene into plant cells.

Answer: D

40. An Artificial herbicide chemical that would kill most crops and plants

A. Glyphosate
B. Glycophosphate
C. Glyceride
D. Glycogen

Explanation: glyphosate is artificially made herbicide chemical.

Answer: A

41. A group of genetically identical individuals are called ____?

A. Tissue culture
B. Clone
C. Transgenic
D. Genetically modified organism

Explanation: a group of genetically identical individuals are called clone.

Answer: B

42. The first successful cloning was carried out on sheep, known as __?

A. Dolly
B. Lamp
C. Ewe
D. Puppy

Explanation: The first successful cloning was carried out on sheep, known as **"Dolly"** the sheep.

Answer: A

43. Transgenic plants are capable of ____ ?

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A. Resistant to disease and drought
B. Producing nutritious food
C. Resistant to herbicide and pesticide
D. All are answer:

Explanation: Transgenic plants are capable of: Resistant to disease, herbicide, and produce nutritious food.

Answer: D

44. The use of pathogenic micro organisms (virus, bacteria, and fungi) or biological toxin to harm or kill human, animals or plants as act of war is known as ____ ?

Explanation: Biological warfare

45. Discus the main concern of using biological weapons?

Explanation: Biological weapons - are pathogenic microoganisms, such as virus, bacteria and fungi that can kill or harm human and other organisms.

- If they are get on the hand of terrorist groups, destructions are encountered.
- The long term treat of pollution by these microorganisms

46. Discuss the disadvantage of using genetically modified plant against insets infestation?

Explanation: The main problem of using genetically modified plants are:

- A resistant gene may jump into weeds
- Can kill other non - targeted insects
- The insect may develop resistant genes, which enable them feed GM plants

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Unit Review Exercise

I. Matching

A	B
1. Cloning	A. Waste water comes from home or industries, containing toxic, organic molecules and pathogenic micro organisms
2. Transgenic	B. A process of mining or leaching of minerals from its ore by aid of micro organisms.
3. Plasmid	C. An electronic monitoring device, which uses a biological material, such as cells, enzymes or antibodies, to detect or measure a chemical compounds
4. Biopesticides	D. Biological treatment of hazardous waste from environment into harmless and non toxic compounds by natural or genetically modified micro organisms
5. Bioremediation	E. The production of many genetically identical copies of an individual by asexual reproduction
6. Sewage	F. A technique in which cells are isolated from plant and grown in nutrient medium
7. Biomining	G. Genetically modified organisms (plant or animals), which contain recombinant DNA
8. Forensic science	H. Biological agents, such as toxin protein products extracted from bacteria, fungi and viruses, which are used to kill insects
9. Biosensor	I. The application of scientific knowledge to questions of civil and criminal law such as paternity case of child, crime investigation
10. Tissue culture	J. Small circular DNA, used to transfer gene

II. Multiple Choices:

11. An elastic nature of dough during microbial fermentation is due to _____?

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C. Alcohol production
D. Break down of glutten

A. CO_2 production
B. Temperature

12. Yogurt fermentation is made by a bacterium called ____ ?
A. *Streptococcus thermophiles*
B. *Streptococcus lactis*
C. *Streptococcus cremoris*
D. *Leuconostoc citrovorum*

13. If an individual has nutritional security, he/she has an access of ____ ?
A. Safe and sufficient food
B. Nutritious food
C. Health care services
D. All are the answer

14. Which of the following is a renewable natural resources?
A. Gold B. Plant C. Natural gases D. Solar energy

15. Which of the following is not true about single cell protein (SCP)?
A. Dead and dry cells of microorganisms
B. Used as a substitute for protein food
C. Are grown on waste materials
D. Potential protein source for developed nation

16. What is the role of plasmid in genetic engineering?
A. To isolate genes from donor cell
B. To transfer gene in to the cell
C. To open up cells
D. To kill bacteria

17. Which of the following is not the importance of genetic engineering?
A. To produce transgenic organisms
B. Increase vulnerability to different disease
C. To produce recombinant DNA
D. To produce antibiotics

18. Which statement correctly describes about non - renewable natural resources?
A. Cannot be easily replaced if once used up
B. Can be easily replaced if managed carefully
C. Are mostly living things and their products

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D. Are produced indefinitely

19. Which of the following is the main product of biogas?

A. Carbon dioxide
B. Methane
C. Hydrogen sulfide
D. Water

20. Which one of the following processes does utilizes biotechnological knowledge?

A. Energy & enzyme production
B. Sewage treatment & antibiotic production
C. Mining process & environmental protection
D. Plastic & well beings

21. Which one of the following is not application of biology in health & well being?

A. Manufacturing of antibiotics
B. Use of bio - sensor
C. Use of forensic science
D. Microbiological mining

22. Which one of the followings is not the application of biology in environment?

A. Waste water treatment
B. Bioremediation & biogas production
C. Solid waster treatment
D. Use of biosesnsor & forensic science

23. Which one of the followings is not industrial use of bacteria?

A. Food processing
B. Recycling of minerals in the soil
C. Pharmaceutical production
D. Production of washing powder

24. Which groups of micrbes are more important in biotechnology applications?

A. Fungi & bacteria
B. Bacteria & protozoan
C. Fungi & Protozoa
D. Virus & fungi

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25. What is the reason that antibiotics did not damage animal cell?
A. Animals have cell wall & have similar cell structures involved in protein synthesis
B. Animals did not have cell wall & have different cell structures involved in protein synthesis
C. Animals have antibiotic resistant protein
D. Antibiotics can equally destroy animals & microbes

26. Which application of biotechnology uses earobic microbes?
A. Processing & producing dairy products
B. Production of alcoholic beverages
C. Producing of fermented meat
D. Bread making

27. Which one of the followings is not main purpose of biotechnology?
A. Increasing yield & obtaining novel products
B. In the diagnosis & treatment of disease
C. Prevention & mitigation of industrial contaminations
D. Making crop plants & livestock animals vulnerable to different diseases

28. Which pairs of enzymes are important in biological washing powder industry?
A. Protease & pectinase
B. Lipase & protease
C. Amylase & Lipase
D. Pectinase & Lipase

29. Which one of the followings is not role of microbes in bread making?
A. Conditioning the dough to be workable
B. Production of ethyl alcohol
C. Imparting flavors & odors
D. Producing fermentation in the dough

30. Which one of the followings is false about sewage treatment?
A. Larga floating materials are separated in the secondary sewage treatment.
B. Secondary sewage treatment is designed to remove almost all the

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BODs.

C. Secondary sewage treatment encourage the growth of aerobic bacteria

D. Sewage is sprayed over the bed of rock in trickling filter method

31. _____ is an animal feed produced by mass culture of *Methylophilus methylotrophus?*

A. Single cell protein
B. Mycoprotein
C. Pruteen
D. Spirulina

32. _____ is made by adding Lactobacillus to cream

A. Cultured butter milk
B. Yogurt
C. Sour cream
D. Cheese

Answer question number 33 based on the given information.

1. Barely is partially germinated.
2. Malt is dried, crushed & mixed with hot water producing germinated
3. Hop is added to the wort and boiled
4. Wort is separated from mash
5. Mixture of hop and wort is fermented by yeast to produce alcohol

33. Which one of the following shows the correct steps in commercial beer production?

A. $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$
B. $1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 4$
C. $2 \rightarrow 1 \rightarrow 4 \rightarrow 3 \rightarrow 5$
D. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

34. Which one of the followings contains the highest alcoholic content?

A. Beer B. Wine C. Spirits D. Tej

35. Which one of the followings is false about tissue culture?

A. It produces quality plant
B. It produces plants genetically different from the original
C. It disrupts the production large quantities of food
D. It produces disease free plants

36. How bactericidal antibiotic attacks disease causing microbes?

A. It directly kills the microbes
B. It interferes in protein synthesis

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C. It disrupts the production of cell wall
D. It inhibits nucleic acid replication

37. Identify the false statement about biosensor?
A. It measures specific components in beer
B. It monitors pollutants in different environments
C. It detects the different compounds in food that add flavors
D. It does not detect pathogens, proteins and DNA.

38. What is the first step in the preparation of genetic finger printing?
A. Transfers DNA fragments from the gel to nylon membrane
B. Cut the DNA in to fragments with restriction enzyme
C. Isolates DNA from a cell
D. Separates the DNA fragments by gel electrophoresis

39. Which one of the following is the positive effect of insect resistance crops made by genetic engineering?
A. The evolution of resistance by the insect pests
B. The added gene is transferred to other species of plants
C. Damaging different on the species of insects
D. Protect the crop from insect pests

40. Which one of the followings is false about cloning?
A. It used to make large numbers of plants and animals
B. It used to produce skin and other tissues for treatment
C. It used to produce identical copies of organisms
D. It forms clones, which shows genetic variability

41. Which one is not correct about genetic finger printing?
A. Is unique to individual
B. Is used for criminal investigation
C. It is made by gelelectrophoresis in the PCR machine
D. Is based on RNA analysis

III. Short Answer

42. A condition in which all the people have physical, and economic access to sufficient, safe, and nutritious food that meets their food dietary needs

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for an active and healthy life at all times is known as _____ ?

43. A situation whereby individuals have access to sufficient, safe and nutritious foods, safe water and adequate sanitation, the ability to access health care services, and knowledge of child care, food storage and preparation and hygiene is called _____ ?

44. A person who place value on being fully human while connecting with a higher purpose is known as _____ ?

45. The use of pathogenic microorganisms (virus, bacteria, and fungi) or biological toxin to harm or kill human, animals or plants as an act of war is called _____ ?

46. The fuel, made from plant materials, containing cellulose and hemicellulose by anaerobic fermentation of microorganisms.

47. Flammable mixture of gases produced by anaeribic break down of organic molecule, such as plants, animal by microorganisms is known as _____ ?

48. The process of transferring genetic material from one organism to another for the purpose of creating transgenic organism is _____ ?

49. A large vessel, which used to culture and grows bacteria in large quantity under controlled condition in the laboratory, is known as _____ ?

50. Dead and dry cells of microorgainsms like yeast, bacteria, fungi and algae, which serves as a food or animal feed supplements is called _____ ?

51. A fungal protein, which is a form of SCP, derived from fungi known as *Fusarium venenatum* for human consumption or meat substitute is known as _____ ?

52. An organism that received genetic material from another organism is known as _____ or_____ ?

53. a mass of cyanobacteria (blue green algae), that is rich in protein (62%), vitamins, mineral and antioxidants is known as _____ ?

IV. Essay type

54. Discuss the main concern of using biological weapons?

55. Discuss the disadvantage of using genetically modified plant against

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56. How do protein toxins produced by genetically modified bacteria affect insect pests?
57. List the three viruses that used to produce biopesticides?
58. How do intracellular enzymes from microorganism are separated after production in the fermenter?
59. List the importance of genetically modified organisms?
60. Why do compost is not a strict fertilizer?
61. Discuss the main objectives of treating sewage (waste water)?
62. What is the difference between trickle filter method and activated sludge method of sewage treatment?
63. How do microorganisms degrade biohazard toxin waste into harmless products in the environment?

Answer to Unit Review Exercise

Matching

1. E	3. J	5. D	7. B	9. C
2. G	4. H	6. A	8. I	10. F

Multiple choice

11. D	15. D	19. B	23. B	27. D	31. C	35. C	39. D
12. A	16. B	20. D	24. A	28. B	32. C	36. A	40. D
13. D	17. B	21. D	25. B	29. B	33. B	37. D	41. D
14. B	18. A	22. D	26. D	30. A	34. C	38. C	

Short Answer:

42. Food security
43. Nutrition security
44. Conscious citizen
45. Biological warfare
46. Biofuel

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47. Biogas

48. Genetic engineering

49. FermentER

50. Single - cell protein

51. Mycoprotein

52. Transgenic/ genetically modified organism

53. Spirolina

Essay type

54. **Biologically weapons** - are pathogenic microorganisms, such as virus, bacteria and fungi that can kill or harm human and other organisms.
   - ➥ If they are get on the hands of terrorist groups, destructions are encountered.
   - ➥ The long term treat of pollution by these microorganisms

55. The main problems of using genetically modified plants are:
   - ➥ A resistant gene may jump into weeds, can kill other non - targeted insects
   - ➥ The insect may develop resistant genes, which enable them feed GM plants

56. The toxin binds in insects gut and damage epithelial cells, that it can't absorb, digested food, and starve to death.

57. a. Nuclear polyhdrosis virus (NPV)

   b. Granulosis virus (GV)

   c. Cytoplasmic polyhedrosis virus (CPV)

58. Cells are separated from the fluid, and then broken open by various means to release their content

59.
   - ➥ To produce crop or livestock with increased yields, growth rate, drought and disease resistant
   - ➥ To produce high quality nutrition food
   - ➥ For the treatment of waste or sewage

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- For diagnosis and treatment of disease

60. Because its nitrogen content is very low.

61. To make waster water safer for use and joins streams, oxidize organic molecules

62. **trickle filter method** uses natural microorganisms in the stone bed to oxidize organic molecules, and has no oxygen blowing system

Activated sludge method uses seeds of sludge to oxidize organic molecules, have oxygen blowing system.

63. The enzyme produced by microorganisms changes the nature of chemicals waste, which is known as bio - transformation

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UNIT 8

8. Microorganisms

8.1 Microorganism

Microorganism are very small organisms that can't be easily seen by necked eyes. An instrument that used to see very small organisms is called **microscope**. Some eukaryotic micro organisms are visible by necked eyes.

Microbiology is the study of microorganisms. There are five major groups of microorganisms.

These are: prokaryotes (bacteria), viruses, fungi, and protoctists (protozoa, and microscopic algae).

Through DNA technology, scientist Carl Woese and his colleagues found that there are two major branches of prokaryotes. One branch, formerly called Eubacteria, which are ordinary bacteria. The other branch is called Archaea, formerly called Archaebacteria. The name archaea means "archaic" Most known species of archaea live extreme environments.

8.2 Archaea

Archaea are not like bacteria in many ways. Archaea differ in the makeup of their call walls. They also differ in their membrane lipids and in their genetics and metabolism.

Genetic analysis of archaea has revealed at least three groups of archaea: **methanogens, halophiles,** and **thermoacidophiles.**

Methanogens - are named for their unique way of getting energy: they

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convert hydrogen gas and carbon dioxide into methane gas.

- Oxygen is poisonous to them, so methanogens can live only in anaerobic environments (environments that lack oxygen), such as in deep fresh water, marine mud, swamp mud, and sewage.
- Methanogens also thrive in the intestinal tracts of organisms such as cows and termites.

Halophiles are "salt - loving" archaea that live in environments that have very high salt concentrations. High salt concentrations would kill most bacteria but favor growth of halophiles because halophiles adapted to live in very salty water.

Thermoacidophiles - live in very acidic environments that have very high temperatures, such as the hot springs. Some thermoacidophiles live as temperature up to 121 ^{o}C (230 ^{o}F) and at a pH of less than 2.

- Hyperthermophylls - lives at higher temperature (up to 121 C^{0})
- Acidophiles - are lives at extreme lower pH
- Psychrophiles - are cold loving, grow best at 4C^{0}

8.2.1 Characteristics of Archae

- Archaeal cell walls do not have peptidoglycan.
- Archaeal cell walls have different amino acids than bacterial cell walls do.
- Archaea also have different types of lipids in their cell membranes.
- Archaea are like eukaryotic cells in that archaeal genes have introns, portions of DNA that do not code for amino acids and that are transcribed into RNA but are removed before being translated into proteins.
- Archaea are reproduce by binary fission, fragmentation, and budding
- Nutritionally they are aerobic, facultative anaerobe or strict anaerobe

Surprisingly, rRNA analysis has shown that archaea are more closely related to eukaryotes than they are to bacteria. This similarity of genes suggests that archaea and eukaryote likely share a more recent common ancestor than eukarya and bacteria do.

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8.2.2 Beneficial Aspects of Archaea

Archaea are living in extreme environment, so that possess an important enzyme that can be exploited in the commercial uses.

- Added to commercial detergents to increase performance at higher temperatures and pH levels
- The protease and lipase from alkaliphilic bacteria are added to detergents to enhance stain removal capability
- Used as a bioremediation, to detoxify or remove oil, gasoline, and other pollutants or toxic chemicals from the environment
- Enzymes from thermophilic archaea are use in polymerase chain reaction (PCR), which used in molecular biology laboratory to amplify a sample DNA
- Another archaeal enzyme has been useful in the food industry to convert corn starch to dextrins.

8.3 Eubacteria

Eubacteria - are prokaryote, which literally means "true bacteria".

- They don't have a nucleus; their DNA is contained within the cytoplasm of the cell.
- They also don't have any membrane - enclosed organelles, and
- They're typically much smaller than eukaryotic cells.

8.3.1 General Characteristics of Bacteria

- Unicellular prokaryote
- Have no true nucleus and membrane bounded organelles.
- Have no major organelles, such as mitochondria, golgi bodies, plastids, and endoplasmic reticulum
- Have rigid call wall, made of peptidoglycan.
- Photosynthetic bacteria have no true chlorophyll, but have bacteriochlorophyll to absorb light

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Bacteria can respiration in different ways

- **Obligate anaerobes** - organisms that cannot live where molecular oxygen is present.
- **Facultative anaerobes** can live with or without oxygen.
- **Obligate aerobes** organism that can't live without oxygen

Get nutrition (carbon) and energy in four different ways:

a. **Chemohetrotrophs** - are organisms that get both energy and carbon from other organisms

b. **Photoheterotrophs** - are organism that get energy from light but can't use carbon directly from CO_2.

c. **Photoautotrophs** - are organisms that get energy from light and carbon from CO_2 to make food.

d. **Chemoautotrophs** - are prokaryote that get energy from inorganic chemicals and carbon from CO_2

Bacteria live in different habitat **(omnipresent)**, means they live in air, water, and soil. Based on the source of electrons, bacteria are; lithotrophs and organotrophs.

Lithotrophs - uses inorganic substance as a source of electron

Organotrophs - uses organic substance as a source of electron

Based on carbon, electron, and energy sources, bacterial nutrition can be: photolitoautotrophs, photoorganoheterotrophs, chemolithoautrophs, and chemoorganoheterotrophs.

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Nutritional Type	Carbon Source	Energy Source	Electron Source	Representative Microorganisms
Photolithoautotroph	CO_2	Light	Inorganic e- donor	Purple and green sulfur bacteria. cyanobacteria
Photoorganoheterotroph	Organic carbon	Light	Organic e- donor	Purple nonsulfur bacteria, green nonsulfur bacteria
Chemolithoautotroph	CO_2	Inorganic chemicals	Inorganic e- donor	Sulfur- oxidizing bacteria, hydrogen-oxidizing bacteria, methanogens, nitrifying bacteria, iron - oxidizing bacteria
Chemolithoheterotroph	Organic carbon	Inorganic chemicals	Inorganic e- donor	*Some sulfur - oxidizing bacteria (e.g Beggiatoa)*
Chemoorgano heterotroph	Organic carbon	*Organic chemicals often same as C source*	*Organic e- donor. Often same as C source*	*Most nonphotosynthetic microbes, including most pathogens, fungi and many protists and archaea*

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8.3.2 The Structure of Bacteria

Bacterial structures are grouped into three parts. These are External, envelope and internal.

☑ External structure includes flagella, capsule, slim layer, pili and fimbrae.

☑ Envelopes are cell wall and cell membrane.

☑ Internal structure includes cytoplasm, ribosome, chromosome, endospore, and cytoskeleton

Cell wall - located outside of the plasma membrane, protects the cell from bursting, and give shape. In bacteria, the cell wall is made of the carbohydrate - protein complex , called peptidoglycan.

Capsule - is a sticky layer composed of proteins and carbohydrates, located outside the cell wall that help bacterial cells attach to surfaces and to other cells. It also helps to avoid capture by certain cells of the human immune system.

A slime layer is more loosely attached to the cell wall than a capsule. These outer layers are made of polysaccharides or protein, used for protection against phagocytosis by other microorganisms or WBC

Pili and fimbriae: is a slender or hair like protein thread, found in some bacteria, which helps attach to surfaces and to other cells during conjugation.

Flagella - Bacteria flagella are filaments of a protein called **flagellin,** which are attached on cell wall of bacteria cells, and used to propel the cell by rotating in a whip like motion.

Plasmid: In addition to their genomic DNA, most bacteria have one or more plasmids, smaller circular fragments of DNA, for genetic exchange, or for resistance to antibiotics.

Endospore - contains DNA, a thick coated, resistant structure.

Chromosome carries genetic information from one generation to the next

Cytoplasm contain DNA, ribosome, and organic compound that are needed for life.

Cell membrane - regulates movements of molecules in and out of the cell.

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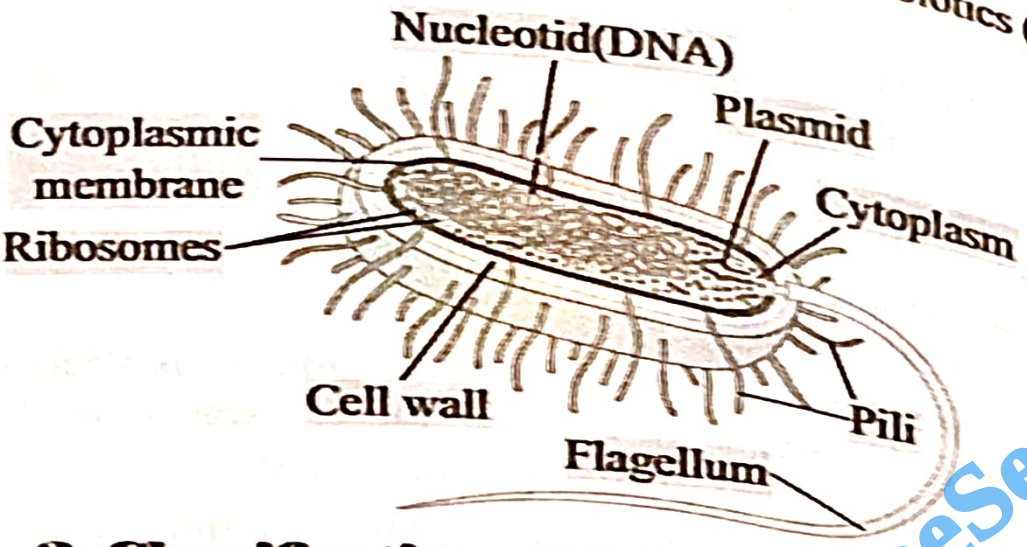
Outer cell membrane - protect cell against antibiotics (only in gram negative bacteria).



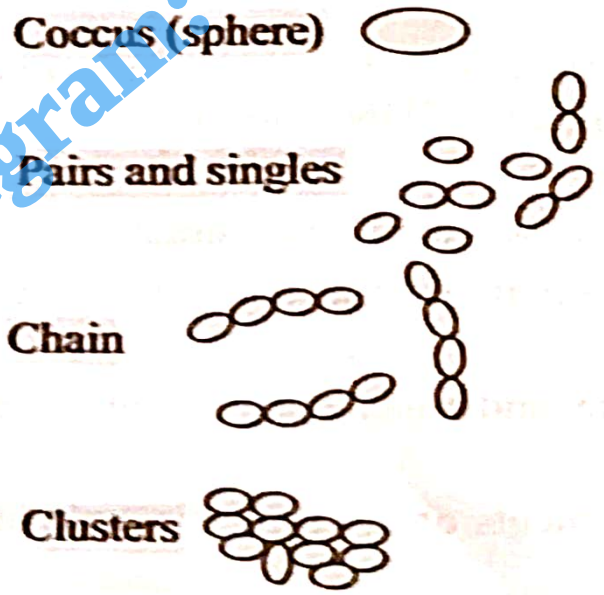
8.3.3 Classification of Bacteria

Bacteria can be classified into two ways; based on shape and Gram stain

I. Based on shape

Bacteria occur in many shapes and size and have distinct biochemical and genetic characteristics. Most bacteria have one of three basic shapes.

1. Bacilli - rod - shaped bacteria.
2. Cocci - Sphere - shaped bacteria. When cocci occur in chains, they are called streptococci; clusters of cocci are called staphylococci;
3. Spirochaetes (corckskrew) - spiral shaped bacteria
4. Comma shaped - such as vibrio cholera



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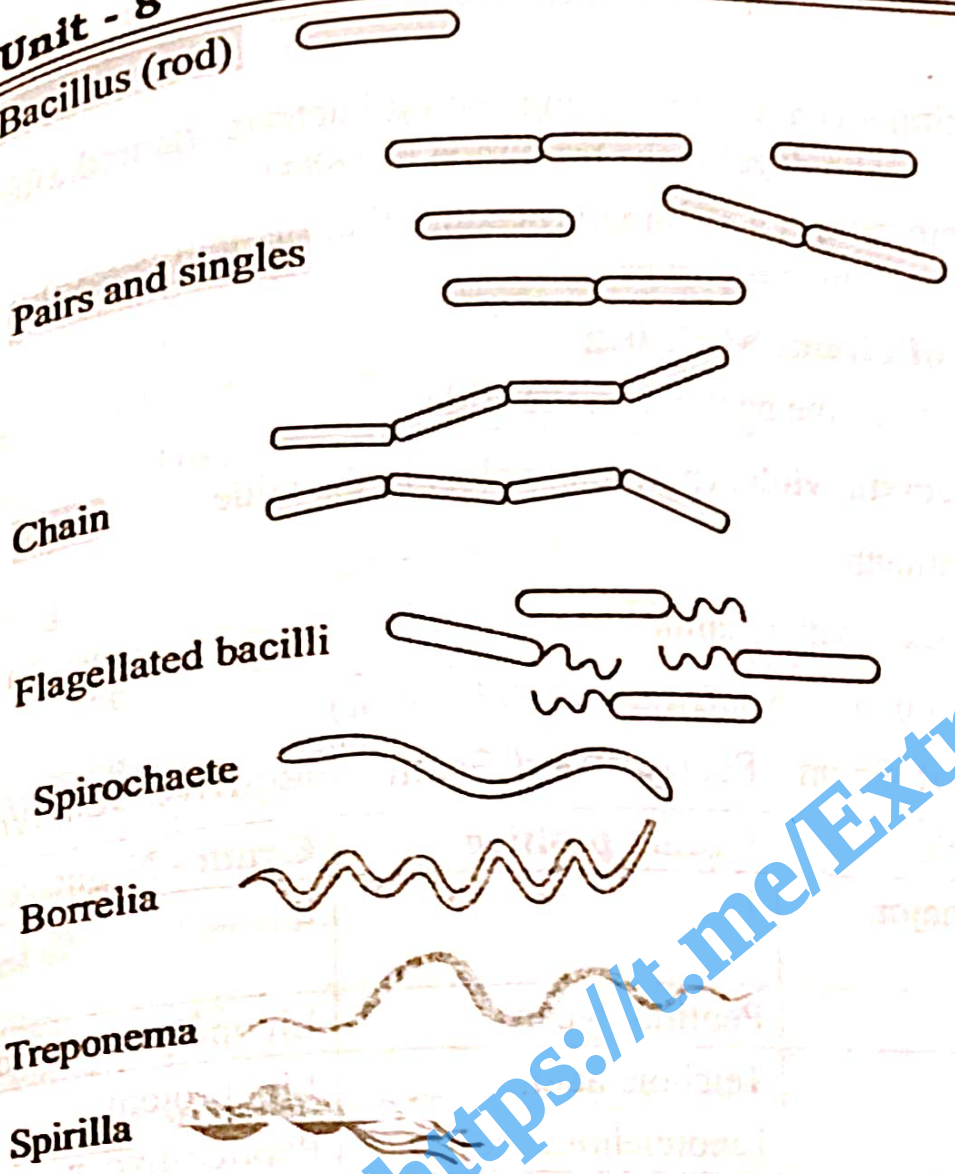

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II. Based on gram stain

Based on the structure of the cell walls and retaining ability of Gram stain, biologists group most species of bacteria into two categories. These are Gram - negative bacteria and Gram - positive bacteria.

☑ **Gram - negative** bacteria appear pink when stained by the safranin (red stain). Crystal violet is decolonized by an alcohol/acetone wash. Because the cell walls of gram - negative bacteria have two layers, a thin peptidoglycan layer and a thick outer membrane,

➥ The membrane produces **endotoxin**, which is used to resist antibiotics

☑ **Gram - positive bacteria** retain the crystal violet stain and appear purple. Because Gram-positive bacteria have cell walls with thick layer of peptidoglycan

Note:

Gram's staining - is a test for distinguishing bacteria (named after Hans Christian Gram, who develop the technique in 1884).

Differential staining is a staining procedure that distinguishes organisms based on their staining properties.

Procedure of Gram Staining

1. Fixation - attach the bacteria on the slide
2. Applying **crystal violet** (purple in color) in the slide
3. **Iodine** treatment
4. Decolonization with ethanol
5. Counter staining with **safranin** (pink in color)

Comparison of Gram - Positive and Gram - Negative Cell Wall

Characteristic	Gram - positive	Gram - Negative
Number of major layers	1	2
Chemical Composition	Peptidoglycan, Teichoic acid, Lipoteichoic acid	Lipopolysaccharide, Lipoprotein, Peptidoglycan
Overall thickness	Thicker (20 - 80nm)	Thinner (8 - 11nm)
Outer membrane	No	Yes
Periplasmic space	Narrow	Extensive
Porin proteins	No	Yes
Permeability to molecules	More penetrable	Less penetrable

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8.3.4 Reproduction in Bacteria

Bacteria can be reproduce by binary fission, budding, fragmentation, and conjugation

i. What is Binary fission

Binary fission is the most common form of asexual reproduction in bacteria, which a cell is divided into identical cells, Binary fission occurs with remarkable speed, some bacteria species divide in less 20 minutes;

one billion bacteria within 10 hours!

Steps of binary fission

a. DNA replicates and the two daughter DNA molecules are pulled to opposite ends of the cell.

b. The cell wall and cell membrane begin to invaginate (fold inward) to divide the cell into two half.

c. Finally, the cell membrane pinches in to two form two identical daughter cells.

ii. **Budding** - is a less common form of asexual reproduction among bacteria. In budding, a cell develops a bulge, or bud, that enlarges, matures, and eventually separates from the mother cell.

iii. **Fragmentation** - walls develop within the cell, which then separates into several new cells. A few species of bacteria (acrinomycetes) divide by fragmentation.

Vertical Gene Transfer

Prokaryotes can get gene from other prokaryotes in three ways; transformation, bacterial conjugation, and transduction.

☑ **Transformation** occurs when a bacterial cell takes up fragments of foreign DNA (or RNA) released by another bacterium.

☑ **Transduction** - a virus obtains a small part of DNA from a host bacteria and transfer it to the other bacteria

☑ **Conjugation** - is the process by which two prokaryotes bind together and transfers DNA to the other cell through a structures called a **sex pilus** (long hair like structure on the cell surface). *Escherichia coli* has been the most extensively studied bacteria

Steps of conjugation on E. coli bacteria

a. In the E. coli, there are donor cells **(F^+ cells),** which have DNA that can be transmitted to recipient cells **(F^- cells).**

b. **F^+ cells** have a DNA sequence known as the **F factor** (a plasmid containing **F genes**) that is necessary for a bacterium to serve as a donor during conjugation.

➥ **F genes** encode sex pili, enzymes essential for transferring DNA, and antibiotic resistant gene.

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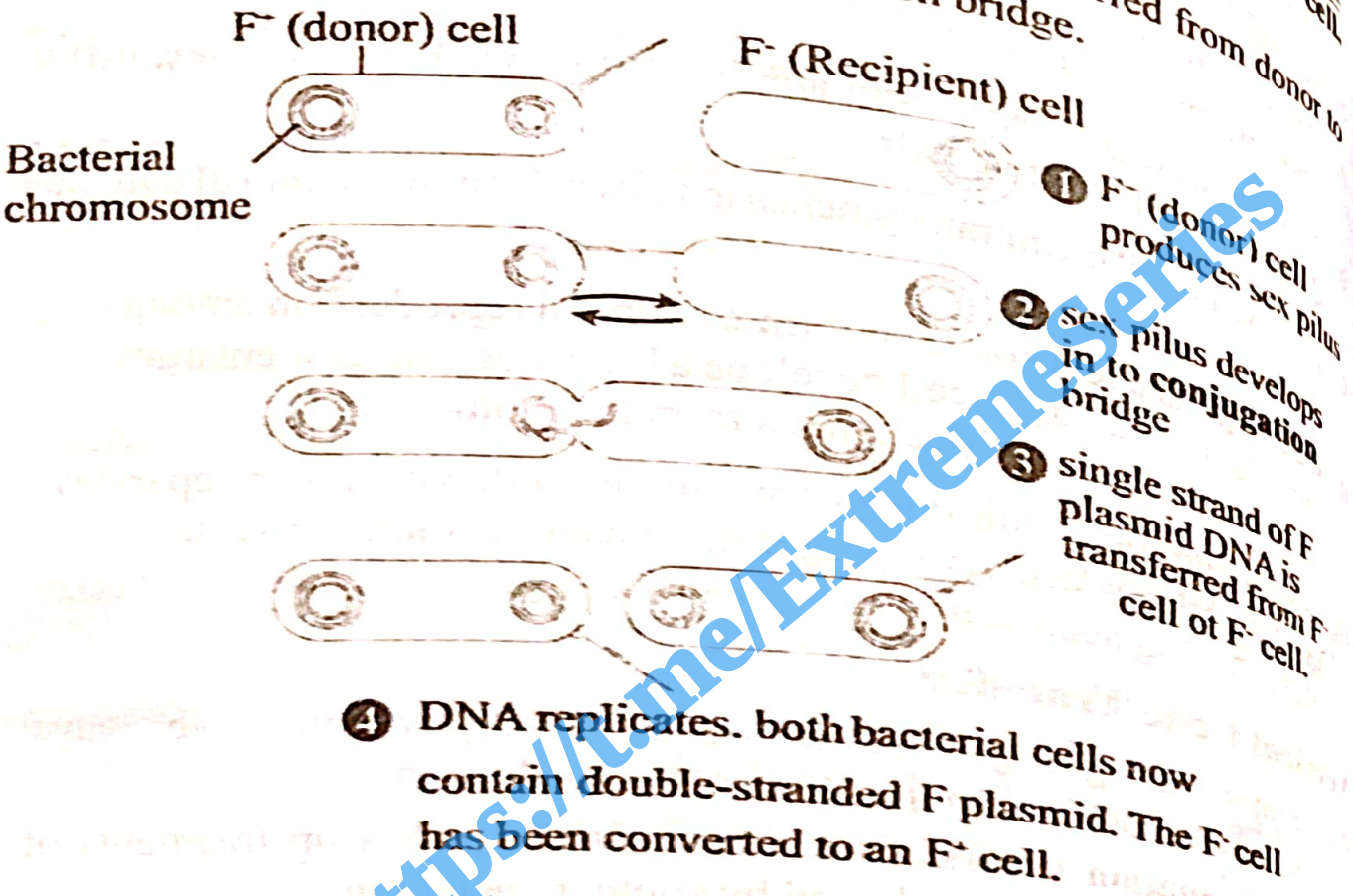
c. The sex pilus of a F^+ cell recognizes and binds to surface of an F^- cell forming a conjugation bridge between the two cells.

d. The **F plasmid** replicates itself, and DNA is transferred from donor to recipient bacterium through the conjugation bridge.



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Examples of common bacteria diseases

Unit - 8

Bacterial Diseases	Causative agent	Description	Organs affected	Transmission /vector
Pertussis	*Bordetella pertussis*	Gram (-) rod	Upper respirator tract,	Air
Meningo coccal Meningitis	*Nesseria meningitidis*	Gram (-) diplococcus	Upper respirator tract, meninges	Air
Tuberculosis	*Mycobacterium tuberculosis*	Acid - fast	Lungs, bones, other organs	Air
Typhiod fever	*Salmonella typhi*	Gm (-) rod	Intestine, blood gall bladder	Food, water
Cholers	*Vibro cholerae*	Gm (-) spore - forming anaerobic rod	intestine	Food, water
Tetanus	*Clostridium tetani*	Gm (+) spore forming anaerobic rod	Nerves as synapse	Soil
Syphilies	*Treponema pallidum*	spirochete	Skin, cardiovascular organs	sexual
Gonorrhea	*Neisseria gonorrhoeae*	Gm (-) diplococcus	Urethra, cervix, fallopian tubes, epidiymis, eyes, pharynx	sexual
Leprosy	*Mycobacterium leprae*	Acid - fast	Skin, bone, peripheral nerves	Contact

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8.3.5. Physical Factors that Affecting Microbial Growths

Growth is an orderly increase in the quantity of all cellular constituents. The growths of micro organisms are influenced by various physical and chemical factors of their environment.

- The chemical factors includes oxygen, carbon, nitrogen, phosphorus, sulfur, etc.
- The physical factors includes temperature, oxygen level, pH, osmotic pressure, hydrostatic pressure, and radiation

i. Temperature

Temperature is the most important factor that determine the rate of growth, multiplication, survival, and death of all living things.

- At optimum temperature, the most rapid rate of growth occurs.
- High temperature denaturing enzymes, disrupt microbial membrane
- Low temperature deactivate enzymes, solidify microbial membranes

Based on the temperature they are living, bacteria can be:

- **Psychrophiles** - capable of growth and reproduce in cold temperature
- **Mesophiles** - grows as moderate temperature
- **Psychrotrophs** - cold tolerant bacteria
- **Thermophiles** - heat loving bacteria
- **Hyperthermopiles** - bacteria that thrives in extremely hot temperature

ii. pH

pH disrupt the plasma membrane, inhibit the activity of enzymes, and membrane transport protein

Based on pH level, bacteria can be:

- Accidophiles (PH = 1 - 0.5)
- Alkaophillus (pH - 7.5 - 14), and
- Neutrophilus (pH - 5.5 - 8)

iii. Osmotic pressure

Osmotic pressure is the minimum pressure, which needs to be applied to a solution to prevent the inward flow of water across a membrane.

Based on osmotic pressure, bacteria can be grouped into two:

i. **Osmotolerant** (can grow at high salt concentration), and

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ii. **Halophiles** (grow in the presence of salt at concentration above 0.2 - 0.6)

iv. **Radiation**

Radiation includes

- **Infrared** (heating the earth),
- **Ionizing ray** (produce mutation and death),
- **Visible light** (for photosynthesis) of sunlight.

v. **Hydrostatic pressure** - is force acting per unit area

Based on hydrostatic pressure, bacteria can be classified into two;

i. **Barotolerant** (doesn't get affected by increased pressure)

ii. **Barophiles** (grow at higher hydrostatic pressure)

Illustrative Example

1. Which of the following statement correctly describes bacterial? (UEE, 2016 E.C)

   A. Single cell prokaryote organism
   B. Multicellular eukaryotic organism
   C. Single celled eukaryotic organism
   D. Multicellular prokaryote organism

   Explanation: The general characteristics of bacteria a includes;

   - Unicellular (single cell) prokaryote
   - Have no true nucleus and membrane bounded organelles, etc.

   Answer: A

2. Bacteria cell wall is made from _____

   A. Peptidoglycan
   B. Cellulose
   C. Lipoprotein
   D. Phospholipid

   Explanation: Bacterial cell wall is made of peptidoglycan, which is composed of carbohydrate and protein.

   Answer: A

3. Which of the following do not belong to domain Archaea?

   A. Prokaryotes that produce methane from carbon dioxide and hydrogen

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B. Thermopiles C. Halophiles D. Bacteriophages

Explanation: Bacteriophage is grouped under eubacteria.

Answer: D

4. Which of the following is rod shaped bacteria? (UEE, 2003)

A. Cocci C. Spirilla

B. Spirochaetes D. Bacilli

Explanation: The collective name of rod - shaped bacteria are called bacilli.

Answer: D

5. Gram positive and gram negative bacteria are differentially stained by gram's stain. This is due to the structural difference on their

A. Cell membrane C. Capsule

B. Ribosome D. Cell wall

Explanation: The staining ability of bacteria is depends on the nature of cell wall. Gram positive bacteria have thick cell wall than gram negative bacteria.

Answer: D

6. Thermophiles bacteria are heat tolerant bacteria, while ___ is cold tolerant.

A. Psychrotrophs C. Hyperthermophiles

B. Mesophiles D. Acidophilus

Explanation: Based on temperature, bacteria can be: Psychrophiles - capable of growth and reproduction in cold temperature. Mesophiles - grows at moderate temperature. Psychrotrophs - cold tolerant bacteria. Thermophiles - heat loving temperature. Hyperthermophiles - thrives very high temp.

Answer: D

7. Surface appendages used by bacteria to attach to one another are:

A. Flagella C. Plasmid

B. Pilli D. Cell wall

Explanation: Pilli is a small hair like structure on the surface of some bacteria used for attachment.

Answer: B

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8. _____ is a sticky layer composed of proteins and carbohydrates, located on the cell wall that help bacterial cells attach to surfaces and to other cells?

A. Capsule
B. Pilli
C. Cell wall
D. Flagella

Explanation: Capsule is a sticky layer composed of proteins and carbohydrates. located outside the cell wall of bacteria, used for attachment.

Answer: A

9. Which one of the following is food born bacterial disease?

A. Typhoid
B. Syphilis
C. Tuberculosis
D. Gonorrhea

Explanation: Bacteria disease can be transmitted through sexual contact (syphilis and gonorrhea), air born (TB), food born (typhoid and cholera).

Answer: A

10. Which one is the correct sequence of Gram staining?

A. Crystal violet → Safranin → Decolorization/ by ethanol/ → Iodine.
B. Safranin → Crystal violet → Iodine → Decolorization / by ethanol/
C. Crystal violet → Iodine → Decolorization/ by ethanol/ → Safranin.
D. Ethanol → iodine → Safranin → Crystal → violet

Explanation: Add crystal violet, then iodine, the decolorized by ethanol, finally counter safranin

Answer: C

11. The most common form of asexual reproduction in bacteria is _____

A. Budding
B. Fragmentation
C. Binary fission
D. Conjugation

Explanation: Binary fission is the most common form of asexual reproduction of bacteria: means a cell divided into two identical cells.

Answer: C

12. Describe asexual reproduction in prokaryotes?

Explanation: Prokaryotes reproduce asexually by:

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- Binary fission (the cell divides, forming two cells),
- Dudding (a bud forms and separates from the mother cell).
- Fragmentation (walls form inside the cell, which then separates into several cells).

13. Summarize three mechanisms (transformation, transduction, and conjugation) that may lead to genetic recombination in bacteria.

Explanation: In bacteria genetic material can be exchanged by transformation, transduction, or conjugation.

- In transformation, a bacteria cell takes in DNA fragment released by another cell.
- In transduction, a phage carries bacteria DNA from one bacterial cell into another.
- In conjugation, a donor bacterium transfers plasmid DNA to a recipient bacterium

14. Compare characteristics of Archaea and Bacteria?

Explanation: Prokaryotes are grouped into the domain of Archaea and domain of Bacteria. The cell walls of archaea do not have peptidoglycan, but bacteria have cell wall made of peptidoglycan.

8.4 Fungi

Fungi are eukaryotic heterotrophs and spore bearing that absorb nutrients from their surroundings.

- The study of fungi is called **mycology.**
- Scientist who studies fungi is **mycologist.**
- The study of fungal toxin and their effect is **Mycotoxicology**
- The disease caused by fungi in animal are known as **mycoses**

Structure of Fungi

Filaments of fungi are called **hyphae** (hypha). The cell walls of hyphae contain **chitin,** it distinguishes cell walls of fungi from those of plants, which have cellulose but no chitin. A network (mass) of hyphae that forms the body of a fungus is a **mycelium.** In some species, the cells that make up hyphae are divided by cross sections called **septa.**

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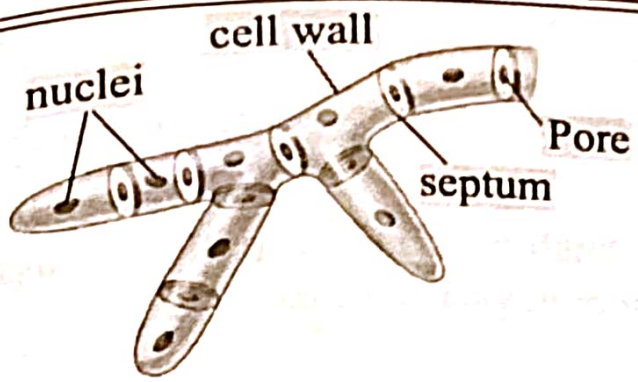

8.4.1 Characteristics of Fungi

- All are eukaryotic - possess membrane bounded nuclei, and a range of membrane bounded organelles (mitochondria vacuoles, endoplasmic reticulum).
- Most are filamentous - composed of individual microscopic filaments called hyphae, which exhibit apical growth and which branch to form a network of hyphea called a mycelium
- Some are unicellular. Example, yeasts
- Protoplasm of a cell is surrounded by a rigid cell wall, made of chitin and glucans, or cellulose
- Many reproduce both sexually and asexually, produces spore
- Their nuclei are haploid, hyphae is multincleated
- Have no chlorophyll (achlorophyllous)
- All are **chemoheterotrophs** (chemo - oragnotrophic) - they utilize carbon from environment, and energy from chemical reaction to synthesize the organic compound they require for growth and energy
- Possess storage compounds such as trehalose, glycogen, sugar alcohol, and lipids.
- May be free living, saprophytic (feed on dead organic matter), parasitic, mutualistic (symbiotic) or form relationship with other organisms

Ecology of fungi

All fungi are heterotrophs (saprophyte, decomposers, parasitic),

- **Saprophytes** are organisms that obtain food from dead organic matter.
- **Decomposers** - are microbes that recycle nutrients by breaking down

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large organic matter into simple molecules.

- **Parasites** - are organisms that depend on another organism to obtain nutrients.
- **Symbiotic relationship** - is a close, prolonged association between two or more different biological species

For example

a. **Lichens** - is a symbiotic relationship between fungus and algae

b. **Mycorrhizae** - is a symbiotic relationship between plant root and fungi

- The plant provides photosynthetic products.
- The fungi helps plants to absorb water and mineral

8.4.2 Classification of Fungi

Fungi have four phylum: these are: **chytridiomycota, zygomycota, Ascomycota,** and **basidiomycota**. The following table shows the 4 phylum of fungi, characteristics feature and representative examples.

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Phylum	Common Name	Features	Examples
Chytridiomycota	Chytrids	Aquatic, gametes have flagella	Allomyces
Zygomycota	Zygote fungi	Zygosporangium, no regularly occurring septa, usually no fleshy fruiting body	Rhizopus
Ascomycota	Sac fungi	Ascus, perforated septa	Neurospora, baker's yeast
Basidiomycota	Club fungi	Basidium, perforated septa	Armillariella, mushrooms

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8.4.3 Reproduction in Fungi

Fungi reproduce by means of asexual and sexual reproduction

Asexual reproduction

Asexual reproduction doesn't involve union of gametes, it is done by production of spore. It is more important than sexual production.

Types of asexual reproduction

i. Sporulation
ii. Fragmentation
iii. Fission
iv. Budding

Sporulation

Spore is a small propagating unit of fungi, functioning as a 'seed'.

Sporulation is the process of producing spore. There are two types of spore; sporangiospores and conidia

a. **sporangiospores:** are spores that are produced in sporangin. The sac like structure on hypae, which produces sporangia is called **sporangium.**
b. **Conida:** are non - motile, dust like asexual spore, which is produced directly from somatic hyphae knower as **conidophore.** It is extremely light and blown by wind.

Fragmentation

Fragmentation is the most common method of asexual reproduction, the hyphae of fungi break into small pieces (fragments), which function as a propagating unit and grow into new mycelium. The spore produced by fragmentation is called **arthrospores.**

Fission

Fission is a kind of asexual reproduction in fungi, where a parent cell elongates, nucleus undergo mitotic division and forms two nuclei, then content divide into two equal halves by the formation of a transverse septum and separates into two daughter cells.

Budding

The type of spore formed through budding is called blastospores. The parent cell initially outgrows small buds, and then its nucleus divides. One daughter nucleus migrates into bud, and the bud cutting off completely from parent cell functioning as an independent propagating unit.

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Sexual reproduction

Sexual reproduction involves the union of two sex cell (opposite mating type, but not male and female). The nucleus of the two opposite sex cells are called **heterokaryon**, which means genetically different. The phases of sexual reproduction are:

i. **Plasmogamy** - is the union of two protoplasm, where the two nuclei come together within the same cell.

ii. **Karyogamy** - is the union of opposite mating nuclei to form diploid nucleus (2n), which is zygote.

iii. **Meiosis** - is the reproduction division, chromosomes reduced by half to produce haploid cell (n).

8.4.4. Economic Importance of Fungi

Fungi plays a significant role in our daily life. Some fungi are prized for their usefulness to mankind, while others are harmful.

Useful aspects of fungi :

Fungi as a food - there are over 200 species of edible fungi.

Examples:

- ☑ *Agaricus campestris* (field mushroom), *pleurotus ostreatus* (oyster mushroom).
- ☑ **Single cell protein** (SCP) - obtained from yeast (Aspergillus, fusarium, etc.)
- ☑ **Mycoprotein** is a meat substitute, is derived from the fungus *fusarium venenatum*
- ☑ To make siljo, data, bula, siljo, shameta, etc.
- ☑ Yeast is an important source of vitamin B and D complex, making injera, cheese, tela, tej, etc.

Fungi as medicine

Several fungi are used in the production of medicine. Examples:

Antibiotics - made from *penicillum notatum.*

Fungi in production of organic acid

Commercial organic acids such as oxalic acid, citric acid gallic acid, fumaric acid, etc. are produced from moulds such as Aspergillus species.

For example: citric acid is produced from molasses through fermentation

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by *Asperillus niger*.

Fungi in enzyme production

Different enzymes are extracted from fungi for commercial purpose. For example: inverstase, amylase, cellulase, lipase, pectinase, protease, etc.

Invertase - is extracted from *Saccharomyces cervisae*

Amylase - is extracted from *Aspergillus niger*

Fungi as decomposers

Fungi play an importance role in recyling nutrients. Saprophytic fungi secret enzymes and decompose dead remain of plant and animals, which increases fertility of soil.

Fungi in biological research

Several fungi are used as an important tool for the study of process, such as aging, cancer, genetic recombination.

Example :- Aspegillus niger has been used as a test in organism to determine Mg, Cu, Zn, and Mo in soil.

Biologists use the yeast *Saccharomyces cerevisiae* and other fungi as model organisms for research in molecular biology and genetics

Harmful Aspects of Fungi

There are several activities of fungi that are harmfull to human kind. Fungi incite many disease of plants, animals, and human.

Fungi as a pathogen, plant diseas

Most parasitic fungi causes diseases in plants. Examples; rusts, smuts, mildew, blight, ruts, and wilts

- *Puccinia graminis* cause ruts disease on wheat stem
- *Erisiphea graminis* cause powdery mildew, plant leaves get covered with a white powder.

Fungi as a pathogen, Animal disease

Several fungi species lives parasitically on an animal and cause various superficial fungal infection. For example:

i. **Ring worm** (by *Tinea corporis*) - it is a common skin infection that looks like circular rash

ii. **Candidiasis** - caused by yeast (caused by *candida albican*), which affects mouth, throat, vagina, and blood stream

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iii. Athlete's foot (caused by *Tinea pedis*) - is a fungal infection, which is common among people whose feet is sweaty.

iv. Disease of hair scalp, eyelashes, and eye brows are caused by a fungus called *Tinea capitis*

v. **Disease of finger nail** is caused by a fungus called **Tinea unguim** (onchomycosis)

Note

The mode of transmission of superficial fungal infection are: skin contact, animal - human contact with inanimate objects, such as carpets, moisture, dust in shower, swimming pool, gyms.

The most frequently affected parts of the body are hair, head, and facial skin.

Fungi can spoil agricultural products if it is improperly stored.

Examples of fungi involved in this aspect are : mucur, rhizopus, aspergillus, yeast, etc.

Molds can cause deterioration of fabrics, leather, electrical, insulation and other goods, especially in warm humid climate.

Mycotoxicoses (ingestion of fungal toxin), mycetismus (mushroom poisoning).

Example :

- ☑ **Aflatoxins** - is a toxin produced from *Aspergillus flavus* and *A. parasitics*. When crops are contacted with this toxin and ingested by human, it has carcinogenic effect in the liver. Contaminated meat and dairy products are also a source of toxin.
- ☑ **Eragrotism** - is a disease of cereals (rye, wheat and barley), caused by a fungus called *Claviceps purpurea*. Sclerotium is a visible, hard, dark purple mass of mycelium that replaces the grain tissue and formed mycotoxins which contain alkaloid substance known as **ergot**.
- ☑ When ergot is ingested by human or animals it causes **nervous system dysfunction**, and **gangrenous** (loss toes or finger)
- ☑ **Mushroom poisoning** - It is due to **mycotoxins**, which is produced by mushroom that affect human body

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Illustrative Example

15. Which of the following statement about fungi is not true?

A. Eukaryotic and heterotrotrophs
B. Most are multicellular filamentous
C. Spore bearing and absorb nutrients
D. Photosynthetic and eukaryotic

Explanation: Fungi are achlorophyllous, don't have chlorophyll for photosynthesis.

Answer: D

16. The study of fungi is called _____

A. Mycosis
B. Mycology
C. Mycotoxicology
D. Microbiology

Explanation

- The study of fungi is called **mycology.**
- The study of microorganism is called **microbiology.**
- The study of fungal toxin and their effect is **Mycotoxicology.**
- The disease caused by fungi in animal are known as **mycoses.**

Answer: B

17. Which of the following statement about fungi structure is not true?

A. Filaments of fungi are hyphea
B. A network (mass) of hypae is a mycelium
C. The cell walls of hypae contain chitin
D. The cell walls of hyphae contain cellulose

Explanation: The cell wall of fungi is made of chitin.

Answer: D

18. The type of nutrition in fungi is

A. Chemohetetrophs
B. Autotrophs
C. Chemoautotrophs
D. B and C

Explanation: Fungi are chemo heterotrophs (chemo - oragnotrophic) they utilize carbon from environment, and energy from chemical reaction to synthesize the organic compound.

Answer: A

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19. The symbiotic association between plant root and fungi is
A. Mycorrhizae
B. Lichens
C. Commensalism
D. Parasitism

Explanation: Symbiotic association between plant and fungus is mycorrhizae. The association b/n fungus and algae is lichens.

Answer: A

20. Which of the following statement about fungal **asexual** reproduction is not true?
A. Conida are non - motile, dust like asexual spore
B. Spores are produced in sporangia
C. Spore is a small propagating unit of fungi
D. Fission is the process of producing spore

Explanation: Fission is a kind of asexual reproduction in fungi, where a parent cell elongates, and nucleus undergoes mitotic division and forms two nuclei, then content divide into equal halves.

Answer: D

21. Which of the following statement about fungal **sexual** reproduction is not true?
A. The two opposite sex cells are heterokaryon
B. There is male and female fungi
C. There is opposite mating type
D. Produces genetically different offspring

Explanation: There is no male and female individuals in fungi, but opposite mating type (heterokayon).

Answer: B

22. Which one of the following is not the importance of fungi as a food?
A. **Agaricus campestris** (field mushroom)
B. Single cell protein (SCP) - obtained from fusarium
C. Antibiotics - is made from **penicillum notatum**
D. To make siljo, data, bulla, siljo, shameta.

Explanation: Antibiotics is a drug not a food, which is made from *penicillum notatum*.

Answer: C

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23. Aspegillus niger has been used as a test in organism to determine Mg, Cu, Zn, and Mo in soi. This is an example of fungi, which is important for:

A. Decomposers
B. Biological research
C. Enzyme production
D. Production of organic acid

Explanation: Several fungi are used as an important tool for the study of process, such as aging, cancer, genetic recombination. Example: Aspegillus niger has been used as a test in organism to determine Mg, Cu, Zn, and Mo in soil

Answer: B

24. Which of the following is not a plant disease due to parasitic fungus?

A. Rusts and smuts
B. Mildew and blight,
C. Ruts and wilts
D. Ring worm and Tinea pedis

Explanation: Ring worm and *Tinea pedis* are fungi infectious diseases in animal.

Answer: D

25. A type of toxin produced from fungi such as *Aspergillus flavus* and *A. Parasiticus* that causes carcinogenic effect in the liver is

A. Aflatoxins
B. Ergot
C. Antibiotics
D. Mycotoxin

Explanation: Aflatoxins - is a toxin produced from *Aspergillus flavus* and *A. parasiticus*. When crops are conated with this toxin and ingested by human, it has carcinogenic effect in the liver.

Answer: A

26. Differentiate among the members of the following pairs of related terms: a. hypha/mycellium

Explanation: Hypha is a filament fungus, while mycelium is a mass (network) of filaments.

27. Summarize the ecological significance of fungi as decomposers?

Explanation: Most fungi are decomposers that break down organic compounds in dead organisms, leaves, garbage, and wasters into simpler nutrients that can recycled.

28. Describe the important ecological role of mycorrhizae?

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Explanation: Mycorrhizae are mutualistic associations between fungi and the roots plants. The fungus supplies water and nutrient minerals to the plant, the fungus obtain organic compounds from the plant.

8.5 Protozoa

The word protozoa is come from Greek work 'protozoon' meaning "first animal". Protozoa are unicellular or multicellular, eukaryotic organisms. Most of them are free living and harmless.

8.5.1 Characteristics of protozoa

- Mostly unicellular that lacks cell wall, eukaryotic (have true nucleus, membrane bounded organelles)
- Mostly free living, but some are parasitic or symbiotic
- Mostly heterotrophic that don't have chlorophyll, but some are autotrophs (photosynthetic), and saprozoic
- They are motile have locomotive organelle (pseudopodia, flagella, and cilia).
- Mostly smaller in size that can't be seen by necked eyes, but some are larger.
- Habitat (aquatic or terrestrial), free living or symbiotic mode of life
- Reproduce sexually (conjugation) and asexually (fission and budding)
- It can be **animal like** (eg. protists - amoeba, Paramecium, trypanosome), **plant like** (eg. green algae and red algae),

8.5.2 Reproduction in protozoan

Protozoa reproduce by both sexual and asexual means

Sexual reproduction in protozoa

Sexual reproduction occurs during the life cycle of most protozoa. A distinctive feature of ciliates is the presence of two types of nuclei. These are:

- ☑ **Micro nuclei** that are involved in sexual reproduction
- ☑ *Macro nuclei* that contain genes to regulate basic cellular functions such as growth, feeding, waste removal and maintaining water balance.

Example

Conjugation in Paramecium is a sexual process of genetic

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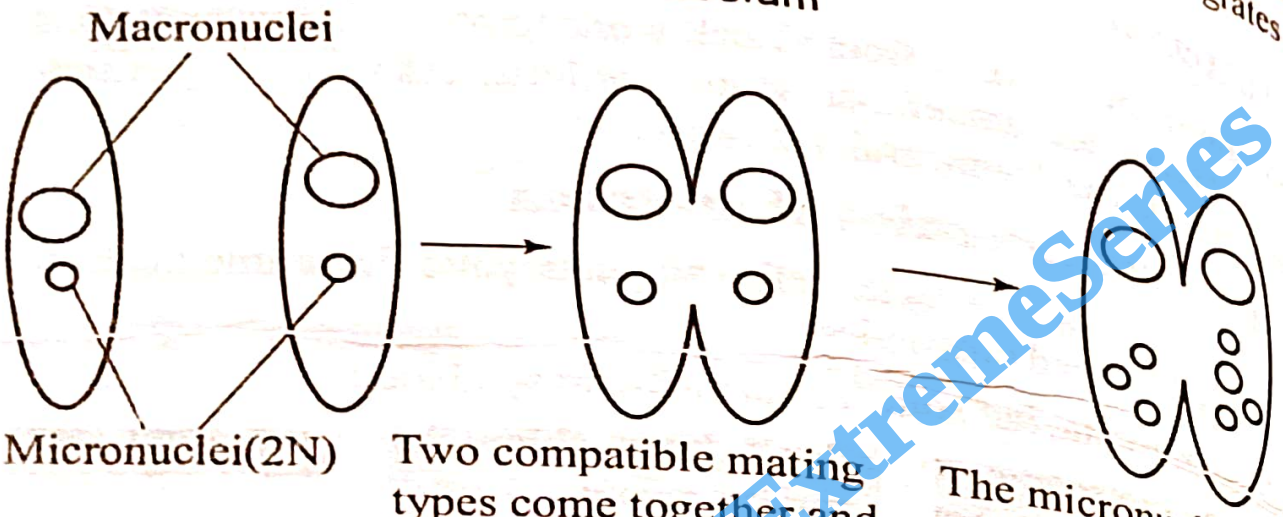
recombination, but it is not a reproductive process. After the exchange of genetic materials ciliates reproduce by **binary fission** and during the process of binary fission, the existing macronucles disintegrates and a new one is formed from cell's micronclei.

Steps of conjugation in Paramecium



i. Two Paramecia conjugate. Each Paramecium has two micronclei, one micronucleus undergoes meiosis, and the other micronucles in each sell disintegrate.
ii. Three of the four haploid micronuclei disintegrate, the remaining micronucles undergroes mitosis.
iii. The Paramecia donate micronuclei to each other. The macro-nuclei disintegrate.
iv. The micronuclei in each cell - each genetically different - fuse
v. The new diploid nuclei divide mitotic ally, eventually giving rise to a macro nucleus and the appropriate number of micronuclei.

Asexual Reproduction in protozoa

Many protists reproduce asexually through binary fission, schizogony, and budding. All of these types of reproduction produce offspring that are genetically identical to the parent cell.

☑ Most protozoa reproduces by **binary fission,** a single protists cell divides into two cells.

☑ Some protists reproduce by multiple fission (suchizogony), a form of

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cell division that produces more than two offspring.

☑ Others are reproducing by budding, which is a self - supportive cells detaches from the parent organism

Nutrition in protozoa

Protists obtain energy in a number of ways. Many protists are autotrophs, organisms that have chlorophyll and can make their own food molecules. These protists make food in much the same way that plants do.

Some protists are **heterotrophs**, organisms that must get their food by eating other organisms or by products and remains. Some heterotrophic protists engulf smaller protists and digest them. Other heterotrophic protists obtain energy by absorbtion (**saprozoic**)

Protozoa take food in to the body by the following methods.:

i. **Absorption** - Protozoa absorb food across their plasma membranes.

ii. **Ingestion** - is the process of taking food in to the body through the mouth like opening. The cilia outside the body of protozoa create a wave like motion to push the food in to the mouth like opening is called cytosome.

iii. **Engulf** - is the process of taking food by using pseudopodia. The process is called phagocytosis. The food is digested in the vacuole and the waste products are removed by the process of exocytosis.

Beneficial protozoa

It helps in sanitation and improve quality of water

- Because, large number of protozoa living in polluted water, and feed upon bacteria and organic matter, then purify it
- Food for aquatic animals
- For biological and medical research
- They make the environment more productive

8.5.3 Common Diseases Caused by Protozoa

Most protozoa living in the environment are not harmful, but some protozoa are disease causing agents. There are a number of disease caused by protozoa, such as: malaria, African trypanosomiasis, chagas disease, leishmaniasis, toxoplasmosis, and cryptosporidiosis etc.

Major protozoal parasites of human and other organisms are listed in the following table.

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Diseases	Causative agents	Organs affected	Transmission	Clinical Features
Amoebiasis	*Entamoeba histolytica*	Large intestine and liver	Ingestion of fecally contaminated food or water	Tssue destruction in the larger intestine and other organs/amoebic abscesses./ Intestinal cramps and diarrhea. Due to intestinal ulceration, amoebic dysentery / bloody diarrhoea/
Giardiasis	*Giardia lamblia*	Small intestine	Ingestion of fecally contaminated food or water	Ingested cyst survives stomach passage. Trophozoits emerge from the cyts in the in the small intestine where some attach to epithelium and others moves freely. Mucosal function is impaired by adherent protozoa and immune response
Trichomoniasis	*Trichomas vaginalis Trichom onas*	genital/ urinary and adductive systems	Sexual contact	Vaginal discharge and odor. Edema, swelling due to excessive accumulation of serum in issue space Erythema, redness on the skin around vagina
African sleeping sickness	*Trypanos oma brucei*	brain vous system	Testsefly / Glossina/	Fever, joint pains followed by neurological disorder and Somnolence / sleepy state/

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Leishmaniasis: ☑ Mucocutaneous ☑ Visceral ☑ Cutaneous	*Leishmania donovani*	blood, Skins cells, intestine, kidney and liver	Female sandy fly / Phlemotomus/	skin ulcers /lesion/Mucocutaneous complications, visceral disease (hepatosple nomegaly)
Malaria	*Plasmodium falciparum* *P. vivax,* *P. ovale and* *p. malariae*	Red blood cells, and brain	Female anopheles mosquito	fever, shivering, cough, respiratory distress, pain in the joints, headache, water diarrhea, vomiting and severe anemia
Toxoplamosis	*Toxoplasma gondii*	lymphatic vessels and eyes	Domestic cat uncooked food /meat/ contact with cat feces	Blindness and mental retardation in congenitally infected children. Immunosuppressed /inhibition of the immune response/ patients show severe symptoms such as: Splenomegaly, polymyositis, dermatomyositis, myocarditis, pneumonitis, hepatitis, and encephalitis
Chagas disease	*Trypanosoma cruzi*	CNS, Heart, Gastro intestinal tract	Kissing bugs	In acute phase the symptoms are fever and splenomegaly. In chronic phase the symptoms are irreversible damage to heart, esophagus and colon

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Illustrative Example

29. Which of the following statement incorrectly describes protozoa?

A. Mostly unicellular eukaryotic
B. Mostly free living
C. Lacks cell wall
D. Mostly photosynthetic

Explanation: Most protozoans are heterotrophic, that don't have chlorophyll to make photosynthesis.

Answer: D

30. Which of the following is not true about sexual reproduction in Paramecium (protozoa)?

A. It is called conjugation
B. It is not strict reproductive process
C. It is a king of genetic recombination
D. It involves macronucles of a cell

Explanation: Conjugation in Paramecium is a sexual process of recombination, but it is not a reproductive process. It involves only micronucles, but the macronucleus regulate basic cellular functions.

Answer: D

31. Asexual reproduction in protozoa involves all, except _____.

A. Binary fission
B. Multiple fission
C. Budding
D. Fragmentation

Explanation: Asexual reproduction in protozoa involves fission (binary fission and multiple fission), and dudding.

Answer: D

32. The process of taking food in protozoa by using phagocytosis process is

A. Engulfing
B. Absorption
C. Ingestion
D. Assimilation

Explanation: Protozoa take food in different ways, absorption, and engulf. Engulfing is the process of taking food by the action of phagocytosis.

Answer: A

33. List the importance of protozoa?

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Explanation

- It helps in sanitation and improving quality of water
- Food for aquatic animals
- For biological and medical research
- They make the environment more productive.

34. Explain how protozoa are useful in sanitation and improving water quality?

Explanation: Because, large number of protozoa living in polluted water, and feed upon bacteria and organic matter, then purify it

Matching

Match the type of protozoal disease listed under column 'A' with its transmission rout listed under column 'B'?

A	B
35. Amoebiasis	A. Bite of tsetse fly
36. African sleeping sickness	B. Ingestion of fecally contaminated food or water
37. Leishmaniasis	C. Sexual contact
38. Trichomoniasis	D. Bite of anopheles mosquito
39. Malaria	E. Bite of female sandy fly
40. Toxoplasmosis	F. Contact with cat feces

Answer

35. B 36. A 37. E 38. C 39. D 40. F

8.6 What are Viruses?

A virus is a small particle consisting of a nucleic acid core surrounded by a protein coat, in order to reproduce, a virus must infect a living cell. Viruses are small obligate intracellular parasites and non - cellular particles. They have genomes composed of either DNA or RNA enclosed in a protein Coates. They can be seen only with electron microscope.

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8.6.1 Characteristics of Virus

- They have nucleic acid (DNA or RNA) surrounded by protein co known as **capsid**.
- They cannot be grown on free media, grow only in living cells.
- They do not have cellular organization (cell organelles).
- They are obligate intracellular parasites, there must be living cell to liv and multiply (bacteria, protozoa, fungi algae, plants and animals).
- They lack enzymes necessary for protein and nucleic acid synthesis They depend on the machinery of host cells
- The are unaffected by antibacterial antibiotics.
- They are inert/nucleoprotein/ filterable agents.
- They occupy a space in between living and non - living, because they are crystallizable and non - living oustide the body of host.
- Most viruses range in sizes from 20 - 450 nm in diameter.
- They do not independently fulfill the characteristics of life. They are inactive outside the host cell and active inside the host cell.
- They consist of protein shell/capsid/ that surround nucleic acids. The nucleic acids can be: Double stranded DNA, Double stranded RNA, Single stranded DNA, and Single stranded RNA.
- They have molecules on their surfaces impart /provide/ high specificity for attachment to the host cells.
- They multiple by taking control of host cells genetic materials and regulating the synthesis and assembly of new viruses.
- They lack machineries for the synthesis of proteins.

8.6.2. Structure of Viruses

The particle of a virus is called **virion**, which is composed of protein coat (capsid) and genome (DNA or RNA). However, the basic (generalized) structure of viruses are:

- ☑ **Core** - it is the genetic material either DNA or RNA but not both/ genome/.
- ☑ **Capsid** - it is protein coat made up of number of morphological similar sub units are called Capsomeres.
- ☑ **Envelope** - some virus contains envelope that surrounds nucleocapsid.

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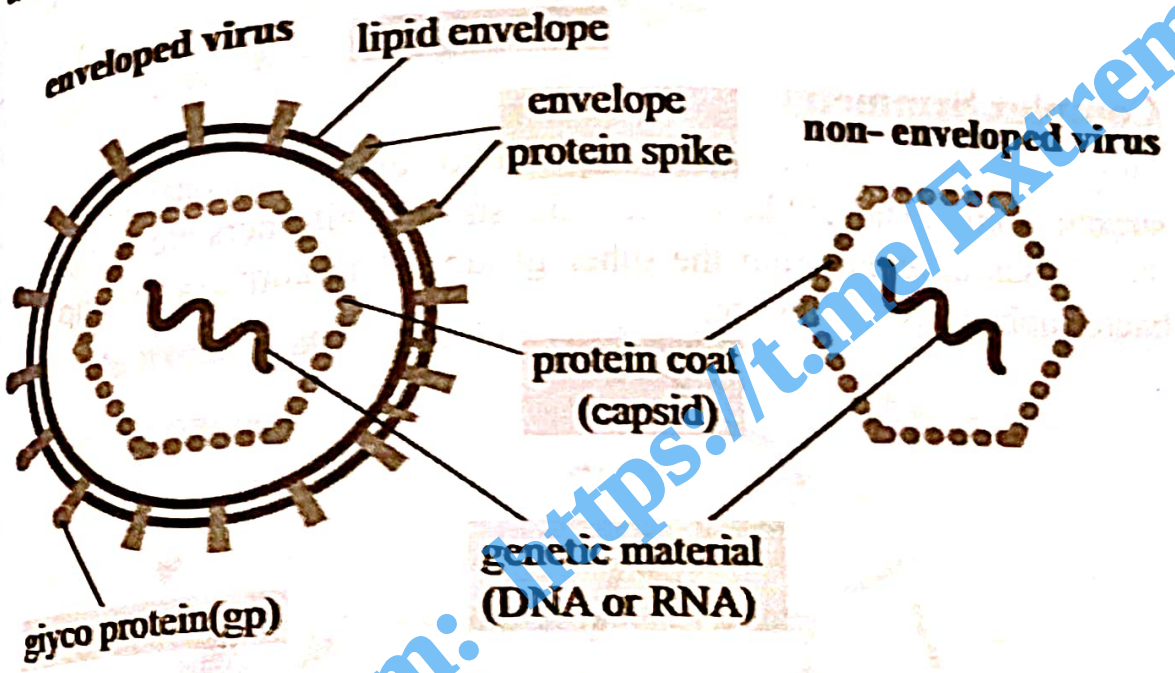
which is made up of glycoroprotein and phospholipid bilayer.

- **Nucleocapsid** - it is the combined structure formed by the core/genome/ and capsid.
- **Capsomeres** - they are identical repeating subunits that made up capsid.
- **Enzymes** - some viruses contain enzymes which play a central role during the infection process.

For example - Some bacteriophage contains enzymes, such as

- **Lysozyme**, which make small hole in bacterial cell that allows viral DNA enter in to bacterium
- **Reverse transcriptase** catalyzes the conversion of RNA to DNA



8.6.3 Viral Symmetry

Viruses can be classified on the basis of morphology (shapes) of capsid. These are: helical symmetry, icosahedral symmetry, and complex symmetry.

Helical or Cylindrical Symmetry

There are several viruses found with a **helical morphology**. These viruses consist of identical protein subunits which assembled in a helical structure around the genomes.

Helical structure provides flexibility to the filaments. The most common example of a helical virus is the **tobacco mosaic virus** /TMV/.

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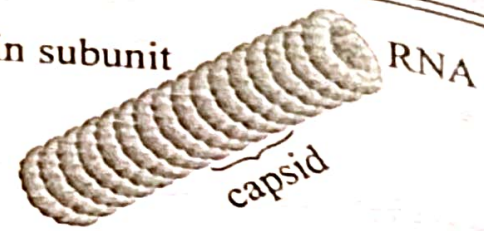
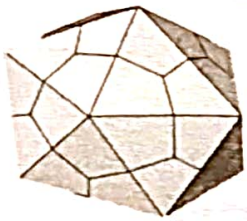


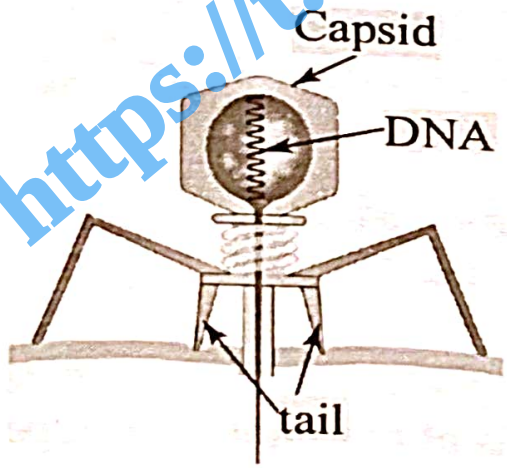
Icosahedral Symmetry

It refers to a shape with 20 triangular faces and 12 corners. It is rigid structure that provides protection to the genome. The common examples papova virus, picorna virus, adenovirus, toga virus, etc.



Complex Symmetry

They have no clearly identifiable capsid. However, they consist of complex structural components like protein tails sheath, tail fibers etc, which make the viruses different from the other groups. The common examples are bacteriophage, pox virus etc.



8.6.4 Classification of Viruses

Virus can be classified based on the following primary criteria

a. The type and character of viral genome (i.e DNA viruses, RNA viruses and retrovirus).

b. The strategy of viral replication (i.e life cycle, such as: lytic cycle lysogenic cycle)

c. The types of organisms they infect (i.e animals infecting virus, plant infecting virus, and bacteria infecting virus). Bacteria infecting viruses

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are called bacteriophage

Difference between DNA and RNA Viruses

The genetic material of a virus can be either DNA or RNA, not both. The virus that contain DNA as their genetic material are called the **DNA viruses. RNA viruses,** on the other hand, contain RNA as their genetic material.

- DNA viruses are mostly double - stranded while RNA viruses are single - stranded.
- RNA mutation rate is higher than the DNA mutation rate. Because, DNA molecule is stable while RNA is unstable.
- DNA replication takes place in the nucleus while RNA replication takes place in the cytoplasm.

8.6.5. Viral Replication

A virus invades a living host cell to replicate, then hijacks /uses/ the metabolism of the host cell to produce copies if it.

Stages of viral replication are shown below using bacteriophage as an example:

i. **Adsorption** - attachment of virus to a host cell

Viral glycoproteins bind to receptors on a host cell's membrane

ii. **Penetration** - entry of viral genome to the host cell

- Viruses enter to the host cell by receptor mediated endocytosis.

iii. **Uncoating** of the viral genome

The Uncoating, breaks up capsid by the action of lysosomal enzyme of the host cell.

iv. **Synthesis** - the virus uses the host cell's machinery (metabolism) to synthesize its capsid and other proteins.

v. **Maturation** - is an assembly of new virus parasitical (components) into complete virion

vi. **Release**, the new viruse particles are departed (leave) from a host cell either by killing (lysis) or without killing. The new viruses released from the host cell again infect another new cell.

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Note: Scientists have gained a better understating of virus replication by studying bacteriophages, viruses that infect bacteria.

Phage capsid are made up of a hexagonal head filled with DNA. Attached to the head is a protein tail with one or more tail fibers. The tail helps the virus inject its genome into the host cell.

Lytic and lysogenic cycles of phage

Phages and other viruses can reproduce either by lytic cycle or lysogenic cycle

Lytic cycle

During the lytic cycle, a virus, such as a T4 phage, invades a host cell, produces new viruses, and released out by killing (lytic) of the host cell. viruses that reproduce only by the lysis cycle are called **virulent**, Viruses destroy the cells that they infect.

Lysogenic Cycle

- In lysogenic cycle the virus's DNA becomes incorporated into the DNA of the host cell and hides for a period of time in the form of a non infective phage called **prophage.**
- Environmental condition triggers the virus DNA to star producing viral proteins, each time the cell divides, the phage DNA, proteins are also replicated.
- Then the whole viruses are assembled, which are then leave the cell either by lysis (splitting) or chronic (ongoing) through exocytosis.
- A virus whose replication includes the lysogenic cycle is called a **temperate viruses**

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The comparison between lytic and lysogenic cycle

Lytic Cycles	Lysogenic cycle
DNA of the virus does not integer ate into the host DNA	DNA of the virus integrate into the host DNA
Results in lysis or bursting of host cell	May or may not kill the host cell
DNA hydrolyzed (degraded)	DNA not hydrolyzed
Abence of prophage stage	Presence of prophage stage
DNA replication of viruse takes place independently from the host	DNA replication of viruse takes place along with the host A replication
Occurs within a short period of time	DNA replication of the viruse take time
Symptoms of Viral replication are evident	Symptoms of Viral replication are not evident
Genetic recombination between the phage and the bacterium is not allowed	Genetic recombination the host bacterium is allowed
Cellular mechanism of the host cell is totally undertaken by the viral genome	The cellular mechanism of the host cell is some what disturbed by the viral genome

8.6.6 HIV and AIDS

Human Immunodeficiency Virus (HIV) is a types of retrovirouses, RNA, as its genetic material. During replication, RNA is transcribed into DNA by an enzyme known as **reverse transcriptase.**

The viral envelope which is made of lipid and glycoprotein (Gp 120 and Gp41) , which is used for attachment on host cell. The main target of HIV is the immune system, CD4-Tlymphocyte or T - helper cell.

The acronym AIDS refers to acquired immune deficiency syndrome,

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8.6.7 Common viral diseases in Ethiopia

Viral diseases	Causative agent	Sign and symptoms	Transmission	Prevention and control
Mumps	Mumps virus	Swollen and painful parotid glands, on chewing swallowing.	Person to person in infected saliva	Vaccine
Measles (rubeola)	Measles virus	Cough, nasal discharge, eye redness, and high fever,	Droplet contact	Vaccine
Rabies	Rabies virus	Tingling, burning, coldness at bite site, fever, dache, increased muscle tension, paralysis hydrophobia /aversion in swallowing ids/	Bite from rabid animals	Avoiding rabid animals, Thoroughly washing bitten area , pre - exposure Vaccination when needed
Polio	Polio virus	Often no sign and symptoms	Fecal - oral - route	Polio Vaccine, good personal hygiene
Common cold (rhinitis)	Rhino virus Adeno virus and others	Sneezing, sore throat, runny and stuffy nose, cough	Respiratory droplets	Practicing good hygienic
Chicken pox (varicella)	Varicella zoster virus	Fever, headache, malaise with red, itchy rash face, scalp, chest and back	Droplet contact	Chicken pox vaccine

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Illustrative Example

41. Which of the following statement is not true about viruses?
A. Have nuclei acid (DNA or RNA)
B. Obligate intracellular parasites
C. Do not have cellular organization
D. They can be seen only with compound microscope

Explanation: A viruse is a small particle consisting of the nucleic acid (DNA or RNA). Viruses are very small obligate intracellular parasites and non- cellular particles. They can be seen only with electron microscope.

Answer: D

42. Which one of the following is not the characteristic of virus?
A. Unaffected by antibacterial antibiotic
B. Independently fulfill the characteristics of life
C. Cannot be grown on free media
D. Lack machineries for the synthesis of proteins

Explanation: Virus do not independent fulfill the characteristics of life. They are inactive outside the host cell and active only inside the host cell.

Answer: B

43. Select the correct combination of viral structure and its function?
A. Core - it is the genetic material
B. Envelop - surround nucleocapsid
C. Capsid - it is protein coat
D. Plasmid - small circular DNA

Explanation - Plasmid is a small circular DNA found in bacteria, not in viruses.

Answer: D

44. An enzyme, which makes small hole in bacterial cell that allows viral DNA enter into the bacterium is
A. Reverse, transcriptase
B. Lipase
C. Lysozyme
D. Polymerase

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Explanation: Some virus contain enzymes which play a central role during the infection process. For examples:

- Lysozyme, which makes small hole in bacterial cell that allows viral DNA enter in to the bacterium and
- Reverse transcriptase catalyses the conversion of RNA to DNA.

Answer: C

45. A viral symmetry, which has a capsid with a shape of 20 triangular faces and 12 corners to provide protection to the genome is

A. Complex Symmetry
B. Icosahedral Symmetry
C. Helical
D. Spiral

Explanation: viruses are classified based on the shape of capsid (helical,icosahedral,and complex symmetry).icosahedral symmetry refers to a shape of 20 triangular faces and12corners.

Answer: B

46. The difference between DNA virus and RNA virus is that,DNA virus is

A. Mostly double - stranded
B. Have higher mutation
C. It more unstable
D. Replication takes place in the nucleus

Explanation: RNA mutation rate is higher than that of DNA mutation rate, Because, DNA molecule is stable while RNA is unstable.

DNA replication takes place in the nucleus while RNA is in the cytoplasm.

Answer: D

47. The bacteriophage replication stage, which breaks up its capsid by the action of lysosomal enzyme of the host cell is

A. Adsorption
B. Uncoating
C. Synthesis
D. Penetration

Explanation: Uncoating, breaks up capsid by the action of lysosomal enzyme of the host cell.

Answer: B

48. Which of the following statement about the lytic cycle of a virus is not

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true?

A. The cycle are called **virulent**
B. It ends with death of a cell
C. Absence of prophage stage
D. The virus genome is incorporated with host cell genome

Explanation: During the lytic cycle, the virus invades a host cell, produces new viruses, and released out by killing (lyses) of the host cell. Viruses that reproduce only by the lytic cycle are called virulent.

Answer: D

49. The difference between lytic cycle of virus and lysogenic cycle is that, lysogenic cycle is:

A. Host DNA is hydrolyzed
B. Occurs within a short period of time
C. Genetic recombination between host DNA
D. Cellular mechanism of the host cell is totally undertaken by the viral genome

Explanation: In lysogenetic cycle DNA replication of the virus takes longer time than lytic cycle, viral genome is integrated with host with DNA.

Answer: C

50. What type of viral infection is occurred as a result of the bite of infected rabid animals?

A. Mumps
B. Rabis
C. Polio
D. Chicken pox

Explanation: Rabis is a viral infection due to rabis virus transmitted by the bite of rabid animals.

Answer: B

8.7 Normal Micro Biota

The normal microbiota is the population of microorganisms that are found in the body of healthy individuals. It is also called normal flora. It is the group of micro-organisms that colonize the host without causing diseases. Microbiotas are localized in a certain regions of the hosts, such

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as;

- ☑ Upper respiratory tract (nose, nasopharynx, aropharnx) are the home for the normal microbiota such as; Streptococcus, Neisseria, haemopilus, staphylococcus and the fungal genera Candida.
- ☑ The skin is the home for:

Staphylococcus, streptococcus, orynebacterium, propioni bacterium, micrococcus and the fungi general Candida.

- ☑ Female reproductive tract contains staphylococcus, lactobacillus, caryne bacterium, streptococcus, enterococcus and the fungal genera Candida
- ☑ Oral cavity contains streptococcus, Neisseria, haemophilius, lactobacillus, staphylococcus, propioni bacterium, triponema
- ☑ **Urinary tract**
- ☑ Small intesine contains bacteriodes, lactobacillus, and streptococcus,
- ☑ Large intestine - contains: E, coil, bacteriods, fuso bacterium lactobacillus, enterobacter, citrobacter, klebsiela, bifidobacterium and the fungal genera. candida.

8.7.1 Types of Normal Microbiota

Microbiota are classified into two major categories,

Resident microbiota - is group of micro-oraginsms that inhabit body sites for extended periods of time.

Transient microbiota - are group of micro-organisms present for several days, weeks, months and then disappear.

What are reasons to acquire the knowledge of the normal human microbiota?

- ➥ To understand the different microorganisms at particular locations because they provide greater insight in to the possible infections.
- ➥ Helps the physican investigatory understand the causes and consequences of colonization and growth by microorganisms normally absent at a specific body site.
- ➥ To increase awareness about the role of normal microbiota in stimulation the host immune response.

How do normal microbiotas exclude pathogens?

- ☑ Covering binding sites that used for pathogenic attachment.

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- Consuming available nutrients
- Producing compounds toxic to other bacteria.
- Stimulating the adaptive immune system.

The Germ theory of Disease and Koch's postulates

Germ theory states that certain disease are caused by invasion of the body by microorganism. This theory was developed by Robert Koch and Louis Pasteur.

Using appropriate controls, Koch demonstrated that when a small amount of blood from a diseased mouse was injected into a healthy mouse disease symptoms appear.

Koch's postulates

- The disease - causing organism must always be present in animals suffering from the disease but not in healthy animals
- The organism must be cultivated in a pure culture away from the animal body.
- The isolated organism must cause the disease when inoculated into healthy susceptible animals.
- The organism must be isolated from the newly infected animals and cultured again in the laboratory, after which it should be seen to be the same original organism.

8.8 Modes of Disease Transmission and Ways of Prevention

Micro -organisms are transmitted in health care settings by the following main routes.

- Contact
- Dróplet
- Airborne
- Common vehicle /Vectors/

The following table shows the mode of diseases transmission

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Method of transmission	Transmission route	Examples
Droplet infection	Many of these diseases are respiratory diseases. The micobes are carried in tiny droplets through the air when an infected person coughs or sneezes. They are inhale by other people.	Common cold, 'flu' pneumonia
Drinking contaminated water	Drinking contaminated water containing the organisms	Cholera, typhoid fever
Eating contaminated food	*The organisms initially infect a region of the gut*	*Salmonellosis, typhoid, listeriosis, botulism*
Direct Contact	*Many skin infections such as athlete's foot are spread by direct contact with an infected person*	*Athlete's foot, ringworm*

Sexual intercourse	From one sexual partner to another during intercourse. Through semen or vaginal secretions.	Candidiasis, syphilis, AIDS, Gonorrhoea
Blood - to - blood contact	Many of the sexually transmitted diseases can be transmitted by blood - to - blood contact. Drug users sharing an infected needle can transmit AIDS	AIDS, hepatitis B
Animal vectors	Diseases are spread through the bites of insects. For example: ➥ Mosquitoes spread malaria and ➥ Tsetse lies spread African sleeping sickness	Malaria, African sleeping sickness

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8.9 Use of microorganisms

Microorganisms play great role in different ways. For example: in agriculture, sewage treatment, bioremediation, medicine, food processing and production, health, and nutrient recycling. Most of these topics are discussed in unit 6 and 7 of this book in more detail.

i. Agriculture

Microorganisms such as bacteria and fungi are decomposers, which break down organic matter into simpler from, so that plant can use it. For example nitrogen fixation (nitrogen gas into nitrate), carbon cycle (organic molecule is decomposed and release C in the form CO_2), etc.

ii. Sewage treatment

Sewage is waste water containing organic molecule and pathogenic microorganisms. Anaerobic bacteria are used in the treatment of sewage.

- I helps to produce methane gas for alternative energy sources.
- It helps to remove phosphorous which is a major pollutant of the water bodies

iii. Bioremediation

Bioremediation is the process of removing contaminants /pollutants from the environment. This process is done by enzymes exacted from microorganisms, or metabolic process of catabolic capabilities to degrade chemicals.

For examples: synthetic hydrocarbons, such as oil, poly chlorinated biphenyl, polyaromatic hydrocarbons, radionuclide and metals are contaminants/ pollutants that need to be degraded and detoxified by using microbes.

iv. Food production

Many species of lactic acid bacteria such as lactobacillus, streptococcus, etc. Produces lactic acid which is used in milk processing, such as in cheese and yoghurt to give characteristics flavor and aroma.

v. Medicine

Microorganisms are used to produce medicines that can target defected genes and make it functional again. Insulin (a drug that is prescribed to diabetic patients) is produced from bacteria or fungi.

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vi. Health

Our body contains ten times more microorganisms than the body cells. These microorganisms are useful, for example: help digestion of food, fight against harmful bacteria, synthesizing vitamins such as vitamin K, biotin, and folic acid.

vii. Biotechnology

Biotechnology is the application of microorganisms to make things that human need. For example: to make insulin, human growth hormone (HGH). Microorganisms can also be used as a vector to deliver drugs to body cells.

Illustrative Example

51. Which of the following is not true about normal microbiota?
   A. Thy are found in the body of healthy individuals
   B. They are normal flora
   C. Colonize the host and causing disease
   D. Localized in different part the body

Explanation: The normal microbiota is the population of microorganisms that are found in the body of healthy individuals. It is also called normal flora. It is the group of micro - organisms that colonize the host without causing disease.

Answer: C

52. Which of the following disease is transmitter by droplet infection?
   A. Cholera
   B. Typhoid
   C. Pneumonia
   D. Ring worm

Explanation: Droplet infections are respiratory disease, such as pneumonia, TB, common cold. etc.

Answer: C

53. Skin infections such as athlete's foot are spread by _____ .
   A. Sexual intercourse
   B. Animal vectors
   C. Direct Contact
   D. Eating contaminated food

Explanation: Man skin infections such as athlete's foot are transmitted through direct skin contacts.

Answer: C

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54. Examples of disease that are spreaded through the bites of insects is

A. Cholera
B. Malaria
C. IDS
D. Ring worm

Explanation: Many diseases are spread through the bites of insects. For example:

- Mosquitoes spread malaria
- Tsetse flies spread sleeping sickness.

Answer: B

55. ___ is group of micro-organisms that inhabit body sites for extended periods of time

Answer: Resident microbiota

56. ___ are group of micro-organisms present for several days weeks, months and then disappears?

Answer: Transient microbiota

57. Explain how the normal microbiota exclude pathogens?

Explanation:

- ☑ By covering binding sites that used for pathogenic attachment
- ☑ By consuming available nutrients
- ☑ By Producing compound toxic to other bacteria
- ☑ By stimulating the adaptive immune system

58. Germ theory of disease are states _____

Answer: Certain diseases are caused by the invasion of the microorganism.

8.10 The Role of Microorganisms in Recycling Minerals through Ecosystem

Many bacteria are decomposer, break down the complex molecules that are found in the bodies of the dead organisms into much simpler molecules. They use some of these for their own metabolism and they release some minerals in to the environment in various forms. For example, S, P, O, N, C, H

The carbon cycle

All organisms are composed of organic molecules such as proteins, lipids,

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and carbohydrates. The carbon travels through the food chain as primary producers are eaten by primary consumers, which are then eaten by secondary consumers. Decomposers then break down remains of producers and consumers to release carbon in the form of CO2

Main processes involved in carbon cycle.

i. Photosynthesis

The process that fixes carbon atoms from an inorganic source (carbon dioxide) into organic compounds (glucose). It reduces the amount of carbon from the air.

ii. Feeding and assimilation

Feeding passes carbon atoms in complex molecules to the next tropic level in the food chain where they are assimilated into the body of the organisms

iii. Respiration/fermentation

Respiration is the process of breaking down of organic food to release energy by organisms. CO2 is released into the air as a by product. This increases the amount of carbon dioxide in the air.

iv. Fossilization

Sometimes living things do not decay fully when they die due to the conditions in the soil as the result fossil fuels are formed. The fossil fuel stores carbon.

v. Combustion

Fossil fuels are burned and releasing carbon dioxide into the atmosphere. It increase the amount of carbon in the air

vi. Methanogenesis and Methane Oxidation

In anaerobic environments, carbondioxide is used by methanogens. They obtain energy by oxidizing hydrogen gas and by using CO_2 as a terminal electron acceptor and generate methane.

- Methane that enters the atmosphere is oxidized by ultraviolet light and chemical ions in to carbon monoxide (CO) and CO_2.
- Methlyotrophs can use methane energy source and oxidizing it to produce CO_2

Nitrogen cycle

All organisms need nitrogen to make proteins and nuclei acids. The complex pathway that nitrogen follows in an ecosystem is called the nitrogen cycle.

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Nitrogen as, N_2, makes up about 78 percent of the atmosphere. Most plants can use nitrogen only in the form of nitrate.

The process of converting N_2 gas into **nitrate** is called **nitrogen fixation**. This process is done by bacteria such as (azotobacter, cynobacteria, clostrodium rhizobium, etc.), which live in the soil and inside root of some kinds of legume plants, such as beans, peas, clover, and alfalfa.

Under nitrogen - limiting conditions legumes form a symbiotic relationship with bacteria known as **rhizobia**. These plants supply carbohydrates for the bacteria, and the bacteria produce usable nitrogen for the plant.

What is the major process that involves in nitrogen cycle?

The bodies of dead organisms contain nitrogen, in the form of proteins and nucleic acids. Urine and dung also contain nitrogen.

i. Nitrogen fixation

The process of converting N_2 gas to nitrate by the action of bacteria is called **nitrogen fixation**

ii. Ammonification

Decomposers break down protein form dead materials and release in the form of ammonia, (NH_3) which in soil becomes ammonium, NH_4^+. This process s known as ammonification.

iii. Nitrification

Ammonium oxidize it into **nitrites** (NO_2^-) by Soil bacteria known as **nitrosomonas,** and nitrates (NO_3^-) by nitrobacter, in a process called nitrification. Plants use nitrates to make amino acids.

iv. Weathering of rock

Erosion of nitrate - rich rocks also releases nitrates into an ecosystem.

v. Denitrification

Nitrogen is returned to the atmosphere through **denitrification.** Denitrification occurs when anaerobic bacteria such as (bacillus, paracoccus, and psydomonas) break down **nitrates** and release **nitrogen gas** into the atmosphere. Decreases the amount of N from the soil.

vi. Feeding and assimilation

Animals obtain nitrogen in the same way they obtain energy - by eating plants and other organisms and then digesting proteins and nucleic acids.

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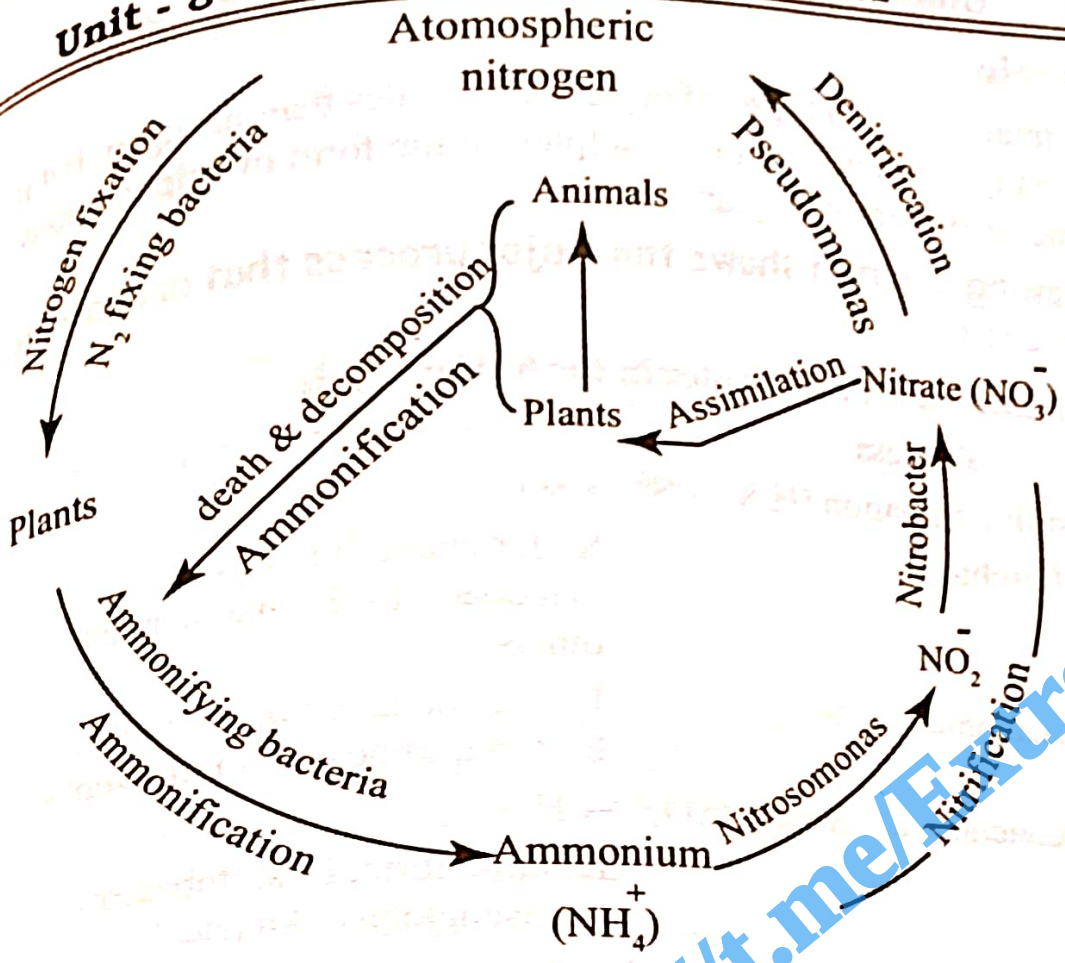

Micro-organism	Process
Nitrogen - fixing bacteria, **Example** Rhizobium	Nitrogen gas is fixed into forms other organisms can use (e.g ammonium)
Ammonifying bacteria (decomposers)	The decomposers break down proteins in dead organisms and animal waste releasing ammonium ions, which can be converted to nitrates.
Nitrifying bacteria, **Example** Nitrosomonas and Nitrobacter	Nitrification is a two - step process. Ammonia or ammonium ions are oxidized first to nitrites (by Nitrosomonas) and then to nitrates (by Nitrobacter) which is the form most usable by plants.
Denitrifying bacteria, **Example** Pseudomonas	Nitrates are reduced to nitrogen gas, returning nitrogen to the air and reducing the amount of nitrogen in the soil.

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Sulfur cycle

Sulfur is found in fewer types of organic molecules than nitrogen, but it is found in many proteins. Plant uses sulphur in the form of sulphate ions in the soil and incorporated in plant tissues.

The following diagram shows the major process that are involved in sulfur cycle.

Key Processes and Prokaryotes in the Sulfur Cycle

Process	Organisms
Sulfide/sulfur oxidation ($H_2S \rightarrow S^0 \rightarrow SO_4^{2-}$)	
Aerobic	Sulfur chemolithotrophs (Thiobacillus, Begiatoa, many others)
Anaerobic	Purple and green phototrophic bacteria, some chemolithotrophs
Sulfate reduction (anaerobic) ($SO_4^{2-} \rightarrow H_2S$)	Desulfovibrio, Desulfobacter Archaeoglobus (Archaea)
Sulfur reduction (anaerobic) ($S^0 \rightarrow H_2S$)	Desulfuromonas, many hyperthermophiles Archaea
Sulfur disproportionation ($S_2O^{2-}_3 \rightarrow H_2S + SO_4^{2-}$)	Desulfovibrio, and others
Organic sulfur compound oxidation or reduction ($CH_3SH \rightarrow CO_2 + H_2S$) (DMSO $\rightarrow$ DMS)	Many organisms can do this
Desulfurylation (organic - S $\rightarrow H_2S$)	Many organisms can do this

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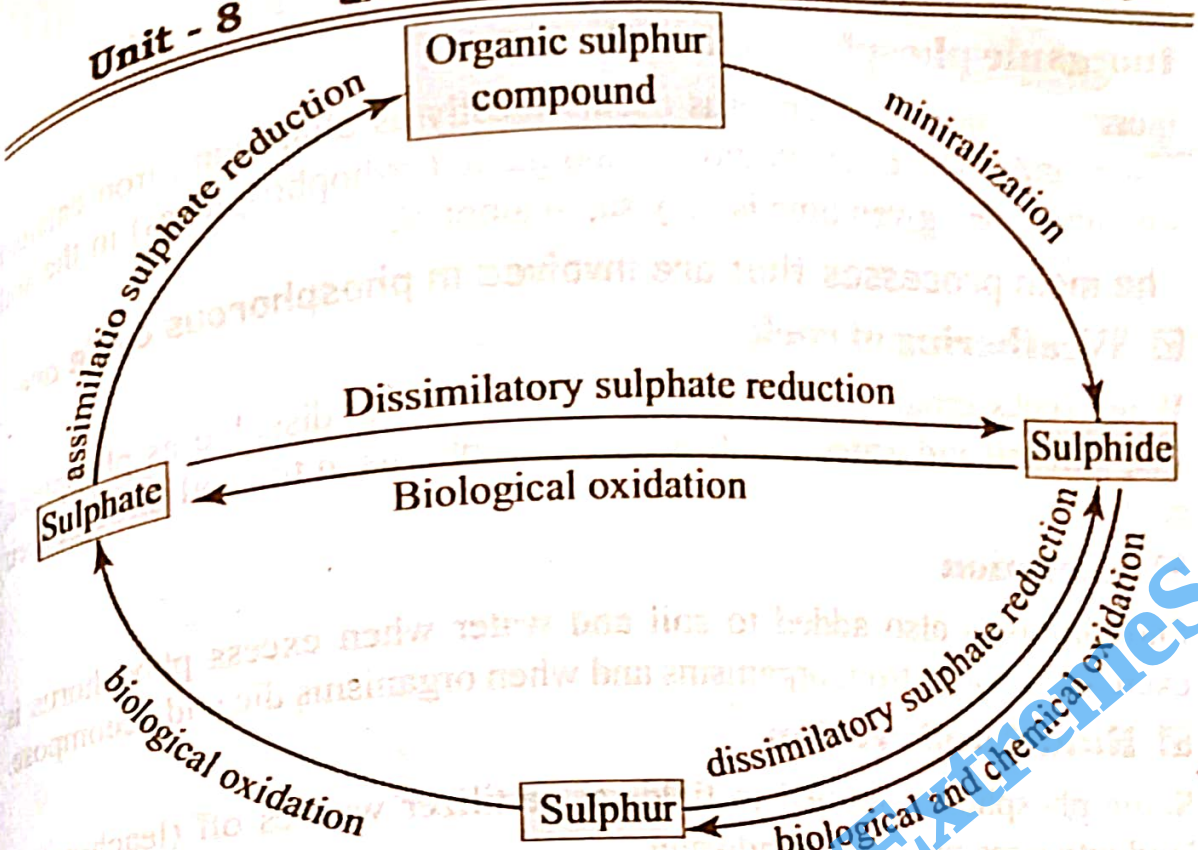

Phosphorus cycle

Phosphorus is an element that is essential material needed by animals to form bones, teeth, molecules, such as DNA and RNA. Plants get the phosphorus they need from soil and water in the form of phosphate, whereas animals get their phosphorus by eating plants or other animals.

The phosphorus cycle is the movement of phosphorus from the environment to organisms and then back to the environment. This cycle is slow and does not normally occur in the atmosphere, because phosphorus rarely occurs as a gas.

Phosphorus occurs in soils as both organic and inorganic forms.

Organic phosphorus in soils

A large number organic phosphorus in soils are microbial origin, but is not available for plant uptake until the organic materials are decomposed and the phosphorus released via the mineralization process, which is carried out by microbes. The rate of phosphors release is affect by factors such as:

- Soil moisture
- Composition of the organic material
- Oxygen concentration and pH.

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Inorganic phosphorus in soil

Inorganic from of phosphorus occurs mostly as aluminum, iron calcium compounds. The concentration of inorganic (orthophosphates) in the soil solution at any given time is very small amount.

The main processes that are involved in phosphorous cycle are:

☑ **Weathering of rock**

When rocks erode, small amounts of phosphorus dissolve as phosphate, PO_4^{-3}, in soil and water. Plants absorb phosphorus in the soil through their roots.

☑ **Excretion**

Phosphorus is also added to soil and water when excess phosphorus is excreted in waste from organisms and when organisms die and decompose.

☑ **Runoff and erosion**

Some phosphorus applied to fields as fertilizer washes off (leaching)the land into streams and groundwater.

Important Terminologies in the Process of Phosphorus Cycle

☑ **Desorption** - is the process when phosphorus is removed from the surface of soil particles induced by the depletion of solution concentration

☑ **Adsorption** - is the process when phosphorus is attached to the surface of soil particles, which is unavailable to plants

☑ **Dissolution** - Is the form of weathering that occurs when the phosphate minerals are dissolved and releases phosphate back to the solution

☑ **Precipitation** - Is the process in which phosphorus reacts with other substance to form solid minerals or secondary compound of phosphorus.

☑ **Atmospheric deposition** - Is the process of precipitating particles, aerosols and gases from the atmosphere to the earth's surface.

☑ **Immobilization** - Is the process of converting inorganic forms of phosphate into organic forms in the living cells of soil microbes. It is the reverse process of mineralization.

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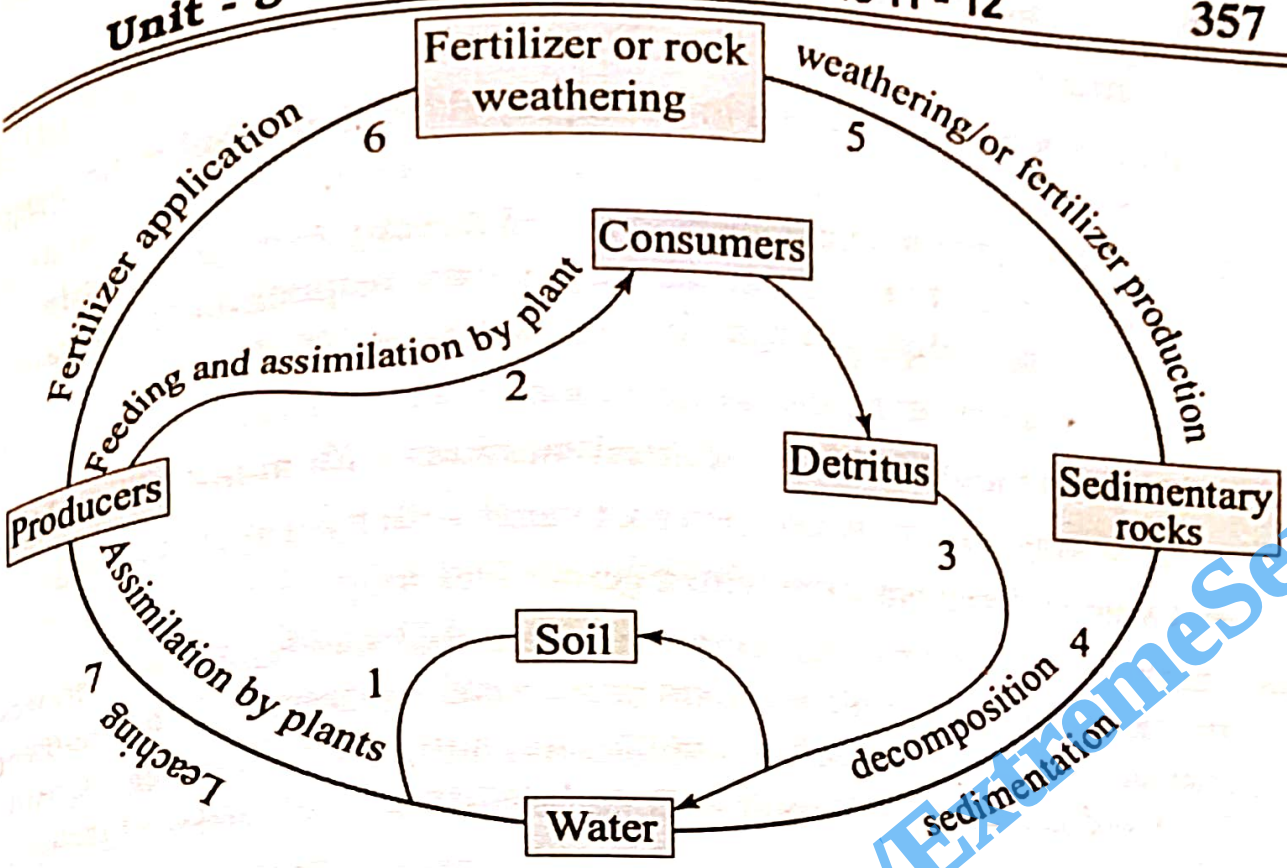

8.11 Controlling Microorganisms

1. **Sterilization:-** It is the removal or destruction of all forms of microbial life. A process by which an article, surface, or medium are free from all microorganisms (vegetative or spore). Any material that has been subjected to this process is said to be sterile.

Sterilization is performed by sterilizing agents such as **heat**, or **chemicals** (germicides, disinfectant, and antiseptics).

i. **Germicide (microbicide)** - is any chemical agent that kills pathogenic microorganisms /germs/, but cannot kill resistant microbial cells. It can be used on inanimate (nonliving) materials or on living tissue.

Examples: Fungicide kills fungi and **virucide** inactivates viruses

ii. **Disinfection** - is the use of a chemical agent that destroys or removes all pathogenic organisms, destroys vegetative pathogens but not bacterial endospores. It is used only on inanimate objects because they can be toxic to living tissues. It can also remove harmful products of microorganisms (toxins) from materials.

Examples of disinfection includes:

➥ Applying a solution of 5% bleach/ sodium hypochlorite/ to

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examining table

- Immersing thermometers in an isopropyl alcohol solution between uses.

iii. **Antiseptics** - are chemical agents applied directly to exposed body surfaces such as skin and mucous membranes, wounds and surgical incisions /cut/ to destroy or inhibit vegetative pathogens.

Examples of antiseptic include:

- Preparing the skin before surgical incisions with iodine compounds
- Swabbing /cleaning/ an open root canal with hydrogen peroxide
- Ordinary hand washing with a germicidal soap.

iv. **Sanitization** - is any cleansing technique that mechanically removes microorganisms to reduce the level of contaminants. It is a treatment intended to lower microbial counts on eating and drinking utensils. Sanitized is a compound (e.g soap or detergent) that is used to perform this task.

v. **Preservation** - measure taken to prevent microbe causing spoilages of susceptible products (pharmaceuticals, foods).

vi. **Decontamination**

Physical Methods of Sterilization

The application of heat is a simple, cheap is a simple, cheap and effective method of killing pathogens.

There are different methods of sterilized by heat such as, pasteurization, and dry heat sterilization.

I. **Pasteurization** - is the antimicrobial treatment used for foods in liquid form (eg. milk). Pasteurization includes:

- Low - temperature pasteurization: 61.5 °C for 30 minutes , 71°C for 15 seconds.
- Hight - temperature pasteurization: brief (seconds) exposure to 80 - 85°C in continuous operation.
- Uperization heating to 150°C for seconds in a pressurized container using steam injection.

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II. Dry heat sterilization

Sterilizing by dry heat includes:

Flaming - is carried out by holding the materials in the flame of the bunsen burner until they become red hot. It is used to sterilized inoculating loop or wire, the tip of forceps, searing spatulas, etc.

☑ **Incineration** - Is an excellent method of destroying infective materials by burning them to ashes, not effective for metals and heat - resistant glass materials. It is used for complete destruction and disposal of infectious material, such as:

- Syringes,, needles, culture material
- Dressings, bandages, bedding
- Animal carcasses (dead body) and pathology samples.

☑ **Hot - air oven** - sterilization which involves heat exposure up to 160 - 180°C for 2hours and 30 minute. It can destroy spores.

III. Moist heat sterilization

Autoclaves are charged with saturated and pressurized steam, which are used for killing microorganisms. It includes;

A. 121 °C, for 15 minutes, one atmosphere of pressure (total: 202 kPa).

B. 134 °C, for three minutes, two atmospheres of pressure (total: 303 kPa).

2. Intermittent sterilization (tyndallization)

Certain heat - labile substances /break down by heat/ such as serum, sugar, egg, etc, that cannot withstand the high temperature of the autoclave can be sterilized by a process of intermittent sterilization. It is carried out a period of 3 days and it requires:

- A chamber to hold the materials
- A reservoir for boiling water.

Items to be sterilized are kept in the chamber are exposed to free flowing steam at 100°C or 20 minutes for each of the three consecutive days.

Bacterial Isolation techniques

Microorganisms can be isolated from food, soil, water or from other materials. For bacterial isolation:

- The soil or food samples are collected from the desired sites.
- Bacteria are separated on artificial media by serial dilution method.

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- Each of the isolates are purified on new media and experimented for the morphological characteristic like shape, gram nature and arrangement of cells, motility etc.
- Finally molecular techniques are used for further identifications.

What is Serial Dilution?

Serial dilution is the process of diluting a sample several times to obtain the appropriate colony number.

Example of Serial Dilutions

Let's say, for example, that a milk has 10,000 bacteria per millilitre. If 1 ml of this sample were plated out, there would theoretically be 10,000 colonies formed, which is not countable.

If 1 ml of this sample were transferred to a tube containing 9ml of sterile water, each millilitre of fluid in this would now contain 1000 bacteria, again. This is still being too many potential colonies to count on a plate.

Therefore, another serial dilution could be made. One millilitre containing 1000 bacteria would be transferred to a second tube of 9ml of water. Each millilitre of this tube would now contain only 100 bacteria, which is easily countable number.

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Illustrative Example

59. What is the role of microorganisms in recycling nutrients in the ecosystem?

Explanation: Many microorganisms break down the complex molecules that are found in the bodies of the dead organisms into much simpler molecules and they release some minerals in to environment in various forms, such as: C, O, N, H, P

60. The process in carbon cycle that decreases the amount of carbon from the atmosphere is

A. Respiration
B. Decomposition
C. Combustion
D. Photosynthesis

Explanation Photosynthesis absorbs CO_2 from the atmosphere. Thus, the amount decrease. But the other processes such as respiration, decomposition, and combustion increases the amount of CO_2.

Answer: D

61. An organism that uses methane as an energy source an oxidizing it to produce CO_2 is called _____

A. Methlyotrophs
B. Photoautothrophs
C. Chemoautotrophs
D. Photoheterotrophs

Explanation: Methlyotrophs can use methane as an energy sources and oxidizing it to produce CO_2.

Answer: A

62. Which of the following is a nitrogen fixing bacterium?

A. Pseudomonas
B. Rhizobium
C. Nitrozomonas
D. Nitrobacter

Explanation: The process of converting N_2 gas to nitrate is called nitrogen fixation. This process is done by bacteria such as (azotobacter, cyanobacter, clostrodium, rhizobium, etc.)

Answer: B

63. The process of breaking down organic materials by bacteria and release it is the from ammonia, NH3, which in soil becomes ammonium, $NH4^+$ is known as _____ ?

A. Nitrification
B. Nitrogen fixation
C. Ammonification
D. Denitrification

Answer: C

64. Ammonium oxidize into nitrites (NO_2^+) by Soil bacteria ____, and nitrates (NO_3^+) by _____, in a process called nitrification, respectively.

A. Nitrobacter, nitrosomonas
B. Nitrosomonas, pseudomona
C. Pesudomonas, nitrobacter
D. Nitrosomonas nitrobacter

Explanation: Soil bacteria, take up ammonium and oxidize it into

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nitrites (NO_2^+) by nitrosomonas, and nitrates (NO_3^+) by nitrobacter, in a process called nitrification.

65. The process that nitrogen is returned to the atmosphere by bacteria such as pseudomonas is called?

A. Nitrification
B. Nitrogen fixation
C. Ammonification
D. Denitrification

Answer: D

66. Which of the following is anaerobic sulfur bacterium that reduces SO_4^{-2} into H_2S?

A. Disulphovibrio
B. Thiobacillus
C. Beggiatoa
D. Nitrifying bacteria

Explanation: Anaerobic sulfur bacterium that reduces $SO4^{-2}$ into H_2S includes desulfovibrio, desulfobacter.

Answer: A

67. Animals uses the mineral phosphorous to make _____

A. Protein
B. Carbohydrate
C. Nucleic acid
D. Lipids

Explanation: Phosphorus is an element that is an essential material needed by animals to from bones, teeth, and nucleic acid (DNA and RNA).

Answer: C

68. Which of the following statement is not true about the phosphorous cycle?

A. A large number organic phosphorus in soils are microbial origin
B. Inorganic form of phosphorus occurs mostly as aluminum, iron or calcium compounds
C. When rocks erode, small amounts of phosphorus dissolve as phosphate, PO_4^{-3}
D. Desorption is the form of weathering where phosphate minerals are dissolved and releases phosphate back to the solution.

Explanation: Desorption - is the process when phosphorus is removed from the surface of soil particles induced by the depletion of solution concentration.

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Answer: D

69. What is the function of an autoclave in sterilization
A. Uses chemicals to remove pathogens from surface
B. Uses ultraviolet lamps to kill microorganisms
C. Sterilizes medical instruments with dry heat
D. Applies heat and pressure to sterilize using saturated steam.
Explanation: Autoclave is charged with saturated and pressurized steam that are used for killing microorganisms.

Answer: D

70. What is the purpose of tyndailization?
A. Complete removal of all microorganisms
B. Sterilization of heat - labile substances
C. Sterilization of medical instruments
D. Disinfection of surfaces
Explanation: Certain heat - labile substance such serum, sugar, egg, etc. that cannot withstand the high temperature of the autoclave can be sterilized by a process of intermittent sterilization (tyndallization).

Answer: B

71. Which method is used for sterilization infective materials by burning them to ashes?
A. Flaming
B. Incineration
C. Pasteurization
D. Tyndallization
Explanation: Incineration is an excellent method of destroying infective materials by burning them to ashes, not effective for metals and heat - resistant glass materials

Answer: B

72. Chemical agents applied directly to exposed body surfaces as skin and mucous membranes, wound and surgical incisions /cut/ to destroy or inhibit vegetative pathogens.
A. Antiseptic
B. Germicide
C. Disinfection
D. Preservation
Explanation: Antiseptic are chemicals applied on body surfaces such

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as skin and mucous membrane, wounds, etc.

73. The process of diluting a sample of bacteria several times to obtain the appropriate colony number is known as

Answer: A

A. Morphological characteristics
B. Molecular techniques
C. Gram's staining
D. Serial dilution

Explanation: Serial dilution is the process of diluting a sample several times to obtain the appropriate colony number.

Review Exercise

Answer: D

I. Multiple Choice Items

1. Which one of the following is not true about fungal reproduction?
   A. Both sexual & asexual reproduction involves sporulation
   B. Opposite mating types of spores come together & produce genetically similarly nuclei
   C. Asexual reproduction produces thousands of genetically identical spores by mitosis
   D. Many asexual spores develop with in sporangium
2. Which one of the following members of microorganisms has plant and animal like characteristics?
   A. Virus B. Protozoa C. Fungi D. Archaea
3. Protozoan engulf food by the process of _____
   A. Phagocytosis C. Endocytosis
   B. Pinocytosis D. Exocytosis
4. The cell wall of a bacteria cell is
   A. Made of peptidoglycan C. Prevent cell rupture
   B. Anchors flagella D. All
5. What is the reason that gram negative bacterium can't retain crystal violet?
   A. Because of thin peptidoglycan on cell wall
   B. The presence thick peptidoglycan

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C. The presence of endotoxins
D. The presence of double membrane

6. Leishmaniasis is transmitted by the bite of
A. Glossina
B. Testes fly
C. Sandy fly
D. Domestic cat

7. Which one is incorrect about the characteristics of protozoa?
A. Have true nucleus
B. Mostly microscopic
C. Are eukaryotic
D. They are prokaryotic

8. Which fungal disease is transmitted through sexual intercourse?
A. Tinea
B. Asperilosis
C. Mycoses
D. Candidiasis

9. Disease of hair scalp, eyelashes, and eye brow are caused by a fungus called
A. Tinea capitis
B. Tinea corporis
C. Tinea pedis
D. Tines unguim

10. Which groups of archaea grow best at cold environment ?
A. Extreme Halophile
B. Methanogen
C. Psychrophilic
D. Termophiles

11. What is the electron source of Chemolithotrophs microbes?
A. Organic donor
B. Inorganic electron donor
C. Light
D. CO_2

12. Gram negative bacteria finally retain a color of ____ with gram stain?
A. Purple B. Pink C. Yellow D. Violet

13. Which of the following microbes cannot be cultured in nutrient medium?
A. Bacteria B. Fungi C. Virus D. Algae

14. Which of the following are spherical ___ shaped bacteria?
A. Bacilli
B. Cocci
C. Spirochetes
D. Spirilla

15. Which one of the following structures in bacteria cell is involved in conjugation?

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A. Flagella
B. Capsules
C. Pilli
D. Peroxisome

16. What is the counter stain used in Gram staining?
A. Ethanol
B. Iodine
C. Crystal violet
D. Safranin

17. What is the primary stain used in Gram staining?
A. Ethanol
B. Iodine
C. Crystal violet
D. Safranin

18. Which structure of bacteria is used for locomotion?
A. Pilli
B. Endospore
C. Flagella
D. Capsules

19. What is the role of the sex pilus in bacteria conjugation?
A. Transfer of genetic material
B. Attachment to surfaces
C. DNA replication
D. Locomotion

20. What is the main different between asexual and sexual reproduction in bacteria?
A. Sexual reproduction involves binary fission.
B. Asexual reproduction involves binary fission
C. Sexual reproduction results in genetically identical daughter cells.
D. Asexual reproduction involves genetic exchange.

21. Which microorganisms are grouped in prokaryotic?
A. Bacteria
B. Protozoa
C. Algea
D. Fungi

22. All of the following are the characteristic of Archaea, except?
A. Their cell wall lacks true peptidoglycan
B. Their have a membrane - bound nucleus
C. They reproduce asexually
D. They are unicellular

23. One of the following is the useful aspects of fungi?
A. Causing ergotism

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B. Production antibiotics
C. Spoiling agricultural products
D. Causing plant diseases

24. In conjugation
A. Two bacterial cells of different mating types come together, and genetic material is transferred from one to another
B. A bacterial cell develops a bulge that enlarges and eventually separates from the mother cell
C. Fragments of DNA released by a broken cell are taken in by another bacterial cell.
D. A phage carries bacterial genes from one bacterial cell another

25. Bacteria that are autotrophs
A. Do not require atmospheric oxygen for cellular respiration
B. Must obtain organic compounds from other organisms
C. Manufacture their own organic molecules
D. Get their nourishment from dead organisms

26. All of the following is the characteristic of true fungi, except _____
A. They have a rigid cell wall composed primarily of chitin
B. Most are filamentous
C. They all contain chlorophyll
D. They are all eukaryotic

27. What are rod - shaped bacteria called?
A. Cocci
B. Bacilli
C. Spirilla
D. Halophiles

28. What are thermoacidophiles?
A. Bacteria
B. Archaea
C. Spirochaetes
D. Cyanobactera

29. Genetic recombination in bacteria can occur during which process?
A. Conjugation
B. Binary fission
C. Heterocyst formation
D. Endospore production

30. Mycotoxicology is ?

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A. The study of fungal diseases in animals
B. The study of fungal toxins and their effects
C. The study of fungal genetics
D. The study of fungal structural

31. What is the name of fungal toxins and their effects the outermost areas of the human bod?

A. Systemic mycoses
B. Superificial mycoses
C. Mycetism
D. Mycotoxicoses

32. What is the unique feature of conjugation in protozoa?

A. Cell budding
B. Production of spores
C. Exchange of halpoid micronuclei
D. Formation of a zygote

33. What makes viruses different from other microorganisms.

A. They possess both DNA and RNA
B. They are larger in size compared to other microorganisms.
C. They have membrane bounded organelles.
D. They cannot carry out metabolic processes by their own

34. Which of the following do not belong to domain Archaea?

A. Methanogen
B. Thermophiles
C. Halophiles
D. Bacteriophages

35. What is the major different between the lytic and lysogenic cycles of bacteriophages?

A. The lytic cycle occurs outside the host cell, while the lysogenic cycle occurs inside the host cell.
B. In the lytic cycle, the phage DNA integrates into the host DNA, while in the lysogenic cycle, it does not.
C. The lytic cycle results in the lysis of host cells, while the lysogenic cycle does not
D. The lytic cycle involves the replication of the viral genome along with the host DNA, while the lysogenic cycle does not.

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36. One of the following is a unique feature of becteriophages?
A. Ability to infect bacteria
B. Lack of a protein coat
C. Replication through binary fission
D. Obligate intracellular parasites

37. Microbiota is characterized by?
A. Protective role against pathogens
B. Can't be changed in an individual's life time
C. Cause diseases under all circumstances.
D. Consist of harmful parasites

38. Peptidoglycan is a chemical compound found in the walls of
A. Most viroids
B. Most archaea
C. All prokaryotes
D. Most bacteria

39. Pasteurizations is
A. Sterilization of foods in liquid from
B. Sterilization of medical instruments
C. Killing all microorganisms from food
D. Destruction of infections material

40. What are fungi that feed on decaying organic matter called?
A. Parasites
B. Mutualists
C. Symbionts
D. Saprophytes

41. Lichens represent, a symbiotic association between a fungus and which other types of organism?
A. An alga B. A plant C. A mold D. A rhizoid

42. In a mycorrhiza, a fungus lives in a symbiotic relationship with which of the following?
A. A virus
B. A plant
C. A bacterium
D. A slime mold

43. Which statement about fungi is not true?
A. A multicellular fungus has a body called a mycelia

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B. Hyphae are composed together & form mycelia.
C. Most fungi are multicellular.
D. Fungi have on environmental benefit

44. The sac like structure on hypha, which produces sporangiospore is called
A. Sporangium
B. Basidiospores
C. Conidia
D. Gametes

45. Plasmid is
A. Circular protein molecules
B. Tine bacteria
C. Required by bacteria
D. Used to confer resistance gene for antibiotics.

46. The capsid of virus consists of
A. Protein subunits
B. Nucleic acid
C. Helical lipids
D. RNA and DNA

47. Viruses that kill host cells are
A. Lysogenic
B. lytic
C. Prophages
D. Temperature

48. Which one of the following structure of bacteria is mismatched?
A. DNA → Circular structure
B. Ribosome → Protein synthesis
C. Cell wall → Made of peptidoglycan
D. Cell wall → Made of cellulose

49. All viruses have
A. DNA
B. RNA
C. Protein coat
D. Both DNA and RNA

50. All of the following structures are common to all bacteria, except
A. DNA
B. Cytoplasm
C. Cell membrane
D. Flagella

51. Fungi uses all mode of nutrition except;

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A. Photosynthesis C. Saprophytic
B. Heterotrophic D. Symbiosis

52. Which one is the correct procedure to differentiate bacteria based on Gram staining?
A. Crystal violet → Safranin → Decolorisation /by ethanol/ → Iodine.
B. Safranin → Crystal violet → Iodine → Decolorisation /by ethanol/.
C. Crystal violet → Iodine → Decolorisation /by ethanol/ → Safranin.
D. Ethanol → Iodine → Safranin → Crystal violet.

53. Viruses cannot reproduce themselves in the absence of a host b/c;
A. All are obligate parasite C. It is acellular
B. It is non-living D. A & C

54. One of following structure is common to bacteria and plant.
A. Plasmid C. Capsule
B. Flagella D. Cell wall

55. Which one of the following pathogens is mismatched with the disease it causes?
A. Protozoa → Influenza C. Viruses → Measles
B. Fungi → Athlete's foot D. Bacteria → Pneumonia

56. Which of the following infection is different from the others.
A. Pneumonia C. Cholera
B. TB D. Candidiasis

57. Droplet infections are transmitted in tiny droplets through:
A. Breathing C. Sneezing
B. Coughing D. All

58. The nitrogen fixing bacteria that fixes free N_2 in to NH_3 or ammonium ion
A. Rhizobium C. Nitrosomonas
B. Pseudomonas D. Nitrobacter.

59. Disease causing bacteria can be transmitted by _____
A. Sexual intercourse B. Eating contaminated food

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C. Droplet infection D. All of the above

60. The bacteria infecting viruses are called ____

A. Insectivores C. Macrophage

B. Bacteriophages D. T - helper

61. Viruses are different from bacteria in that:

A. All are parasite

B. Incapability to reproduce by themselves.

C. Smaller in size

D. All of the above

62. Which of the following cannot be recycled?

A. Carbon C. Nitrogen

B. Energy D. Phosphorus

63. Saprophytic fungi;

A. Obtain nutrients from live organisms

B. Feeding on dead organic matter

C. Are chemosynthetic

D. Can photosynthesize

64. During nitrogen cycle, nitrate converts into nitrogen gas by;

A. Denitrifying Bacteria C. Decomposers

B. Nitrogen fixed fixing bacteria D. Nitrifying bacteria.

65. A viral life cycle where the viral genome is finally kills the host cell is

A. Lytic cycle C. Lysopenic cycle

B. Chronic release D. Plasmid

66. The structure of bacteria, which help to acquire antibiotic resistance gene is

A. Pilli B. Flagella C. Capsule D. Plasmid

67. Gram negative bacteria differ from gram positive bacteria in that gram negative bacteria has

A. Thick cell wall C. Easily curable

B. Endotoxin D. Purple color gram stain

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68. All are true protozoans, except
A. They are unicellular
B. They have cell wall
C. They are eukaryotes
D. Some are photosynthetic

69. Which group of bacteria in nitrogen cycle is responsible for decreasing the amount of nitrogen in the soil?
A. Nitrogen fixing bacteria
B. Ammonifying bacteria
C. Denitrifying bacteria
D. Nitrifying bacteria

70. Which one of the following process is not sulphur cycle?
A. Photochemical oxidation
B. Combustion
C. Acidic rain
D. Photosynthesis

71. Which of the following process Carbon cycle doesn't contribute for the increments of CO_2 in the atmosphere
A. Combustion
B. Respiration
C. Decomposition
D. Photosynthesis

72. Which one of the followings is the harmful aspect of fungi? Their role:
A. Biodegradation
B. In the manufactures of antibiotics
C. Deterioration of usable materials
D. As a sources of food

73. Fungi disease such as dermatomycosis can be transmitted through:
A. Direct Contact
B. Droplet infection
C. Sexual intercourse
D. Contaminated water and food

74. Microbes that are sensitive to oxygen are called
A. Obligate anaerobes
B. Obligate aerobes
C. Facultative
D. Micriaerophiles

75. Which of the following in not the characteristic feature of fungi?
A. Most of them are multicellular
B. All are heterotrophs
C. Most of the are fillamentous
D. Some or them are prokaryotic

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76. Which statement is true about gram positive bacteria?

A. Have outer membrane
B. Stained pink by Grams stain
C. Stained purple by Grams stain
D. Have thin peptidoglycan cell wall

77. Why do you think that microorganisms such as bacteria & fungi decompose dead organic matter?

A. To clean up the ecosystem
B. To release mineral nutrients for plants
C. To keep the balance of atmospheric gases
D. To derive energy for their own use

78. Choose the organism which belongs to eukaryotic

A. Blue - green algae
B. Amoeba
C. Bacteria
D. Viruses

79. Which of the following disease is correctly matched with its causative agent?

A. Malaria - fungus
B. AIDS - virus
C. Ringworm - protozoa
D. Syphilis - worm

80. Among the following organism, which one belongs to the prokaryotes?

A. Paramecium
B. Streptococcus
C. Spirogyra
D. Tapeworm

81. What is the role of microorganisms such as bacteria and fungi in the ecosystem?

A. Antibiotic production
B. Supplying energy to the ecosystem
C. Recycling of nutrients
D. Synthesis organic substance

82. What is the main mode of transmission of disease such as cholera and typhoid fever?

A. Bites of animal vectors
B. Sexual intercourse
C. Drinking contaminated water
D. Blood - to - blood contacts

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83. What are the individual strands of a fungal filament called?

A. Fungus B. Spores C. Hyphae D. Hydra

84. In which par of the cell do Gram - positive and Gram - negative bacteria differ regarding their staining property with gram's stain?

A. Cytoplasm C. Cell wall

B. Cell membrane D. Nucleus

85. Among the following human diseases, identify the one that is more associated with skin infection?

A. Leprosy C. Gonorrhoea

B. Chancroid D. Leishmaniasis

86. In which type of agricultural crops do nitrogen fixing bacteria is more associated?

A. Legumes C. Vegetables

B. Cereals D. Fruits

II. Short and Essay Types

87. In the lytic cycle, the host cell become ___, releasing phage particles.

88. A bacterium can transfer its genes to another bacterium by ___, ___, & ____.

89. The type of reproduction, in which a bacterium attaches to another bacterium and passes fragment of its DNA to the recipient cell?

90. The small bacterial chromosomes that are dependent of the main chromosome is called _____ ?

91. Summarize the ecological significance of fungi as decomposers?

92. Describe the important ecological role of mycorrhizae?

93. Explain why viruses are using biochemical machinery of a living cell in order to reproduce.

94. There are many types of viruses, classified by their _____

95. Discuss about lysogenic cycle of a phage?

96. Describe the structure and common shapes of prokaryotic cells?

97. Contrast the bacterial cell wall in gram - positive and gram - negative bacteria.

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98. Describe the function of fimbriae and pili for bacteria ?

99. Describe asexual reproduction in prokaryotes?

100. Compare characteristics of Archaea and Bacteria.

101. Imagine that you discover a new microorganisms. After careful study you determine that it should be classified in domain Archaea. What characteristics might lead you to this decision?

Answer Key to review Exercise

I. Multiple choices

1. B	10.C	19.A	28.B	37.A	46.A	55.A	64.A	73.A	82.C
2. B	11.B	20.B	29.A	38.D	47.B	56.D	65.A	74.B	83.C
3. A	12.B	21. A	30.B	39.A	48.D	57.D	66.A	75.D	84.C
4. D	13.C	22.B	31.B	40.D	49.C	58.A	67.B	76.B	85.D
5. A	14.B	23.B	32.C	41.A	50.D	59.D	68.B	77.D	86.A
6. C	15.C	24.A	33.D	42.B	51.A	60.B	69.C	78.B	
7. D	16.D	25.C	34.D	43.D	52.C	61.D	70.D	79.B	
8. D	17.C	26.C	35.C	44.A	53.D	62.B	71.D	80.B	
9. A	18.C	27.B	36.A	45.D	54.D	63.B	72.C	81.C	

II. Short answer and essay

87. Bursts

88. Conjugation, transformation, or transduction

89. Conjugation

90. Plasmid

91. **Most fungi** are decomposers that break down organic compounds in dead organisms, leaves, garbage, and wastes into simpler nutrients that can be recycled.

92. **Mycorrhizae** are mutualistic associations between fungi and the roots of plants. The fungus supplies water and nutrient minerals to plants and fungus obtain food from plants .

93. Because virus don't have cellular organelles to make metabolism to reproduce.

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94. Size and shape, by their genetic material (RAN or DNA), or by their host organism.

95. Some phage undergoes a lysogenic cycle, in which their DNA is inserted into the host chromosome, where it replicates for generations. When conditions are appropriate, the phage DNA exits the host chromosome and enters a lytic cycle

96. **Prokaryotic cells** are very small and do not have membrane enclosed organelles such as nuclei mitochondria. Prokaryotic cells have several common shapes: spherical (cocci), rod - shaped (bacilli), and spiral.

97.
- The walls of gram - positive bacteria are very thick and consist mainly of peptidoglycan.
- The cell walls of gram - negative bacteria consist of a thin peptidoglycan layer and an outer membrane

98. Some prokaryotes have hairlike appendages called fimbriae (pili). Both fimbriae and pili help cells adhere to one another or to certain other surfaces, including cells they infect.

99. Prokaryotes reproduce asexually by
- **Binary fission** (the cell divides, forming two cells)
- **Budding** (a bud forms and separates from the mother cell),
- **Fragmentation** (walls form inside the cell, which then separates into several cells).

100.
- Prokaryotes are assigned to domain Archaea and domain Bacteria.
- The cell wall of archaea do not have peptidoglycan, but bacteria have cell wall made of peptidoglycan.

101.
- Archaea cell walls do not have peptidoglycan.
- Archaea cell walls have different amino acids than bacteria cell walls do.
- Archaea also have different types of lipids in their cell membranes, etc.

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UNIT 9

9. Energy Transformation

Energy transformation is the conversion of energy from one form to another form by living organisms. For example:

Respirations is the process by which living things convert the chemical energy in glucose into other form of energy (ATP), that is used to fuel cellular activities through metabolism.

Photosynthesis: is the process of transforming energy of the sun into chemical energy.

Metabolism is the sum of chemical reactions that takes place within each cell of an organism.

Metabolism can be anabolism and catabolism.

1. **Catabolism** - is reaction that are involved in the breakdown of complex molecules into smaller pieces.
2. **Anabolism** - is reactions that are involved in the synthesis of complex molecules from smaller pieces.

The comparison between anabolism and catabolism:

No	Anabolism	Catabolism
1	It is the construction phase of metabolism	It is the destructive phase of metabolism
2	It is the process where by simpler substance are joined together to form complex macromolecules.	It is the process where by complex macromolecules are broken down to form simpler substance or monomers.

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3	The process requires energy to construct substances	The process releases energy as a result of the breakdown of molecules
4	It is an endergonic (energy absorbing) reaction	It is an exergonic (energy - releasing) reaction.
5	It occurs during photosynthesis	It occurs during cellular respiration

9.1 Photosynthesis

Photosynthesis is the sequence of events by which light energy is converted into the stored chemical energy of organic molecules

It is series of chemical reaction that light energy to assemble CO_2 into glucose ($C_6H_{12}O_6$) and other carbohydrates.

$$6\,CO_2 + 12\,H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Carbon dioxide — Water — Glucose — Oxygen — Water

Photosynthesis is the first step in the flow of energy through most of the living world, capturing the vast majority of the energy that living organisms use. Leaf is major site of photosynthesis in plants.

External and Internal Structure of the Leaf

A. External Structures of the Leaf

The outer leaf layer is known as the epidermis.

- It is one cell thick.
- It secretes a wax coating called the **cuticle** that prevent water loss
- It provides protection
- It has pores called **stomata** which allow entry of carbondioxide for photosynthesis.
- The opening and closing of the stomata is controlled by the **guard cells.**
- Most leaves are flat in shape, which **increases the surface area** to absorb light energy from the sun
- The external structures of a leaf contain three main parts. These are: Leaf base, Leaf lamina, and petiole

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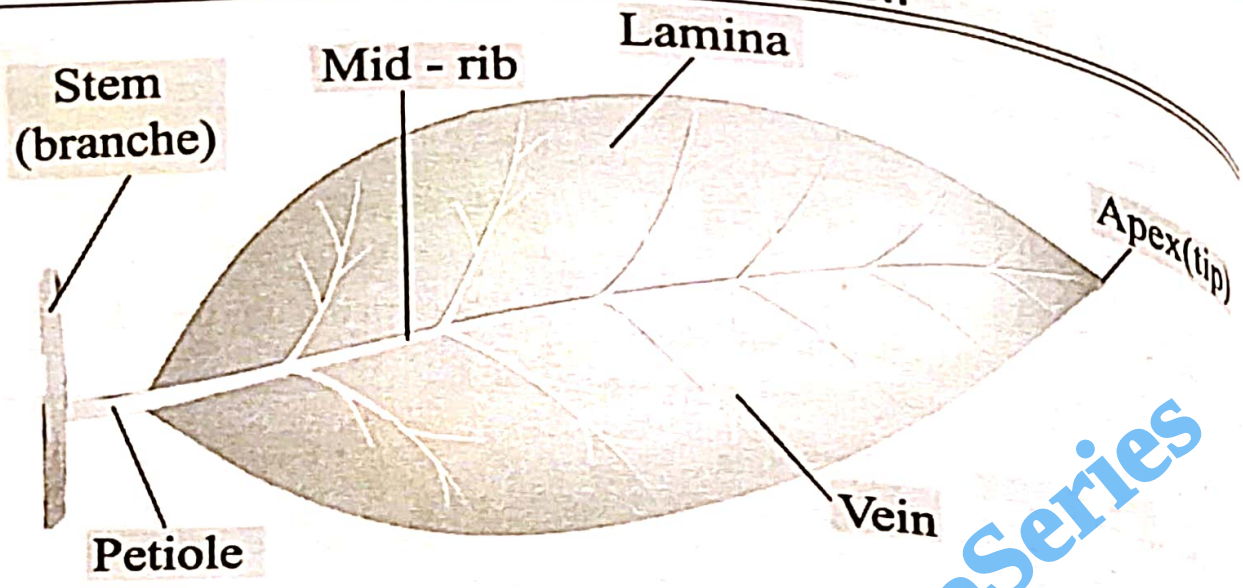

B. Internal structure of the leaf - the mesophylls

The mesophylls are made up of specialized **parenchyma cells**

- It is found between the lower and upper epidermis of a leaf.
- They are specialized for **photosynthesis** because it contains **chloroplast.**
- They include: palisade mesophylls, spongy mesophylls

i. Palisade mesophylls

- They are cells closely packed together near the surface of the leaf where they receive more sunlight
- They contain **more chloroplasts** than spongy mesophyll cells.
- They are the main photosynthetic sites of the plant.

ii. Spongy mesophylls

In three dimensions they are spongy in appearance because they have **many large air spaces** between the cells.

They have fewer cells with fewer chloroplasts as the result some photosynthesis takes place.

Here carbon dioxide needed for photosynthesis moves into the cells and the oxygen moves out.

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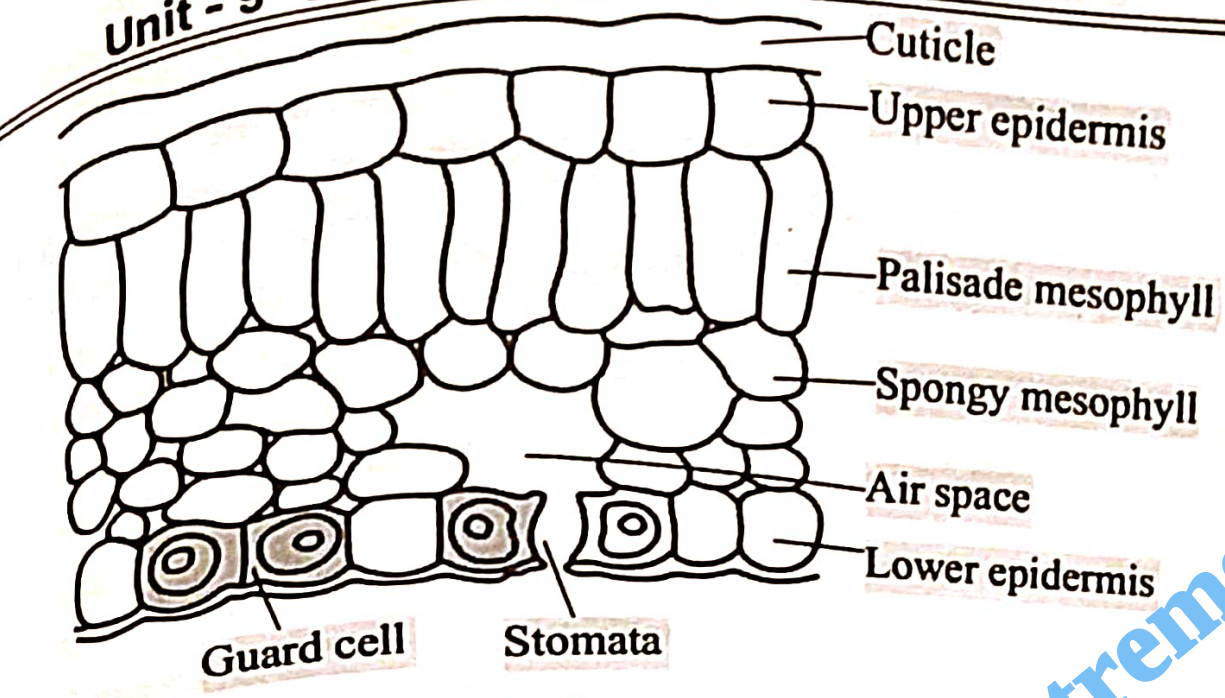

The site of photosynthesis

The highest density of chloroplasts is found in the mesophyll cells of leaves. Chloroplast is an organelle containing photosynthetic pigment called chlorophyll

Each chloroplast is surrounded by a pair of membranes. Outer and inner membrane.

- Inside the inner membrane is another system of membranes called **thylakoids** that are arranged as flattened sacs.
- Within the thylakoids there are photosynthetic pigment called **chlorophylls.**

The thylakoids are connected and layered to form stacks called **grana** (singular, granum), where the light - dependent stage of photosynthesis takes place.

Surrounding the grana is a solution (fluid) called the **stroma,** where the light - independent stage of photosynthesis takes place.

The diagram below the parts of chloroplast.

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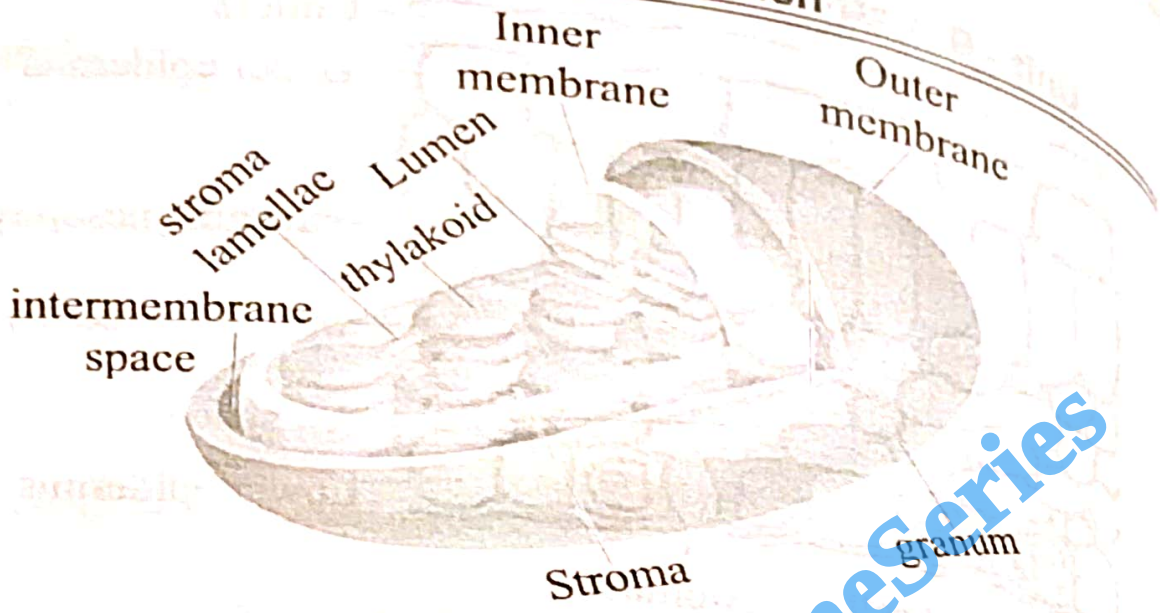

9.1.1 Absorption Spectra and Photosynthetic Pigment

Absorption spectrum is a graph that shows absorption of light from a spectrophotometer.

Light is composed of small particles called **photons** that move as waves

- Photons with shorter wavelengths have more energy than those with longer wavelength

The visible spectrum of light contains all the color of rainbow

- The visible light spectrum ranges within the wave length of 380 to 760nm
- The colors of visible spectrum of light according to increasing their wave length can be arranged as: violet, Indigo, Blue, Green, Yellow, Orange & Red

Photosynthetic cells contain special pigments that absorb light energy

There are three basic classes of pigments. There are: **chlorophyll**, and **accessory pigments** (Caratenoid, and phycobilins)

1. Chlorophyll

Chlorophylls are located on the membrane of the thylakoids

- They are greenish pigments which contain a porphyrin ring.
- The ring provides energized electrons to other molecules.

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There are several deferment types of chlorophylls.

- The two most common types are know as **chlorophyll a** and **chlorophyll b**.
- Chlorophyll a, the pigment that initiates the light - dependent reaction of photosynthesis.
- Chlorophyll b is an accessory pigment that also participates in photosynthesis
- Neither chlorophyll a nor chlorophyll b absorbs much green light. Instead, they allow green light to be reflected or transmitted.
- For this reason, leaves and other plants structure that large amounts of chlorophyll look green

Chlorophyll 'a'

- It absorbs violet - blue and reddish orange wavelengths.
- It is a green pigment found in all plants, algae and cyanobacteria.

2. Accessory pigments

Absorb colors that chlorophyll a cannot absorb, they enable plants to capture more of the energy in light. These are chlorophyll b, carotenoid and phycobilins

i. Chlorophyll "b"

- It occurs only in green algae and in plants
- It absorbs mostly blue and yellow light

ii. Chlorophyll "c"

It is is found only in the photosynthetic members of the Chromista and dinoflagellates.

iii. Carotenoids

Carotenoids are accessory photosynthetic pigments, they cannot transfer sunlight energy directly to the photosynthetic pathway.

- They pass their absorbed energy to chlorophyll a
- ☑ They are usually red, orange or yellow pigments.

Carotenoids absorb different wavelengths of light than chlorophyll

- They absorb in the blue - green region of the solar spectrum
- ☑ They include the familiar compound carotene which gives carrots their color.

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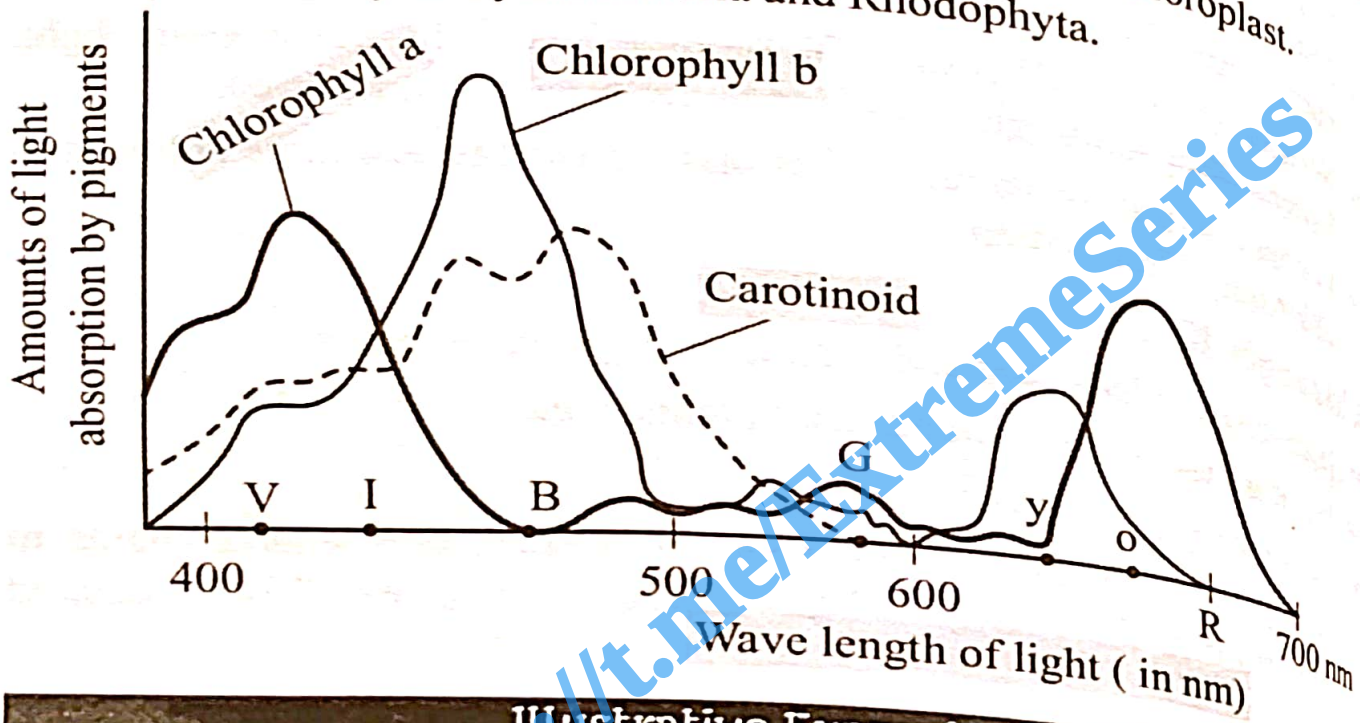
One very visible accessory pigment is **fucoxanthin** the brown pigment whose colors keep other brown algae as well as the diatoms.

iv. Phycobilins

- They are water - soluble pigment
- They found in the cytoplasm or in the stroma of the chloroplast.
- They occur only in Cyanobacteria and Rhodophyta.



Illustrative Example

3. What term is used to describe the chemical process of living organism that involves both the joining up of monomers to produce polymers and the splitting of polymers to monomers?

A. Anabolism
B. Catabolism
C. Metabolism
D. Polymorphism

Explanation

- **Metabolism** is the sum of all the chemical reactions that occur inside the body. It involves anabolism and catabolism.
- **Anabolism** involves the joining up of monomer units to form polymers
- **Catabolism** is splitting of polymers to monomers.

Answer: C

4. Which of the following pairs of organelles are concerned with energy transformation?

A. Nucleus and nucleolus
B. Mitochondria and nucleus

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